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SOLID-VAPOR EQUILIBRIUM OF THE
CARBON DIOXIDE-NITROGEN SYSTEM
AT PRESSURES TO 200 ATMOSPHERES

Gene Everett Smith

A dissertation submitted in partial fulfillment
of the requirements for the degree of
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1963

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To my wife, Darlene Rupp Smith

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NOMENCLATURE

Symbols

A	Helmholtz function
A	Beattie-Bridgeman function defined by (4-3)
A	Benedict-Webb-Rubin function defined by (5-85)
A	Martin-Hou constant
A_0	Beattie-Bridgeman constant
A_0	Benedict-Webb-Rubin constant
a	Beattie-Bridgeman constant
a	Benedict-Webb-Rubin constant
B	Martin-Hou constant
B	Beattie-Bridgeman function defined by (4-4)
B	Benedict-Webb-Rubin function defined by (5-86)
B	Second virial coefficient
B_0	Beattie-Bridgeman constant
B_0	Benedict-Webb-Rubin constant
B^*	Reduced second virial coefficient
b	Beattie-Bridgeman constant
b	Benedict-Webb-Rubin constant
b	Martin-Hou constant
b_0	Lennard-Jones force constant
C	Martin-Hou constant
C	Third virial coefficient
C	Beattie-Bridgeman function defined by (4-5)
C_0	Benedict-Webb-Rubin constant
C^*	Reduced third virial coefficient
c	Beattie-Bridgeman constant

NOMENCLATURE (CONT'D)

c	Benedict-Webb-Rubin constant
D	Fourth virial coefficient
D	Martin-Hou function defined by (5-130)
F	Martin-Hou function defined by (4-12)
F	Kihara function defined by (4-36)
$\bar{F}$	Martin-Hou function defined by (5-124) and (5-127)
f	Fugacity
f	Function defined by (4-17)
$\bar{F}$	Fugacity of a mixture component
G	Gibbs function
g	Molar Gibbs function
H	Enthalpy
K	Dummy constant for writing general mixture rules
k	Martin-Hou constant
k	Boltzmann's constant
M_0	Kihara parameter
N	Avogadro's number
n	Number of moles
P	Pressure
Q	Quadrupole moment
Q^*	Reduced quadrupole moment
R	Gas constant
r	Molecule separation
S	Entropy
S_0	Kihara parameter
s	Molar entropy

NOMENCLATURE (CONT'D)

T	Temperature
T^*	Reduced temperature
U	Internal energy
U_0	Kihara force constant
u	Molar internal energy
V	Volume
V_0	Kihara parameter
v	Molar volume
y	Mole fraction
Z	Dimensionless Kihara parameter
α	Benedict-Webb-Rubin constant
β	Second virial in Beattie-Bridgeman equation
γ	Third virial in Beattie-Bridgeman equation
γ	Benedict-Webb-Rubin constant
$\bar{\gamma}$	Beattie-Bridgeman function defined by (5-46) and (5-50)
δ	Fourth virial in Beattie-Bridgeman equation
$\bar{\delta}$	Beattie-Bridgeman function defined by (5-47) and (5-51)
ϵ	Lennard-Jones force constant
Φ	Enhancement factor
ϕ	Intermolecular potential function
ρ	Molecular core separation
σ	Lennard-Jones force constant

Subscripts

1, 11, 111	Pure component 1 (Carbon Dioxide)
2, 22, 222	Pure component 2 (Nitrogen)

NOMENCLATURE (CONT'D)

12, 112, 122	Interaction subscripts
2, 3, 4, 5	Martin-Hou constant subscripts
c	Critical property
i, j, k	Indices
m	Gas mixture
q	Quadrupole correction

Superscripts

g	Gas phase
s	Solid phase
o	Saturation state
*	Ideal gas state
—	(over property)-Partial molar property

I. INTRODUCTION

In many present day purification processes impurities are removed by condensing them from pressurized gas systems. However, in order to design the equipment required for such processes, it is necessary to have access to phase equilibrium data which often are not available. It becomes desirable, therefore, to have some means for analytically predicting equilibrium compositions with a fair degree of accuracy. For this reason, the two-phase binary equilibrium system in which the concentration of one of the components in the condensed phase is negligibly small is of particular interest because such a system affords a comparatively easy means for analytically determining the composition of the gas phase. An example of such a binary system is carbon dioxide-nitrogen over a wide pressure and temperature range, since the solubility of nitrogen in solid carbon dioxide may be assumed negligibly small.

A considerable number of experimental and analytical investigations have been performed with two-phase binary systems in which the condensed phase is essentially a pure component. However, in many of these previous studies, the pressure and temperature range was such that the condensed-phase component was present in the gas phase in only small amounts and the analytical calculations were simplified accordingly. Only limited data have been collected and analytical calculations performed in regions where the concentration of the condensed-phase component in the gas phase is sufficiently

large so that such simplifications result in appreciable errors, thus necessitating a more general and complicated solution.

It would be desirable to obtain additional data in this latter region so that various methods employed in the analytical calculations and any assumptions made therein could be evaluated by direct comparison with the experimental data. Such an investigation would not only add to the sum total of available phase equilibrium data but could well lead to an accurate method for analytically predicting equilibrium compositions for systems for which little or no data are known. For these reasons it is anticipated that the investigation reported here will be of value in further phase equilibrium studies.

II. PREVIOUS WORK

A. Experimental

Several investigators have collected phase equilibrium data for two-phase binary systems containing an essentially pure condensed phase. One of the earliest such works is that of Hannay and Hogarth ⁽¹⁾ who studied the solubility of solid potassium iodide in alcohol and other vapors. Several years later Pollitzer and Strebel ⁽²⁾ analyzed the effects of nitrogen and hydrogen on the liquid-vapor equilibrium of carbon dioxide, as well as the effect of air, nitrogen and hydrogen on water liquid-vapor equilibrium. In addition, Bartlett ⁽³⁾ has presented his results on a study of the solubility of water in hydrogen and other gases. Lurie and Gillespie ⁽⁴⁾ used the solid amine $\text{Ba Cl}_2 \cdot 8 \text{NH}_3$ with nitrogen to study the solubility of ammonia in nitrogen. In 1929, the effect of carbon dioxide on the vapor pressure of iodine up to fifty atmospheres pressure was reported by Braune and Strassmann ⁽⁵⁾. Saddington and Krase ⁽⁶⁾ analyzed the liquid-vapor equilibrium of water in the presence of nitrogen at pressures to three hundred atmospheres. Field ⁽⁷⁾ investigated the behavior of hydrogen and ammonia, his system being similar to that employed in (4) except that hydrogen was used rather than nitrogen. A few years later Perkins ⁽⁸⁾ extended the works of (4) and (7) with his studies of the effect of helium, argon and nitrogen on the decomposition of the solid amine $\text{Ba Cl}_2 \cdot 8\text{NH}_3$. More recently, Kritchevsky and Koroleva ⁽⁹⁾ determined the effect of hydrogen,

nitrogen, carbon dioxide and methane on the liquid-vapor equilibrium of methanol at pressures as high as seven hundred atmospheres. At approximately the same time, Wiebe and Gaddy⁽¹⁰⁾ investigated the water-carbon dioxide system. The following year, Kritchevsky and Gamburg⁽¹¹⁾ analyzed the behavior of benzene in nitrogen. Gratch⁽¹²⁾ studied the effect of air on the solid-vapor equilibrium of carbon dioxide at pressures up to fifty atmospheres. Shortly thereafter, Ipatev, et al.⁽¹³⁾ investigated the equilibrium of benzene, as in (11), but with hydrogen rather than nitrogen. In addition, Diepen and Scheffer⁽¹⁴⁾ studied the influence of ethylene gas on the vapor pressure of naphthalene. In 1950, Webster⁽¹⁵⁾ reported his results on the effect of air on the liquid-vapor and solid-vapor equilibria of water. An additional study was made by Webster⁽¹⁶⁾ on the carbon dioxide-air system analyzed by Gratch⁽¹²⁾ but at pressures up to two hundred atmospheres. Robin^(17,18) investigated the solubility of phenanthrene in compressed nitrogen as well as other gases. In 1954, Donnelly and Katz⁽¹⁹⁾ reported their findings on the liquid-vapor equilibrium of the carbon dioxide-methane system. In a recent study, Dokoupil, et al.⁽²⁰⁾ determined the effect of hydrogen on the solid-vapor equilibria of the nitrogen, carbon monoxide, and nitrogen-carbon monoxide systems at pressures up to fifty atmospheres. In another recent investigation Ewald⁽²¹⁾ determined the solubility of solid xenon in helium, hydrogen, and nitrogen, as well as that of solid carbon dioxide in helium and hydrogen. Jepson, et al.⁽²²⁾ investigated the solubility of liquid

mercury in several gases at pressures to 30 atmospheres. Shortly thereafter Pikaar⁽²³⁾ studied the solid-vapor and liquid-vapor equilibria of the carbon dioxide-methane system. Prausnitz and Benson⁽²⁴⁾ have reported their investigations on the liquid-vapor equilibria of various liquids in hydrogen, nitrogen and carbon dioxide. The effect of nitrogen on the solid-vapor equilibrium of carbon dioxide has been investigated by Sonntag⁽²⁵⁾ and reported by Sonntag and Van Wylen⁽²⁶⁾. This system is identical to the one reported here. However, the investigation was performed at a maximum pressure of one hundred atmospheres. In 1961, Dokoupil⁽²⁷⁾ reported his findings on the solid-vapor and liquid-vapor equilibria of the nitrogen-hydrogen system.

Many investigations have been made on binary systems in which both components were present in appreciable amounts in the condensed phase. A few of these investigations are of interest in that one of the components is carbon dioxide, this being one of the components in the system under study. One such investigation is that of Poettman and Katz⁽²⁸⁾ who have presented phase diagrams for carbon dioxide with butane, pentane, and propane. Clark and Din⁽²⁹⁾ studied the solid-liquid and liquid-vapor equilibria of the carbon dioxide-acetylene system among others. In 1951, Haselden, et al.⁽³⁰⁾ presented their investigative results on the carbon dioxide-ethylene and carbon dioxide-propylene systems. Later Bierlein and Kay⁽³¹⁾ studied the carbon dioxide-hydrogen sulfide system. This system was further analyzed by Sobocinski and Kurata⁽³²⁾. More recently Sterner⁽³³⁾ extended the work of Donnelly and

Katz⁽¹⁹⁾ on the carbon dioxide-methane system. In 1962, Davis, et al.⁽³⁴⁾ presented their results on the three phase equilibrium of the carbon dioxide-methane system. Haselden and Snowden⁽³⁵⁾ have studied the solid-vapor and liquid-vapor equilibria of the carbon dioxide-cyclopropane and carbon dioxide-propene systems. Finally, in 1963 Vonderheiden and Eldridge⁽³⁶⁾ reported their findings on the liquid-vapor equilibrium of the carbon dioxide-methylene chloride system.

Only a few investigations have been made to determine the properties of the carbon dioxide-nitrogen system. Abdullaev^(37,38) and Mills and Miller⁽³⁹⁾ attempted to determine the liquid-vapor phase diagram for this system but their results were rather incomplete. However, Dana and Zenner⁽⁴⁰⁾ have recently reported their findings on the liquid-vapor phase diagram for the carbon dioxide-nitrogen system as well as the carbon dioxide-oxygen system. Their investigation on the CO_2 - N_2 system was conducted along three isotherms at pressures to one hundred-thirty atmospheres. Ishkin and Burbo⁽⁴¹⁾ as well as Fedorova⁽⁴²⁾ studied the solubility of solid carbon dioxide in liquid nitrogen over a narrow pressure and temperature range. The single phase behavior of gaseous mixtures of carbon dioxide and nitrogen has been investigated by Kritchevsky and Markov⁽⁴³⁾ Michels and Boerboom⁽⁴⁴⁾, Pfefferle, et al.⁽⁴⁵⁾, Zaalishvili⁽⁴⁶⁾, and Cottrell, et al.⁽⁴⁷⁾. However, the majority of these investigations were conducted at or above room temperature. In addition to the preceding investigations, the only other known study performed on the CO_2 - N_2 system is that of Sonntag⁽²⁵⁾

which has been mentioned previously. Sonntag conducted a rather extensive investigation on the solid-vapor equilibrium of this system over six isotherms from 140°K to 190°K and at pressures to one hundred atmospheres. The present study is an extension of that investigation to higher pressures.

B. Analytical

Several investigators have utilized various analytical methods in an attempt to predict the gas phase composition of binary systems having an essentially pure condensed phase. One such study was performed by Gratch⁽¹²⁾ who analyzed his data on the CO₂ - Air system using the Beattie-Bridgeman equation of state. A few years later, Robin, et al.⁽⁴⁸⁾ used a statistical mechanics virial equation of state through the second virial coefficient in conjunction with the Lennard-Jones potential in an attempt to predict the data of Robin^(17,18). In 1953, Robin and Vodar⁽⁴⁹⁾ utilized the same method employed in the preceding reference in an effort to analytically predict the data of references 3, 5, 9, 10, 11, 13, and others. In the same year, Ewald, et al.⁽⁵⁰⁾ compared the data of references 14, 16 and others with a theoretical solution using a virial equation through the third virial coefficient along with the Lennard-Jones potential. An empirical combination of critical constants was used in this investigation to estimate Lennard-Jones force constants for evaluating interaction coefficients. At a later date, Ewald⁽²¹⁾ improved upon this method by utilizing an empirical combination of the pure substance Lennard-Jones force

constants rather than an empirical combination of critical constants to determine interaction coefficients. Ewald⁽²¹⁾ then compared his experimental data with the analytical results based upon this latter procedure for calculating interaction coefficients. The results of these comparisons were quite favorable. Dokoupil, et al.⁽²⁰⁾ used the Beattie-Bridgeman equation of state in a simplified solution to predict their experimental data. Their analytical predictions were quite accurate over a moderate pressure range. In 1959, Prausnitz⁽⁵¹⁾ developed a generalized procedure for evaluating interaction coefficients based on the 3-parameter theory of corresponding states proposed by Pitzer, et al^(52,53). Prausnitz applied his method to the data for one isotherm taken from reference 14. Ziegler⁽⁵⁴⁾ and Kirk, et al.⁽⁵⁵⁾ used the methods of references 20, 21 and others along with the empirical Benedict-Webb-Rubin equation of state to perform calculations for the solid-vapor and liquid-vapor equilibria of the methane-hydrogen system. Prausnitz and Myers⁽⁵⁶⁾ have recently calculated second virial pure substance and interaction coefficients based on the Kihara potential for several cryogenic fluids and their mixtures. In addition, Prausnitz and Myers utilized these pure substance and inter-action coefficients along with third virial pure substance coefficients based on the Lennard-Jones potential and interaction coefficients determined by the method of Sutton⁽⁵⁷⁾ to analytically predict the data of Gratch⁽¹²⁾ and Webster⁽¹⁶⁾. Their analytical results compared favorably with the experimental data.

In 1960, Sonntag⁽²⁵⁾ performed extensive calculations on the solid-vapor equilibrium of the carbon dioxide-nitrogen system and compared the results with his experimental data. Inasmuch as this system is identical to that under investigation here, Sonntag's analytical results are of considerable interest. The calculations were made using the Beattie-Bridgeman equation of state as well as a statistical mechanics virial equation of state through the third virial coefficient with the pure substance coefficients evaluated using the Lennard-Jones potential and interaction coefficients determined by the methods of both Ewald⁽²¹⁾ and Prausnitz⁽⁵¹⁾. The analytical results of this investigation agreed closely with the experimental data in, and in some cases beyond, the density range for which the equations of state can be expected to accurately represent pressure-volume-temperature data. Of the three methods employed by Sonntag, Ewald's was in general the best, followed closely by that using the Beattie-Bridgeman equation. The method of Prausnitz gave much poorer results compared to the experimental data than the other two. Recently Smith, et al.⁽⁵⁸⁾ extended the analytical investigation conducted by Sonntag by using the Benedict-Webb-Rubin and Martin-Hou equations of state in their analytical calculations. The analytical results based on these two equations were superior at high densities to any of those based on the three methods employed by Sonntag in his initial work. This is to be expected since both the Benedict-Webb-Rubin and Martin-Hou equations of state have been shown to accurately predict pressure-

volume-temperature data at densities far exceeding the range of applicability of the three equations used by Sonntag. Regarding the relative merit of the two equations in predicting experimental data, Smith, et al.⁽⁵⁸⁾ indicate that the Martin-Hou equation is considerably better than the Benedict-Webb-Rubin for this particular system.

III. DERIVATION OF THEORETICAL EQUATIONS

A. General

In order to analytically predict the gas-phase composition of a binary solid-vapor equilibrium system in which the solid phase is essentially a pure component, it is necessary to obtain a theoretical equilibrium equation which is applicable to the system under consideration. Such an equation has been previously developed by various authors and the following derivation closely parallels the work of these authors, especially that of Sonntag⁽²⁵⁾ and Beattie and Stockmayer⁽⁵⁹⁾.

B. Thermodynamic Property Relations

Consider the following thermodynamic property relation written for a binary mixture

$$dU = TdS - PdV + \bar{G}_1 dn_1 + \bar{G}_2 dn_2 \quad (3-1)$$

which becomes

$$dU = TdS - PdV + gdn \quad (3-2)$$

for a pure substance. Next consider the Helmholtz function

$$A = U - TS \quad (3-3)$$

which becomes

$$dA = dU - TdS - SdT \quad (3-4)$$

after differentiating. Substitution of (3-2) into (3-4) yields

$$dA = -SdT - PdV + gdn \quad (3-5)$$

Similarly, differentiation of the Gibbs function

$$G = H - TS = U + PV - TS \quad (3-6)$$

yields

$$dG = dU + PdV + VdP - TdS - SdT \quad (3-7)$$

and substitution of (3-2) into (3-7) produces

$$dG = -SdT + VdP + gdn \quad (3-8)$$

Substituting (3-1) into (3-4) and (3-7) yields

$$dA = -SdT - PdV + \bar{G}_1 dn_1 + \bar{G}_2 dn_2 \quad (3-9)$$

$$dG = -SdT + VdP + \bar{G}_1 dn_1 + \bar{G}_2 dn_2 \quad (3-10)$$

which are applicable for binary mixtures. An additional relation may be obtained by expanding (3-8) to give

$$dG = d(ng) = ndg + gdn = -(ns)dT + (nv)dP + gdn$$

or

$$dg = -sdT + vdP \quad (3-11)$$

C. Pure Substance Solid-Vapor Equilibrium

The requirement for solid-vapor equilibrium of a pure substance at a given temperature and pressure (which will be the sublimation pressure P^0) is

$$g^S = g^G \quad (3-12)$$

However, if the vapor behaves ideally then

$$v = \frac{RT}{P} \quad (3-13)$$

If (3-13) is substituted into (3-11) along with the restriction that the temperature be held constant then

$$dg = \frac{RT}{P} dP = RTd(\ln P) \quad (3-14)$$

It is desirable to retain this form for a real gas and therefore the quantity fugacity is introduced by the definition

$$dg = RTd(\ln f) \quad (3-15)$$

Furthermore, since a real gas approaches ideal behavior as

$P \rightarrow 0$ then

$$\lim_{P \rightarrow 0} (f/P) = 1 \quad (3-16)$$

The criterion for equilibrium can now be written as

$$f^S = f^G = f^O \quad (3-17)$$

where the superscript o denotes the value at the sublimation pressure of the pure substance (or vapor pressure in the case of liquid-vapor equilibrium).

D. Solid-Vapor Equilibrium in the Presence of an Inert Gas

The requirement for solid-vapor equilibrium in the presence of an inert gas is

$$g_1^S = \bar{G}_1^G \quad (3-18)$$

where the subscript 1 signifies the substance present in both phases. For component 1 of the mixture, an equation analagous to (3-15) is written as

$$d\bar{G}_1 = RT d(\ln \bar{f}_1) \quad (3-19)$$

with the requirement that the fugacity of component 1 approach its partial pressure as the mixture approaches ideal behavior, or

$$\lim_{P \rightarrow 0} \left(\frac{\bar{f}_1}{y_1 P} \right) = 1 \quad (3-20)$$

Therefore, the requirement for solid-vapor equilibrium in the presence of an inert gas can now be written as

$$f_1^S = \bar{f}_1^G \quad (3-21)$$

If equations (3-11) and (3-15), both written for solid 1 at constant temperature, are equated then

$$dg_1 = RT d(\ln f_1^S) = v_1^S dP \quad (3-22)$$

Integration of (3-22) from P_1^O to the pressure of the system gives

$$RT \ln \frac{f_1^S}{f_1^O} = \int_{P_1^O}^P v_1^S dP \quad (3-23)$$

Substitution of (3-21) into (3-23) yields

$$RT \ln \frac{F_1^g}{f_1^o} = \int_{P_1^o}^P v_1^s dP \quad (3-24)$$

which is the theoretical equilibrium equation for the system in question. Since F_1^g is dependent on the composition of the gas phase, equation (3-24) may be solved to determine the composition provided the values of F_1^g and f_1^o are known. Therefore, it is necessary to determine expressions for F_1^g as a function of the behavior of the gas mixture and f_1^o as a function of the behavior of pure component 1. Such expressions are determined in the next two sections.

E. Pure Gas Fugacity

Taking a cross partial differential in equation (3-5) produces

$$\left(\frac{\partial g}{\partial V}\right)_{T,n} = - \left(\frac{\partial P}{\partial n}\right)_{T,V} \quad (3-25)$$

which reduces to

$$dg = - \left(\frac{\partial P}{\partial n}\right)_{T,V} dV \quad (3-26)$$

while holding T and n constant. Substitution of (3-15) into (3-26) yields

$$RT d(\ln f) = - \left(\frac{\partial P}{\partial n}\right)_{T,V} dV \quad (3-27)$$

which may also be written as

$$RT d(\ln f) + RT d(\ln V) = - \left(\frac{\partial P}{\partial n}\right)_{T,V} dV + \frac{RT}{V} dV$$

or

$$RT d(\ln fV) = - \left[\left(\frac{\partial P}{\partial n}\right)_{T,V} - \frac{RT}{V} \right] dV \quad (3-28)$$

Equation (3-28) is now written for component 1, component 1 being that present in both phases, and is integrated from

some arbitrarily low pressure P^* to the sublimation pressure P_1^0 to yield

$$RT \int_{(f_1^* V_1^*)}^{(f_1^0 V_1^0)} d(\ln fV) = - \int_{V_1^*}^{V_1^0} \left[\left(\frac{\partial P}{\partial n} \right)_{T,V} - \frac{RT}{V} \right] dV \quad (3-29)$$

However, as $P^* \rightarrow 0$ component 1 will behave ideally and

$$\lim_{P^* \rightarrow 0} (f_1^* V_1^*) = P_1^* V_1^* = n_1 RT \quad (3-30)$$

In addition

$$V_1^* \rightarrow \infty \text{ as } P^* \rightarrow 0 \quad (3-31)$$

so that equation (3-29) may be rewritten as

$$RT \ln \frac{f_1^0 V_1^0}{n_1 RT} = \int_{V_1^0}^{\infty} \left[\left(\frac{\partial P}{\partial n} \right)_{T,V} - \frac{RT}{V} \right] dV \quad (3-32)$$

However

$$V_1^0 = V_1^0 / n_1 \quad (3-33)$$

so that equation (3-32) becomes

$$RT \ln \frac{f_1^0 V_1^0}{RT} = \int_{V_1^0}^{\infty} \left[\left(\frac{\partial P}{\partial n} \right)_{T,V} - \frac{RT}{V} \right] dV \quad (3-34)$$

or, finally

$$RT \ln \frac{f_1^0}{P_1^0} = \int_{V_1^0}^{\infty} \left[\left(\frac{\partial P}{\partial n} \right)_{T,V} - \frac{RT}{V} \right] dV$$

$$- RT \ln \frac{P_1^0 V_1^0}{RT} \quad (3-35)$$

which relates f_1^0 to the behavior of pure substance 1.

It is possible to obtain an expression analagous to equation (3-35) which has V rather than P as the independent variable by taking a cross partial differential of equation

(3-8) rather than (3-5) to obtain

$$\left(\frac{\partial g}{\partial P}\right)_{T,n} = \left(\frac{\partial V}{\partial n}\right)_{T,P} \quad (3-36)$$

and, at constant T and n, this becomes

$$dg = \left(\frac{\partial V}{\partial n}\right)_{T,P} dP \quad (3-37)$$

Substitution of (3-15) into (3-3) gives

$$RT d(\ln f) = \left(\frac{\partial V}{\partial n}\right)_{T,P} dP \quad (3-38)$$

which may be rewritten as

$$RT d(\ln f/P) = \left[\left(\frac{\partial V}{\partial n}\right)_{T,P} - \frac{RT}{P} \right] dP \quad (3-39)$$

As before, equation (3-39) may be written for component 1 and integrated from a pressure P^* , which approaches zero, to the sublimation pressure P_1^0 to yield

$$RT \ln \frac{f_1^0}{P_1^0} = \int_0^{P_1^0} \left[\left(\frac{\partial V}{\partial n}\right)_{T,P} - \frac{RT}{P} \right] dP \quad (3-40)$$

which is analagous to equation (3-35).

The evaluation of the integrals on the right hand side of equations (3-35) and (3-40) will require a pressure-explicit and volume-explicit equation of state, respectively. However, since pressure-explicit equations of state are much more accurate than those explicit in volume, equation (3-35) will be used to express f_1^0 as a function of the behavior of pure substance 1.

F. Fugacity of a Component in a Gas Mixture

Taking a cross partial differential in equation (3-9)

yields

$$\left(\frac{\partial \bar{G}_1}{\partial V}\right)_{T,n} = - \left(\frac{\partial P}{\partial n_1}\right)_{T,V,n_2} \quad (3-41)$$

which may be written as

$$d\bar{G}_1 = - \left(\frac{\partial P}{\partial n_1}\right)_{T,V,n_2} dV \quad (3-42)$$

while holding T and n constant. Substitution of equation (3-19) written for the gas phase into (3-42) gives

$$RT d(\ln \bar{F}_1^g) = - \left(\frac{\partial P}{\partial n_1}\right)_{T,V,n_2} dV \quad (3-43)$$

which may also be expressed as

$$RT d(\ln \bar{F}_1^g) + RT d(\ln V) = - \left(\frac{\partial P}{\partial n_1}\right)_{T,V,n_2} dV \\ + \frac{RT}{V} dV$$

or

$$RT d(\ln \bar{F}_1^g V) = - \left[\left(\frac{\partial P}{\partial n_1}\right)_{T,V,n_2} - \frac{RT}{V} \right] dV \quad (3-44)$$

which is analogous to equation (3-28). Equation (3-44) is now integrated from some arbitrarily low pressure P^* to the pressure P of the mixture to yield

$$RT \int_{(\bar{F}_1^g V^*)}^{(\bar{F}_1^g V)} d(\ln \bar{F}_1^g V) = - \int_{V^*}^V \left[\left(\frac{\partial P}{\partial n_1}\right)_{T,V,n_2} - \frac{RT}{V} \right] dV \quad (3-45)$$

As $P^* \rightarrow 0$ the mixture will behave ideally so that

$$\lim_{P^* \rightarrow 0} (\bar{F}_1^* V^*) = y_1 P^* V^* = y_1 nRT \quad (3-46)$$

Also

$$V^* \rightarrow \infty \quad \text{as } P^* \rightarrow 0 \quad (3-47)$$

and equation (3-45) becomes

$$RT \ln \frac{\bar{F}_1^{gV}}{y_1 nRT} = \int_V^\infty \left[\left(\frac{\partial P}{\partial n_1} \right)_{T,V,n_2} - \frac{RT}{V} \right] dV \quad (3-48)$$

However,

$$v = V/n \quad (3-49)$$

so that equation (3-48) may be written in the form

$$RT \ln \frac{\bar{F}_1^{gV}}{y_1 RT} = \int_V^\infty \left[\left(\frac{\partial P}{\partial n_1} \right)_{T,V,n_2} - \frac{RT}{V} \right] dV \quad (3-50)$$

or, finally, as

$$RT \ln \frac{\bar{F}_1^g}{y_1 P} = \int_V^\infty \left[\left(\frac{\partial P}{\partial n} \right)_{T,V,n_2} - \frac{RT}{V} \right] dV - RT \ln \frac{Pv}{RT} \quad (3-51)$$

As indicated in the preceding section, it is possible to obtain an expression analogous to equation (3-51) which has V rather than P as the independent variable. This is accomplished by taking a cross partial differential of equation (3-10) rather than (3-9) to obtain

$$\left(\frac{\partial \bar{G}_1}{\partial P} \right)_{T,n} = \left(\frac{\partial V}{\partial n_1} \right)_{T,P,n_2} \quad (3-52)$$

which reduces to

$$d\bar{G}_1 = \left(\frac{\partial V}{\partial n_1} \right)_{T,P,n_2} dP \quad (3-53)$$

at constant temperature and composition. Substitution of (3-19) written for the gas phase into (3-53) yields

$$RT d(\ln F_1^g) = \left(\frac{\partial V}{\partial n_1} \right)_{T,P,n_2} dP \quad (3-54)$$

which may be rewritten as

$$RT d\left(\ln \frac{F_1^g}{P}\right) = \left[\left(\frac{\partial V}{\partial n_1} \right)_{T,P,n_2} - \frac{RT}{P} \right] dP \quad (3-55)$$

Equation (3-55) may now be integrated from a pressure P^* , which approaches zero, to the pressure P of the mixture to give

$$RT \ln \frac{F_1^g}{y_1 P} = \int_0^P \left[\left(\frac{\partial V}{\partial n_1} \right)_{T,P,n_2} - \frac{RT}{P} \right] dP \quad (3-56)$$

which is analagous to equation (3-51).

In order to evaluate the right hand sides of equations (3-51) and (3-56), equations of state for the gas mixture explicit in pressure and volume, respectively, will be required. However, as indicated previously, pressure-explicit equations are superior in accuracy to those explicit in volume and, therefore, equation (3-51) will be used to express F_1^g as a function of the behavior of the gas mixture.

G. Enhancement Factor

Equations (3-24), (3-35), and (3-51) may be combined to give

$$\begin{aligned}
 RT \ln \left(\frac{y_1 P}{P_1^0} \right) &= RT \ln \frac{Pv}{P_1^0 v_1^0} + \int_{P_1^0}^P v_1^S dP \\
 &+ \int_{v_1^0}^{\infty} \left[\left(\frac{\partial P}{\partial n} \right)_{T,V} - \frac{RT}{V} \right] dv - \int_V^{\infty} \left[\left(\frac{\partial P}{\partial n_1} \right)_{T,V,n_2} \right. \\
 &\left. - \frac{RT}{V} \right] dV
 \end{aligned} \tag{3-57}$$

in which the first integral refers to pure solid 1, the second to pure gaseous 1, and the third to the gas mixture.

Before proceeding further, a dimensionless parameter called the enhancement factor ϕ is introduced, ϕ being defined as the ratio of the actual mole fraction of component 1 to that predicted for ideal behavior, or

$$\phi = y_1 / y_1 \text{ (ideal)} \tag{3-58}$$

If the system is assumed to behave ideally then f_1^S and F_1^G may be expressed as

$$f_1^S = f_1^0 = P_1^0 \tag{3-59}$$

$$F_1^G = y_1 \text{ (ideal)} P \tag{3-60}$$

Substituting (3-59) and (3-60) into (3-21) yields

$$y_1 \text{ (ideal)} = \frac{P_1^0}{P} \tag{3-61}$$

which in turn may be substituted into (3-58) such that the enhancement factor is now expressed as

$$\phi = \frac{y_1^P}{P_1^0} \quad (3-62)$$

It should be noted that the enhancement factor has the value of unity for ideal behavior and other than that for non-ideal behavior. Thus the enhancement factor is a convenient parameter for determining deviation from ideal behavior.

With the definition of enhancement factor as given in (3-62), equation (3-57) may be rewritten as

$$RT \ln \phi = RT \ln \frac{Pv}{P_1^0 v_1^0} + \int_{P_1^0}^P v_1^S dP + \int_{v_1^0}^{\infty} \left[\left(\frac{\partial P}{\partial n} \right)_{T,V} - \frac{RT}{V} \right] dv - \int_V^{\infty} \left[\left(\frac{\partial P}{\partial n_1} \right)_{T,V,n_2} - \frac{RT}{V} \right] dv \quad (3-63)$$

In order to evaluate equation (3-63) a pressure-explicit equation of state for pure component 1 (second integral) and for the gas mixture (third integral) are required. In general an equation of state for the gas mixture is obtained by determining coefficients for the mixture as some function of composition and the pure substance constants. Thus the equation of state for the mixture is a function of pressure, volume, temperature, and composition.

At any given temperature and pressure equation (3-63) contains two unknowns, volume and composition. However, the equation of state for the gas mixture affords a second

relation in the same two unknowns and the set may be solved simultaneously to determine composition for the given temperature and pressure.

IV. EQUATIONS OF STATE

A. General

There are two general classes of equations of state widely used at the present time, the theoretical or virial equation of state and the empirical equation of state. The virial equation based on statistical mechanics requires an intermolecular potential function for evaluation of the virial coefficients. Various potential functions have been proposed in the past and two of these shall be considered in the present investigation, the Lennard-Jones 6-12 potential and the Kihara potential. In addition, several authors have proposed various forms of empirical equations of state, three of which will be considered here, the Beattie-Bridgeman, Benedict-Webb-Rubin, and Martin-Hou equations.

In order to utilize the foregoing equations of state to evaluate the integrals in equation (3-63) it will be necessary to apply them to pure carbon dioxide as well as to a mixture of carbon dioxide and nitrogen. Therefore, the equations of state will first be discussed as they apply to a pure component and thereafter as they apply to a gas mixture.

B. Pure Component

1. Beattie-Bridgeman Equation of State

The Beattie-Bridgeman equation of state (60, 61, 62) is presented as

$$P = \frac{RT}{v^2} \left(1 - \frac{c}{vT^3}\right) \left(v + B_0 - \frac{b B_0}{v}\right) - \frac{A_0}{v^2} \left(1 - \frac{a}{v}\right) \quad (4-1)$$

or alternately as

$$P = \frac{RT}{v^2} (1-C) (v+B) - \frac{A}{v^2} \quad (4-2)$$

where

$$A = A_0 \left(1 - \frac{a}{v}\right) \quad (4-3)$$

$$B = B_0 \left(1 - \frac{b}{v}\right) \quad (4-4)$$

$$C = c/vT^3 \quad (4-5)$$

in which A_0 , B_0 , a , b , and c are the five empirical constants that have been evaluated for several pure substances. The Beattie-Bridgeman equation of state may also be presented in a virial form as

$$P = \frac{RT}{v} + \frac{\beta}{v^2} + \frac{\gamma}{v^3} + \frac{\delta}{v^4} \quad (4-6)$$

where

$$\beta = B_0 RT - A_0 - c R/T^2 \quad (4-7)$$

$$\gamma = A_0 a - B_0 b RT - B_0 c R/T^2 \quad (4-8)$$

$$\delta = B_0 bc R/T^2 \quad (4-9)$$

β , γ , and δ are often referred to as the second, third, and fourth virial coefficients, respectively. However, they should not be confused with the virial coefficients in the equation of state written in terms of the compressibility factor Pv/RT .

The Beattie-Bridgeman equation has been utilized in accurately predicting the behavior of many gases up to 0.8 critical density. Beyond this density range the

equation deviates largely from actual behavior.

2. Benedict-Webb-Rubin Equation of State

The Benedict-Webb-Rubin equation of state (63) is given as

$$P = \frac{RT}{v} + \frac{B_0 RT - A_0 - C_0/T^2}{v^2} + \frac{bRT - a}{v^3} + \frac{a \alpha}{v^6} + \frac{c(1 + \gamma/v^2) \exp(-\gamma/v^2)}{T^2 v^3} \quad (4-10)$$

which contains eight empirical constants. This equation is capable of accurately predicting gas behavior at densities much larger than possible with the Beattie-Bridgeman equation of state, which is to be expected since the Benedict-Webb-Rubin equation contains three additional constants. In general, the Benedict-Webb-Rubin equation of state is very accurate to 1.5 critical density and for particular gases has been used with good accuracy to 2.0 critical density.

3. Martin-Hou Equation of State

The Martin-Hou equation of state (64) is written as

$$P = \frac{RT}{v-b} + \frac{F_2(T)}{(v-b)^2} + \frac{F_3(T)}{(v-b)^3} + \frac{F_4(T)}{(v-b)^4} + \frac{F_5(T)}{(v-b)^5} \quad (4-11)$$

where

$$F_i(T) = A_i + B_i T + C_i \exp(-kT/T_c) \quad (i=2,3,4,5) \quad (4-12)$$

and

$$B_4 = C_4 = 0 \quad (4-13)$$

Combining equations (4-11), (4-12) and (4-13) the Martin-Hou equation of state may be expressed as

$$\begin{aligned} P = \frac{RT}{v-b} &+ \frac{A_2+B_2T+C_2\exp(-kT/T_c)}{(v-b)^2} \\ &+ \frac{A_3+B_3T+C_3\exp(-kT/T_c)}{(v-b)^3} + \frac{A_4}{(v-b)^4} \\ &+ \frac{A_5+B_5T+C_5\exp(-kT/T_c)}{(v-b)^5} \end{aligned} \quad (4-14)$$

The Martin-Hou equation of state, like that of Benedict-Webb-Rubin, is capable of accurately predicting gas behavior at densities as high as 1.5 to 2.0 critical density. An earlier form of this equation was proposed in 1955⁽⁶⁵⁾ in which $A_5 = C_5 = 0$. The constants A_5 and C_5 were added in 1959⁽⁶⁴⁾ to produce the equation as it appears in (4-14), the form to be employed in the present work.

4. Virial Equation of State

The virial equation of state is often presented in terms of the compressibility factor Pv/RT as a power series in reciprocal volumes, or

$$\frac{Pv}{RT} = 1 + \frac{B}{v} + \frac{C}{v^2} + \frac{D}{v^3} + \dots \quad (4-15)$$

where B , C , and D are termed the second, third, and fourth virial coefficients, respectively. To date there has been no success in determining virial coefficients beyond the

third for accurate potential functions. Therefore, the virial equation of state as applied in the present investigation shall be truncated after the third virial coefficient, or

$$\frac{Pv}{RT} = 1 + \frac{B}{v} + \frac{C}{v^2} \quad (4-16)$$

In terms of r_{ij} , the distance between two molecules i and j , and the function

$$f_{ij}(r_{ij}) = \left[e^{-\phi_{ij}/kT} - 1 \right] \quad (4-17)$$

the second and third virial coefficients, as derived from statistical mechanics^(66,67), may be expressed as

$$B = - 2 \uparrow N \int_0^{\infty} \left[e^{-\phi(r)/kT} - 1 \right] r^2 dr \quad (4-18)$$

$$C = - \frac{8 \uparrow^2 N^2}{3} \iiint f_{12} f_{13} f_{23} r_{12} r_{13} r_{23} dr_{12} dr_{13} dr_{23} \quad (4-19)$$

where the last expression is to be integrated over all r_{12} , r_{13} , r_{23} which form a triangle. ϕ_{ij} is a potential

function, also dependent on r_{ij} , which represents the intermolecular potential energy between two molecules i and j . Thus, evaluation of equations (4-18) and (4-19) for the virial coefficients will require an expression for this potential function. Various molecular models have been proposed to determine intermolecular potential functions, two of which shall be considered here, the Lennard-Jones (6-12) potential and the Kihara potential. These two potential functions will be discussed in more detail in the following two sections.

a. Lennard-Jones (6-12) Potential

The Lennard-Jones (6-12) intermolecular potential function^(67, 68, 69) is expressed as

$$\varphi(r) = 4 \epsilon \left[\left(\frac{\sigma}{r} \right)^{12} - \left(\frac{\sigma}{r} \right)^6 \right] \quad (4-20)$$

where σ is the separation distance between two molecules for which the potential energy is zero, and ϵ is the maximum attractive energy between two molecules, occurring at $r = 2^{1/6} \sigma$. An alternate expression for the Lennard-Jones (6-12) potential is

$$\varphi(r) = \epsilon \left[\left(\frac{r_0}{r} \right)^{12} - 2 \left(\frac{r_0}{r} \right)^6 \right] \quad (4-21)$$

where ϵ has the same interpretation as before, and $r_0 = 2^{1/6} \sigma$ is the separation distance for which the potential energy is a minimum, that is, equal to $-\epsilon$.

The Lennard-Jones potential assumes the molecules to be point centers with spherically symmetric force fields. For this potential, the inverse sixth-power attractive component is dominant at large separations ($r \gg \sigma$), and the molecules are attracted to one another with a force inversely proportional to r^7 . At small separations ($r \ll \sigma$) the inverse twelfth-power repulsive power dominates, and the molecules are repelled from one another with a force inversely proportional to r^{13} .

Equation (4-20) may be substituted into (4-18) and integrated analytically to obtain the second virial coefficient^(67, 68). The result of this integration is

$$B^*(T^*) = \sum_{j=0}^{\infty} b_j T^{*- \left(\frac{2j+1}{4} \right)} \quad (4-22)$$

where

$$b_j = - \frac{2^{j+\frac{1}{2}}}{4 j!} \Gamma\left(\frac{2j-1}{4}\right) \quad (4-23)$$

and

$$T^* = kT/\epsilon \quad (4-24)$$

$$B^*(T^*) = \frac{B}{b_0} \quad (4-25)$$

$$b_0 = \frac{2}{3} \pi N \sigma^3 \quad (4-26)$$

Values of $B^*(T^*)$ as given by equation (4-22) have been evaluated and compiled.⁽⁶⁷⁾ Note that b_0 as defined by equation (4-26) is not to be confused with that obtained by setting the index j equal to zero in (4-23).

Equations (4-22) thru (4-26) may now be combined to give an expression for the second virial coefficient as

$$B = - \frac{2}{3} \pi N \sigma^3 \sum_{j=0}^{\infty} \frac{2^{j+\frac{1}{2}}}{4 j!} \Gamma\left(\frac{2j-1}{4}\right) \left(\frac{kT}{\epsilon}\right)^{-\left(\frac{2j+1}{4}\right)} \quad (4-27)$$

The third virial coefficient may be obtained in a similar fashion by using equations (4-17) and (4-20) in (4-19) and integrating to obtain

$$C^*(T^*) = \sum_{j=0}^{\infty} c_j T^{* -\left(\frac{j+1}{2}\right)} \quad (4-28)$$

where

$$C^*(T^*) = \frac{C}{b_0^2} \quad (4-29)$$

and T^* and b_0 are as defined in (4-24) and (4-26). The expansion coefficients c_j in (4-27) are not explicit functions, as are the b_j of equation (4-22), but are complicated integrals. The values of c_j have been determined by Kihara^(70, 71) and have been tabulated in reference 67. Reference 67 also contains a tabulation of $C^*(T^*)$ as determined by equation (4-28) in conjunction with the c_j of Kihara. Rowlinson, et al,⁽⁷²⁾ have also evaluated and tabulated values of c_j which differ somewhat from those of Kihara and, therefore, values of $C^*(T^*)$ determined with the c_j of Rowlinson, et al, will differ from those tabulated in reference 67. A comparison of the different $C(T)$ at selected values of T will be made in Chapter V.

Recently, Pople⁽⁷³⁾ has suggested that a correction term be added to the Lennard-Jones (6-12) virial coefficients for gases which exhibit significant quadrupole moments. Pople, however, has considered only the second virial coefficient and the corresponding correction is

$$B_q = -\frac{2}{3} \pi N \sigma^3 \frac{7}{320} Q^{*2} H_{10}(2/\sqrt{T^*}) \quad (4-30)$$

where

$$Q^* = \frac{Q^2}{\epsilon \sigma^5} \quad (4-31)$$

and Q is the quadrupole moment. The function H_{10} is given as

$$H_{10}(2/\sqrt{T^*}) = (2/\sqrt{T^*})^{17/6} \sum_{j=0}^{\infty} \frac{(2/\sqrt{T^*})^j}{j!} \Gamma\left(\frac{6j+7}{12}\right) \quad (4-32)$$

Values of H_{10} have been tabulated in references 73 and 74. The correction as given by equation (4-30) is to be added to the value obtained from (4-27).

Since carbon dioxide and nitrogen, which are under investigation here, exhibit significant quadrupole moments, the values of the second virial coefficients used here will be those obtained from equations (4-27) and (4-30). In addition, since no quadrupole corrections have been established for third virial coefficients, the values used here will be those determined from equations (4-28) and (4-29).

b. Kihara Potential

The Kihara intermolecular potential function (75, 76) is given as

$$\Phi(\rho) = U_0 \left[\left(\frac{\rho_0}{\rho} \right)^{12} - 2 \left(\frac{\rho_0}{\rho} \right)^6 \right] \quad (4-33)$$

and is identical in mathematical form to that of the Lennard-Jones (6-12) potential as defined by equation (4-21). However, there is a significant difference between the two potential functions in that Kihara assumes the molecule to have an impenetrable core whose configuration and dimensions are suggested by the geometry and internuclear distances of the molecule, whereas Lennard-Jones assumes the molecule to be a point center. Thus, in equation (4-33), ρ is the separation distance between molecular cores rather than molecular centers, ρ_0 is the separation distance at which the potential energy between molecules is a minimum, and U_0 represents the maximum (negative) potential

energy between two molecules.

Using equation (4-33) and the methods of statistical mechanics, Kihara obtains the second virial coefficient as

$$B = \frac{2}{3} \pi N \rho_o^3 F_3(Z) + N M_o \rho_o^2 F_2(Z) + N \left(S_o + \frac{M_o^2}{4 \pi} \right) \rho_o F_1(Z) + N \left(V_o + \frac{M_o S_o}{4 \pi} \right) \quad (4-34)$$

where

$$Z = U_o/kT \quad (4-35)$$

and M_o is the surface integral of the mean curvature of the core, S_o the surface area of the core, and V_o the volume of the core. The F functions in equation (4-34) are given as

$$F_i(Z) = - \frac{i}{12} \sum_{j=0}^{\infty} \frac{2^j}{j!} \Gamma\left(\frac{6j-i}{12}\right) Z^{\left(\frac{6j+i}{12}\right)} \quad (4-36)$$

Values of F_1 , F_2 , and F_3 have been tabulated in references 75 and 77. In the special case of a vanishing core, that is, assuming the molecule to be a point center, M_o , S_o , and V_o all equal zero and equation (4-34) reduces to

$$B = \frac{2}{3} \pi N \rho_o^3 F_3(Z) \quad (4-37)$$

Furthermore, for a vanishing core, comparison of equations (4-21) and (4-33) gives

$$\rho_o = r_o = 2^{1/6} \sigma \quad (4-38)$$

$$U_0 = \epsilon \quad (4-39)$$

In addition, $F_3(Z)$, as defined by (4-36), is given as

$$F_3(Z) = -\frac{1}{4} \sum_{j=0}^{\infty} \frac{2^j}{j!} \Gamma\left(\frac{2j-1}{4}\right) Z^{\left(\frac{2j+1}{4}\right)} \quad (4-40)$$

Finally, equations (4-35) and (4-37) thru (4-40) may be combined to give

$$B = -\frac{2}{3} \pi N \sigma^3 \sum_{j=0}^{\infty} \frac{2^{j+\frac{1}{2}}}{4j!} \Gamma\left(\frac{2j-1}{4}\right) \left(\frac{kT}{\epsilon}\right)^{-\left(\frac{2j+1}{4}\right)} \quad (4-41)$$

Comparison of equations (4-27) and (4-41) show the two expressions to be identical. Thus, it is seen that the second virial coefficients calculated by the Kihara potential and Lennard-Jones (6-12) potential are equivalent for the special case of a vanishing core.

The Kihara potential has not yet been extended to the evaluation of third virial coefficients owing to mathematical complexities. Nevertheless, the second virial coefficient as given by (4-34) will be used in the present investigation. All third virial coefficients employed here shall be determined from equations (4-28) and (4-29).

Prausnitz and Myers⁽⁷⁸⁾ have proposed that the quadrupole correction to the Lennard-Jones second virial coefficient determined by Pople be extended to that of Kihara as

$$B_q = -\frac{2}{3} \pi N \rho_o^3 \frac{7}{320} Q^{*2} H_{10}(2\sqrt{Z}) \quad (4-42)$$

where

$$Q^* = \frac{Q^2}{U_o \sigma^5} \quad (4-43)$$

$$\sigma = 2^{-1/6} \rho_o + \frac{M_o}{2\pi} \quad (4-44)$$

and Q is the quadrupole moment while Z is that given by equation (4-35). As previously indicated, equation (4-42) is to be added to (4-34). The value of σ as given in (4-44) is identical to that of the Lennard-Jones potential in the case of a spherical core and is an approximation for other cores.

C. Gas Mixtures

1. Empirical Equation of State

The common practice by which an empirical equation of state is used to represent a gas mixture is to determine mixture constants as functions of composition and the pure substance constants. These constants are then substituted for the pure substance constants in any given equation of state. Several polynomial expressions of the mole fraction y have been proposed for determining mixture constants. The common second degree forms for binary mixtures are given as

$$K_m = y_1^2 K_{11} + 2y_1 y_2 K_{12} + y_2^2 K_{22} \quad (4-45)$$

where

$$K_{ij} = \frac{1}{2} (K_i + K_j) \quad (4-46)$$

or

$$K_{ij} = (K_i K_j)^{\frac{1}{2}} \quad (4-47)$$

or

$$K_{ij} = \frac{1}{8} (K_i^{1/3} + K_j^{1/3})^3 \quad (4-48)$$

The common third degree forms for binary mixtures are

$$K_m = y_1^3 K_{111} + 3y_1^2 y_2 K_{112} + 3y_1 y_2^2 K_{122} + y_2^3 K_{222} \quad (4-49)$$

where

$$K_{ijk} = \frac{1}{3} (K_i + K_j + K_k) \quad (4-50)$$

or

$$K_{ijk} = (K_i K_j K_k)^{1/3} \quad (4-51)$$

or

$$K_{ijk} = \frac{1}{27} (K_i^{1/3} + K_j^{1/3} + K_k^{1/3})^3 \quad (4-52)$$

Equations (4-45) and (4-46) may be combined to yield

$$K_m = y_1 K_1 + y_2 K_2 \quad (4-53)$$

Likewise, (4-45) and (4-47) may be combined to give

$$K_m = y_1^2 K_1 + 2y_1 y_2 (K_1 K_2)^{\frac{1}{2}} + y_2^2 K_2 \quad (4-54)$$

and (4-45) and (4-48) give

$$K_m = y_1^3 K_1 + \frac{1}{4} y_1 y_2 (K_1^{1/3} + K_2^{1/3})^3 + y_2^3 K_2 \quad (4-55)$$

Similarly, the combination of (4-49) and (4-50) gives the same result as (4-53). Equations (4-49) and (4-51), when combined, give

$$K_m = y_1^3 K_1 + 3y_1^2 y_2 K_1^{2/3} K_2^{1/3} + 3y_1 y_2^2 K_1^{1/3} K_2^{2/3} + y_2^3 K_2 \quad (4-56)$$

and (4-49) and (4-52) yield

$$K_m = y_1^3 K_1 + \frac{1}{9} y_1^2 y_2 (2K_1^{1/3} + K_2^{1/3})^3 + \frac{1}{9} y_1 y_2^2 (K_1^{1/3} + 2K_2^{1/3})^3 + y_2^3 K_2 \quad (4-57)$$

Finally, equations (4-53) thru (4-57) may be generalized for an n component mixture as

$$K_m = \sum_i^n y_i K_i \quad (4-58)$$

$$K_m = \left(\sum_i^n y_i K_i^{1/2} \right)^2 \quad (4-59)$$

$$K_m = \frac{1}{8} \sum_i^n \sum_j^n y_i y_j (K_i^{1/3} + K_j^{1/3})^3 \quad (4-60)$$

$$K_m = \left(\sum_i^n y_i K_i^{1/3} \right)^3 \quad (4-61)$$

$$K_m = \frac{1}{27} \sum_i^n \sum_j^n \sum_k^n y_i y_j y_k (K_i^{1/3} + K_j^{1/3} + K_k^{1/3})^3 \quad (4-62)$$

a. Beattie-Bridgeman Equation of State

Several investigations have been conducted in the past in an attempt to determine which of the foregoing expressions, (4-53) thru (4-57), should be used to determine mixture constants for the Beattie-Bridgeman equation of state. References 20, 25, 79, and 80 indicate that equation (4-53) should be used to determine the constants a and b for a gas mixture, (4-54) for A_0 and c , and (4-55) for B_0 .

b. Benedict-Webb-Rubin Equation of State

A number of investigations^(81, 82, 83) have been made in which several of equations (4-53) thru (4-57) were utilized to evaluate mixture constants for the Benedict-Webb-Rubin equation of state in order to test the ability of the equation to represent gas mixture behavior. The results of these investigations indicate that equation (4-54) should be used to determine the constants A_0 , C_0 , and γ for a mixture, (4-55) for B_0 , and (4-56) for a , b , c , and α .

c. Martin-Hou Equation of State

Whereas several investigations have been made in applying the Beattie-Bridgeman and Benedict-Webb-Rubin equations of state to gas mixtures, only one such investigation involving the Martin-Hou equation has been reported, that of Smith, Sonntag, and Van Wylen⁽⁵⁸⁾. These authors investigated several combining rules used with the earlier form of the Martin-Hou equation, that is, equation (4-14)

with $A_5 = C_5 = 0$. The combining rules were evaluated by testing their accuracies in representing known mixture P-V-T data for several binary mixtures of various composition. The results of this investigation indicated that an appropriate method for determining mixture constants is to use equation (4-53) to determine b , (4-54) for B_2 , B_3 , B_5 , and T_c , and (4-55) for A_2 , A_3 , A_4 , C_2 , and C_3 .

Attempts by the preceding authors to perform a similar investigation for the Martin-Hou equation as given in (4-14) were hindered by the lack of pure component constants which prevented testing the combining rules against various binary mixtures. Nevertheless, using the CO_2 ⁽⁶⁴⁾ and N_2 ⁽⁵⁸⁾ constants for equation (4-14), Smith, et al., checked the accuracies of various sets of combining rules against known P-V-T mixture data for the $\text{CO}_2 - \text{N}_2$ system and concluded that a highly satisfactory method for determining mixture constants is to use equation (4-53) to determine b , (4-54) for B_2 , B_3 , B_5 , and T_c , and (4-55) for A_2 , A_3 , A_4 , A_5 , C_2 , C_3 , and C_5 . This set of combining rules was used in conjunction with equation (4-14), to fit the solid-vapor equilibrium data of Sonntag^(25, 26), thus providing an additional test on the accuracy of the proposed set of combining rules. The resulting analytical values agreed very closely to the experimental, indicating that the proposed set of combining rules are valid. It is this set of combining rules that will be used with equation (4-14) in the present investigation.

2. Virial Equation of State

The virial equation of state for a gas mixture is identical in form to that of equation (4-16) but the virial coefficients must now represent the mixture. The second and third virial coefficients for an n component mixture, as derived from statistical mechanics⁽⁸⁴⁾, are given as

$$B_{\text{mix}} = \sum_{i=1}^n \sum_{j=1}^n y_i y_j B_{ij} \quad (4-63)$$

$$C_{\text{mix}} = \sum_{i=1}^n \sum_{j=1}^n \sum_{k=1}^n y_i y_j y_k C_{ijk} \quad (4-64)$$

where the B_{ij} and C_{ijk} with subscripts not all the same are referred to as interaction coefficients, and those with all subscripts alike are merely the coefficients of the pure substances. For a binary mixture equations (4-63) and (4-64) reduce to

$$B_{\text{mix}} = y_1^2 B_{11} + 2y_1 y_2 B_{12} + y_2^2 B_{22} \quad (4-65)$$

$$C_{\text{mix}} = y_1^3 C_{111} + 3y_1^2 y_2 C_{112} + 3y_1 y_2^2 C_{122} + y_2^3 C_{222} \quad (4-66)$$

The next step in determining the second and third virial coefficients as defined in (4-65) and (4-66) is to determine values for the interaction coefficients B_{12} , C_{112} , and C_{122} . Expressions for evaluating these interaction coefficients will depend upon the potential function utilized in determining the virial coefficients for the pure components. Since two potential functions are being considered in

the present investigation, the interaction coefficients resulting from each will be considered separately in the following two sections.

a. Lennard-Jones (6-12) Potential

The method commonly used with the Lennard-Jones (6-12) potential for determining the interaction coefficient B_{12} in (4-65) is to combine the pure substance intermolecular force constants, ϵ and σ , by some semi-empirical means. The most widely used combination, as suggested in references 21, 47, 49, 50, 67, and 85 thru 90, is

$$\sigma_{12} = \frac{1}{2} (\sigma_1 + \sigma_2) \quad (4-67)$$

$$\epsilon_{12} = (\epsilon_1 \epsilon_2)^{\frac{1}{2}} \quad (4-68)$$

By using equation (4-26), which defines b_o , and equation (4-68) it may be seen that

$$b_{o12} = \frac{1}{8} \left[b_{o1}^{1/3} + b_{o2}^{1/3} \right]^3 \quad (4-69)$$

Expressions (4-68) and (4-69) may now be used in (4-22), (4-24) and (4-25), along with the tabulated values of $B^*(T^*)^{(67)}$, to determine the second virial interaction coefficient. Alternately, equations (4-67) and (4-68) may be substituted into (4-27) in order to evaluate B_{12} .

Ewald ⁽²¹⁾ has suggested that a similar procedure be used for determining the third virial interaction coefficients C_{112} and C_{122} where

$$\epsilon_{1n2} = (\epsilon_1^{3-n} \epsilon_2^n)^{1/3} \quad (4-70)$$

$$b_{o_{1n2}} = -\frac{1}{27} \left[(3-n) b_{o_1}^{1/3} + n b_{o_2}^{1/3} \right]^3 \quad (4-71)$$

where n is either 1 or 2. Equation (4-71) is equivalent to using

$$\sigma_{1n2} = \frac{1}{3} \left[(3-n) \sigma_1 + n \sigma_2 \right] \quad (4-72)$$

where, again, n is either 1 or 2. The values for ϵ_{112} , $b_{o_{112}}$, ϵ_{122} , and $b_{o_{122}}$ as fixed by (4-70) and (4-71)

may now be used as for a pure substance in equations (4-28) and (4-29), along with tabulated values of $C^*(T^*)^{(67)}$, to determine the two third virial interaction coefficients C_{112} and C_{122} .

If the quadrupole correction to the Lennard-Jones potential, discussed previously, is to be employed for a gas mixture, some means must also be used to determine a quadrupole moment which will permit determination of $B_{q_{12}}$. Prausnitz and Myers⁽⁷⁸⁾ suggest that the expression

$$Q_{12}^* = \frac{Q_1 Q_2}{\epsilon_{12} \sigma_{12}^5} \quad (4-73)$$

be used in (4-30) to determine $B_{q_{12}}$ where ϵ_{12} and σ_{12} are as given by equations (4-67) and (4-68). The resulting value for $B_{q_{12}}$ is to be added to the value of B_{12} determined by the methods of this section.

Since the quadrupole corrections have been applied only

to second virial coefficients, no third-order quadrupole corrections are included here.

b. Kihara Potential

Kihara⁽⁹¹⁾ has indicated that the second virial interaction coefficient for his potential be determined by

$$\begin{aligned}
 B_{12} = & \frac{2}{3} \pi N \rho_{o_{12}}^3 F_3(Z_{12}) + \frac{1}{2} (M_{o_1} + M_{o_2}) N \rho_{o_{12}}^2 \\
 & F_2(Z_{12}) + \frac{1}{2} (S_{o_1} + S_{o_2} + \frac{M_{o_1} M_{o_2}}{2 \pi}) N \rho_{o_{12}} F_1(Z_{12}) \\
 & + \frac{1}{2} N (V_{o_1} + V_{o_2} + \frac{M_{o_1} S_{o_2} + M_{o_2} S_{o_1}}{4 \pi}) \quad (4-74)
 \end{aligned}$$

in which

$$\rho_{o_{12}} = \frac{1}{2} (\rho_{o_1} + \rho_{o_2}) \quad (4-75)$$

$$Z_{12} = U_{o_{12}} / kT \quad (4-76)$$

and

$$U_{o_{12}} = (U_{o_1} U_{o_2})^{\frac{1}{2}} \quad (4-77)$$

The values for the F functions appearing in (4-74) are determined in the same manner as for the pure substance where the arguments for the functions are now given by (4-76).

A quadrupole correction to Kihara second virial coefficient may also be determined in a manner similar to that of the preceding section by using equation (4-42) where Q^*

for the pure substance is replaced by

$$Q_{12}^* = \frac{Q_1 Q_2}{U_{o12} \sigma_{12}^5} \quad (4-78)$$

as suggested by Prausnitz and Myers⁽⁷⁸⁾. In this expression U_{o12} is given by (4-77) and

$$\sigma_{12} = 2^{-1/6} \rho_{o12} + \frac{M_{o1} + M_{o2}}{4 \pi} \quad (4-79)$$

where ρ_{o12} is determined from equation (4-75). The quadrupole correction to the second virial interaction coefficient as determined above should be added to the value of B_{12} as given by (4-74).

No third-order interactions are considered here since the Kihara potential and quadrupole corrections have not been extended to third virial coefficients at the present time.

V. SOLUTION OF THEORETICAL EQUATIONS

A. General

The equilibrium equation derived in Chapter III and repeated here for convenience is

$$\begin{aligned}
 RT \ln \phi = RT \ln \frac{Pv}{P_1^0 v_1^0} + \int_{P_1^0}^P v_1^s dP \\
 + \int_{v_1^0}^{\infty} \left[\left(\frac{\partial P}{\partial n} \right)_{T,V} - \frac{RT}{V} \right] dv - \int_V^{\infty} \left[\left(\frac{\partial P}{\partial n_1} \right)_{T,V,n_2} \right. \\
 \left. - \frac{RT}{V} \right] dV
 \end{aligned} \tag{5-1}$$

where the enhancement factor ϕ is

$$\phi = \frac{y_1^P}{P_1^0} \tag{5-2}$$

The equations of state discussed in the preceding chapter for a pure component and for a gas mixture may now be used to evaluate the second and third integrals, respectively, appearing in (5-1). However, as indicated in Chapter III, at any given temperature and pressure the evaluation of equation (5-1) will result in an expression containing two unknowns, volume and composition. The equation of state for the gas mixture offers a second relation in the same two unknowns and the resulting set of two equations may be solved simultaneously to determine the composition and volume. This set of equations as obtained for each of the equations of state employed here will be presented in the

following section.

It is possible to obtain a simplified solution for the composition of the gas phase as fixed by (5-1) by neglecting terms involving y_1 , P_1^0 and $1/v_1^0$, all of which are small at low temperatures. However, terms involving ratios of these quantities may not be small and should not be neglected. Using these simplifying conditions, the equation of state for the gas mixture reduces to that for pure component 2, and may be solved for volume at a given temperature and pressure. With this value for the volume, (5-1) may then be evaluated for the composition. Thus, under these simplifying conditions the set of equations discussed above is modified in that one of the equations may be solved explicitly for one of the unknowns. This modified set of equations as obtained for each of the equations of state discussed previously will also be presented in the following section.

Before proceeding to the determination of the sets of equations described in the preceding paragraphs, it will be convenient to perform the integration indicated in the first integral of (5-1). This may be accomplished by using the relation

$$v_1^S(P, T) = v_1^S(0, T) \left[1 - k_1 P \right] \quad (5-3)$$

where k_1 is the isothermal compressibility factor of solid 1. Thus,

$$\int_{P_1^0}^P v_1^S dP = v_1^S \left[(P - P_1^0) - \frac{1}{2} k_1 (P^2 - P_1^{02}) \right] \quad (5-4)$$

However, if k_1 is negligible, as assumed in this investigation, then

$$\int_{P_1^0}^P v_1^S dP = v_1^S (P - P_1^0) \quad (5-5)$$

B. Equations to be Solved

1. Beattie-Bridgeman Equation of State

The virial form of the Beattie-Bridgeman equation, repeated here as

$$P = \frac{RT}{V} + \frac{\beta}{V^2} + \frac{\gamma}{V^3} + \frac{\delta}{V^4} \quad (5-6)$$

where

$$\beta = B_0 RT - A_0 - c R/T^2 \quad (5-7)$$

$$\gamma = A_0 a - B_0 b RT - B_0 c R/T^2 \quad (5-8)$$

$$\delta = B_0 b c R/T^2 \quad (5-9)$$

is written for pure 1 as

$$P = \frac{nRT}{V} + \frac{n^2 \beta_1}{V^2} + \frac{n^3 \gamma_1}{V^3} + \frac{n^4 \delta_1}{V^4} \quad (5-10)$$

Now,

$$\left(\frac{\partial P}{\partial n} \right)_{T,V} = \frac{RT}{V} + \frac{2n\beta_1}{V^2} + \frac{3n^2 \gamma_1}{V^3} + \frac{4n^3 \delta_1}{V^4} \quad (5-11)$$

so that the second integral in (5-1) becomes

$$\begin{aligned} \int_{V_1^0}^{\infty} \left[\left(\frac{\partial P}{\partial n} \right)_{T,V} - \frac{RT}{V} \right] dV &= \frac{2\beta_1}{V_1^0} + \frac{3\gamma_1}{2(V_1^0)^2} \\ &+ \frac{4\delta_1}{3(V_1^0)^3} \end{aligned} \quad (5-12)$$

Equation (5-6) is written for the gas mixture as

$$P = \frac{RT}{V} + \frac{\beta_m}{V^2} + \frac{\gamma_m}{V^3} + \frac{\delta_m}{V^4} \quad (5-13)$$

or

$$P = \frac{nRT}{V} + \frac{n^2 \beta_m}{V^2} + \frac{n^3 \gamma_m}{V^3} + \frac{n^4 \delta_m}{V^4} \quad (5-14)$$

so that

$$\begin{aligned} \left(\frac{\partial P}{\partial n_1} \right)_{T,V,n_1} &= \frac{RT}{V} + \frac{1}{V^2} \left[\frac{\partial}{\partial n_1} (n^2 \beta_m) \right] + \frac{1}{V^3} \\ &\quad \left[\frac{\partial}{\partial n_1} (n^3 \gamma_m) \right] + \frac{1}{V^4} \left[\frac{\partial}{\partial n_1} (n^4 \delta_m) \right] \end{aligned} \quad (5-15)$$

and the third integral in (5-1) becomes

$$\begin{aligned} \int_V^\infty \left[\left(\frac{\partial P}{\partial n_1} \right)_{T,V,n_2} - \frac{RT}{V} \right] dV &= \frac{1}{V} \left[\frac{\partial}{\partial n_1} (n^2 \beta_m) \right] \\ &+ \frac{1}{2V^2} \left[\frac{\partial}{\partial n_1} (n^3 \gamma_m) \right] + \frac{1}{3V^3} \left[\frac{\partial}{\partial n_1} (n^4 \delta_m) \right] \end{aligned} \quad (5-16)$$

The derivatives on the right hand side of (5-16) may be expanded to obtain

$$\begin{aligned} \int_V^\infty \left[\left(\frac{\partial P}{\partial n_1} \right)_{T,V,n_2} - \frac{RT}{V} \right] dV &= \frac{1}{V} \left[2 \beta_m + n \frac{\partial \beta_m}{\partial n_1} \right] \\ &+ \frac{1}{2V^2} \left[3 \gamma_m + n \frac{\partial \gamma_m}{\partial n_1} \right] + \frac{1}{3V^3} \left[4 \delta_m + n \frac{\partial \delta_m}{\partial n_1} \right] \end{aligned} \quad (5-17)$$

However, since

$$\beta_m = B_{O_m} RT - A_{O_m} - c_m R/T^2 \quad (5-18)$$

$$\gamma_m = A_{O_m} a_m - B_{O_m} b_m RT - B_{O_m} c_m R/T^2 \quad (5-19)$$

$$\delta_m = B_{O_m} b_m c_m R/T^2 \quad (5-20)$$

equation (5-17) may be expressed as

$$\begin{aligned} \int_V^{\infty} \left[\left(\frac{\partial P}{\partial n_1} \right)_{T,V,n_2} - \frac{RT}{V} \right] dV = \frac{1}{V} \left[RT(2B_{O_m} + n \frac{\partial B_{O_m}}{\partial n_1}) \right. \\ \left. - (2A_{O_m} + n \frac{\partial A_{O_m}}{\partial n_1}) - \frac{R}{T^2} (2c_m + n \frac{\partial c_m}{\partial n_1}) \right] \\ + \frac{1}{2V^2} \left[(3A_{O_m} a_m + A_{O_m} n \frac{\partial a_m}{\partial n_1} + a_m n \frac{\partial A_{O_m}}{\partial n_1}) \right. \\ \left. - RT (3B_{O_m} b_m + B_{O_m} n \frac{\partial b_m}{\partial n_1} + b_m n \frac{\partial B_{O_m}}{\partial n_1}) \right. \\ \left. - \frac{R}{T^2} (3B_{O_m} c_m + B_{O_m} n \frac{\partial c_m}{\partial n_1} + c_m n \frac{\partial B_{O_m}}{\partial n_1}) \right] \\ + \frac{1}{3V^3} \frac{R}{T^2} \left[4B_{O_m} b_m c_m + B_{O_m} b_m n \frac{\partial c_m}{\partial n_1} \right. \\ \left. + B_{O_m} c_m n \frac{\partial b_m}{\partial n_1} + b_m c_m n \frac{\partial B_{O_m}}{\partial n_1} \right] \end{aligned} \quad (5-21)$$

In order to solve (5-21) it will be necessary to reduce A_{O_m} , B_{O_m} , a_m , b_m , c_m and their derivatives to expressions involving the pure substance constants. This may be accomplished by using the set of combining rules discussed in Chapter IV, this set being

$$A_{O_m} = y_1^2 A_{O_{11}} + 2y_1y_2 A_{O_{12}} + y_2^2 A_{O_{22}} \quad (5-22)$$

$$B_{O_m} = y_1^2 B_{O_{11}} + 2y_1y_2 B_{O_{12}} + y_2^2 B_{O_{22}} \quad (5-23)$$

$$a_m = y_1^2 a_{11} + 2y_1y_2 a_{12} + y_2^2 a_{22} \quad (5-24)$$

$$b_m = y_1^2 b_{11} + 2y_1y_2 b_{12} + y_2^2 b_{22} \quad (5-25)$$

$$c_m = y_1^2 c_{11} + 2y_1y_2 c_{12} + y_2^2 c_{22} \quad (5-26)$$

where

$$A_{O_{ij}} = (A_{O_i} A_{O_j})^{\frac{1}{2}} \quad (5-27)$$

$$B_{O_{ij}} = \frac{1}{8} (B_{O_i}^{1/3} + B_{O_j}^{1/3})^3 \quad (5-28)$$

$$a_{ij} = \frac{1}{2} (a_i + a_j) \quad (5-29)$$

$$b_{ij} = \frac{1}{2} (b_i + b_j) \quad (5-30)$$

$$c_{ij} = (c_i c_j)^{\frac{1}{2}} \quad (5-31)$$

Before proceeding to the evaluation of the derivatives required in (5-21) it will be convenient to note that

$$y_1 = \frac{n_1}{n_1 + n_2} \quad (5-32)$$

so that

$$\frac{\partial}{\partial n_1} = \frac{\partial y_1}{\partial n_1} \quad \frac{\partial}{\partial y_1} = \frac{n_2}{(n_1 + n_2)^2} \quad \frac{\partial}{\partial y_1} = \frac{y_2}{n} \frac{\partial}{\partial y_1} \quad (5-33)$$

or

$$n \frac{\partial}{\partial n_1} = y_2 \frac{\partial}{\partial y_1} \quad (5-34)$$

The partial derivatives required in (5-21) are now determined as

$$n \frac{\partial A_{o_m}}{\partial n_1} = y_2 \frac{\partial A_{o_m}}{\partial y_1} = 2y_2 \left[y_1 A_{o_1} + (y_2 - y_1) A_{o_{12}} - y_2 A_{o_2} \right] \quad (5-35)$$

and, similarly

$$n \frac{\partial B_{o_m}}{\partial n_1} = 2y_2 \left[y_1 B_{o_1} + (y_2 - y_1) B_{o_{12}} - y_2 B_{o_2} \right] \quad (5-36)$$

$$n \frac{\partial a_m}{\partial n_1} = 2y_2 \left[y_1 a_1 + (y_2 - y_1) a_{12} - y_2 a_2 \right] \quad (5-37)$$

$$n \frac{\partial b_m}{\partial n_1} = 2y_2 \left[y_1 b_1 + (y_2 - y_1) b_{12} - y_2 b_2 \right] \quad (5-38)$$

$$n \frac{\partial c_m}{\partial n_1} = 2y_2 \left[y_1 c_1 + (y_2 - y_1) c_{12} - y_2 c_2 \right] \quad (5-39)$$

Equations (5-24), (5-25), (5-37), and (5-38) can be simplified somewhat since a_m and b_m are merely linear combinations of the pure substance constants. However, it will be convenient to leave these expressions in their present form so that (5-35) thru (5-39) will be similar.

Equations (5-35) thru (5-39) are now substituted into (5-21) to obtain

$$\begin{aligned}
 & \int_V^{\infty} \left[\left(\frac{\partial P}{\partial n_1} \right)_{T,V,n_2} - \frac{RT}{V} \right] dV = \frac{2}{V} \left[y_1 (B_{O_{11}}^{RT} \right. \\
 & \quad \left. - A_{O_{11}}^{-c_{11}} R/T^2) + y_2 (B_{O_{12}}^{RT} - A_{O_{12}}^{-c_{12}} R/T^2) \right] + \frac{1}{2V^2} \\
 & \quad \left\{ 2y_1 \left[(A_{O_m} a_{11} + A_{O_{11}} a_m) - RT(B_{O_m} b_{11} + B_{O_{11}} b_m) \right. \right. \\
 & \quad \left. \left. - (B_{O_m} c_{11} + B_{O_{11}} c_m) R/T^2 \right] + 2y_2 \left[(A_{O_m} a_{12} + A_{O_{12}} a_m) \right. \right. \\
 & \quad \left. \left. - RT(B_{O_m} b_{12} + B_{O_{12}} b_m) - (B_{O_m} c_{12} + B_{O_{12}} c_m) R/T^2 \right] - (A_{O_m} a_m \right. \\
 & \quad \left. - B_{O_m} b_m RT - B_{O_m} c_m R/T^2) \right\} + \frac{2R}{3T^2 V^3} \left[y_1 (B_{O_m} b_m c_{11} \right. \\
 & \quad \left. + B_{O_m} b_{11} c_m + B_{O_{11}} b_m c_m) + y_2 (B_{O_m} b_m c_{12} + B_{O_m} b_{12} c_m + \right. \\
 & \quad \left. B_{O_{12}} b_m c_m) - B_{O_m} b_m c_m \right] \quad (5-40)
 \end{aligned}$$

Equation (5-40) is further simplified by eliminating y_2 in accordance with the relation

$$y_1 + y_2 = 1 \quad (5-41)$$

This simplification is incorporated in the final expression for the equilibrium equation, obtained by substituting equations (5-5), (5-12), (5-40) and (5-41) into (5-1) to yield

$$\begin{aligned}
 RT \ln \frac{y_1 P}{P_1^0} &= RT \ln \frac{Pv}{P_1^0 v_1^0} + v_1^s (P - P_1^0) + \frac{2\beta_{11}}{v_1^0} \\
 &+ \frac{3\gamma_{11}}{2(v_1^0)^2} + \frac{4\delta_{11}}{3(v_1^0)^3} - 2 [y_1(\beta_{11} - \beta_{12}) + \beta_{12}] / v \\
 &- [2y_1 (\bar{\gamma}_{m,11} - \bar{\gamma}_{m,12}) + 2\bar{\gamma}_{m,12} - \gamma_m] / 2v^2 \\
 &- 2 [y_1 (\bar{\delta}_{m,m,11} - \bar{\delta}_{m,m,12}) + \bar{\delta}_{m,m,12} - \delta_m] / 3v^3
 \end{aligned} \tag{5-42}$$

where

$$\beta_{ij} = B_{o_{ij}} RT - A_{o_{ij}} - c_{ij} R/T^2 \tag{5-43}$$

$$\gamma_{ij} = A_{o_{ij}} a_{ij} - B_{o_{ij}} b_{ij} RT - B_{o_{ij}} c_{ij} R/T^2 \tag{5-44}$$

$$\delta_{ij} = B_{o_{ij}} b_{ij} c_{ij} R/T^2 \tag{5-45}$$

$$\begin{aligned}
 \bar{\gamma}_{m,ij} &= (A_{o_m} b_{ij} + A_{o_{ij}} b_m) - (B_{o_m} b_{ij} + B_{o_{ij}} b_m) RT \\
 &- (B_{o_m} c_{ij} + B_{o_{ij}} c_m) R/T^2
 \end{aligned} \tag{5-46}$$

$$\bar{\delta}_{m,m,ij} = (B_{o_m} b_m c_{ij} + B_{o_m} b_{ij} c_m + B_{o_{ij}} b_m c_m) R/T^2 \tag{5-47}$$

and γ_m and δ_m are as given by (5-19) and (5-20), respectively.

Thus, at a given temperature and pressure equations (5-13) and (5-42) form a set of two equations in two unknowns, specific volume and composition, which may be solved simultaneously. Since this set of equations is highly non-linear some method other than substitution or elimination of one unknown in terms of the other must be employed in order to obtain a solution. Such a method is discussed briefly in section V-C.

For the simplified case, where terms involving y_1 , P_1^0 , and $1/v_1^0$ (with the exception of terms involving ratios of these quantities) are neglected (5-13) reduces to

$$P = \frac{RT}{v} + \frac{\beta_2}{v^2} + \frac{\gamma_2}{v^3} + \frac{\delta_2}{v^4} \quad (5-48)$$

and (5-42) becomes

$$\begin{aligned} RT \ln \frac{y_1 P}{P_1^0} &= RT \ln \frac{Pv}{P_1^0 v_1^0} + v_1^s P - 2 \beta_{12} / v \\ &- (2 \bar{\gamma}_{2,12} - \gamma_2) / 2v^2 - 2 (\bar{\delta}_{2,2,12} - \delta_m) / 3v^3 \end{aligned} \quad (5-49)$$

where

$$\begin{aligned} \bar{\gamma}_{2,12} &= (A_{o_2} b_{o_{12}} + A_{o_{12}} b_2) - (B_{o_2} b_{12} + B_{o_{12}} b_2) RT \\ &- (B_{o_2} c_{12} + B_{o_{12}} c_2) R/T^2 \end{aligned} \quad (5-50)$$

$$\bar{\delta}_{2,2,12} = (B_{o_2} b_2 c_{12} + B_{o_2} b_{12} c_2 + B_{o_{12}} b_2 c_2) R/T^2 \quad (5-51)$$

At a given temperature and pressure (5-48) may be solved directly for the specific volume which in turn may be used in (5-49) to solve for the composition. Thus, in the simplified case the necessity of solving two equations simultaneously is eliminated.

2. Benedict-Webb-Rubin Equation of State

The Benedict-Webb-Rubin equation of state, equation (4-10), is repeated here for convenience as

$$P = \frac{RT}{v} + \frac{B_o RT - A_o - C_o/T^2}{v^2} + \frac{bRT - a}{v^3} + \frac{a\alpha}{v^6} + \frac{c(1+\gamma/v^2)\exp(-\gamma/v^2)}{T^2 v^3} \quad (5-52)$$

and for pure component 1 is written as

$$P = \frac{nRT}{V} + \frac{n^2(B_{o1} RT - A_{o1} - C_{o1}/T^2)}{V^2} + \frac{n^3(b_1 RT - a_1)}{V^3} + \frac{n^6 a_1 \alpha_1}{V^6} + \frac{n^3 c_1 (1 + n^2 \gamma_1/V^2) \exp(-n^2 \gamma_1/V^2)}{T^2 V^3} \quad (5-53)$$

Equation (5-53) is now differentiated to obtain

$$\left(\frac{\partial P}{\partial n}\right)_{T,V} = \frac{RT}{V} + \frac{2n(B_{o1} RT - A_{o1} - C_{o1}/T^2)}{V^2} + \frac{3n^2(b_1 RT - a_1)}{V^3} + \frac{6n^5 a_1 \alpha_1}{V^6} + \frac{3n^2 c_1}{T^2 V^3} \exp(-n^2 \gamma_1/V^2) + \frac{3n^4 c_1 \gamma_1}{T^2 V^5} \exp(-n^2 \gamma_1/V^2) - \frac{2n^6 c_1 \gamma_1^2}{T^2 V^7} \exp(-n^2 \gamma_1/V^2) \quad (5-54)$$

which in turn is used to evaluate the second integral in

(5-1), the result being

$$\int_{v_1^0}^{\infty} \left[\left(\frac{\partial P}{\partial n} \right)_{T,V} - \frac{RT}{V} \right] dv = \frac{2(B_{o1} RT - A_{o1} - C_{o1}/T^2)}{v_1^0} \\ + \frac{3(b_1 RT - a_1)}{2(v_1^0)^2} + \frac{6a_1 \alpha_1}{5(v_1^0)^5} + \frac{c_1}{T^2 \gamma_1} + \frac{c_1}{T^2} \exp \left[-\gamma_1 / (v_1^0)^2 \right] \\ \left[\frac{\gamma_1}{(v_1^0)^4} + \frac{1}{2(v_1^0)^2} - \frac{1}{\gamma_1} \right] \quad (5-55)$$

Equation (5-52) is written for the gas mixture as

$$P = \frac{RT}{v} + \frac{B_{om} RT - A_{om} - C_{om}/T^2}{v^2} + \frac{b_m RT - a_m}{v^3} \\ + \frac{a_m \alpha_m}{v^6} + \frac{c_m (1 + \gamma_m/v^2) \exp(-\gamma_m/v^2)}{T^2 v^3} \quad (5-56)$$

or

$$P = \frac{nRT}{V} + \frac{n^2(B_{om} RT - A_{om} - C_{om}/T^2)}{V^2} + \frac{n^3(b_m RT - a_m)}{V^3} \\ + \frac{n^6 a_m \alpha_m}{V^6} + \frac{n^3 c_m (1 + n^2 \gamma_m/V^2) \exp(-n^2 \gamma_m/V^2)}{T^2 V^3} \quad (5-57)$$

and (5-57) is differentiated and algebraically reduced to yield

$$\left(\frac{\partial P}{\partial n_1} \right)_{T,V,n_2} = \frac{RT}{V} + \frac{2n(B_{om} RT - A_{om} - C_{om}/T^2)}{V^2}$$

$$\begin{aligned}
 & + \frac{3n^2(b_m RT - a_m)}{v^3} + \frac{6n^5 a_m \alpha_m}{v^6} + \frac{n^2}{v^2} \left[RT \frac{\partial B_{om}}{\partial n_1} - \frac{\partial A_{om}}{\partial n_1} \right. \\
 & \left. - \frac{1}{T^2} \frac{\partial C_{om}}{\partial n_1} \right] + \frac{n^3}{v^3} \left[RT \frac{\partial b_m}{\partial n_1} - \frac{\partial a_m}{\partial n_1} \right] + \frac{n^6}{v^6} \left[a_m \frac{\partial \alpha_m}{\partial n_1} \right. \\
 & \left. + \alpha_m \frac{\partial a_m}{\partial n_1} \right] + \frac{n^2 \exp(-n^2 \gamma_m / v^2)}{T^2 v^3} \left[3c_m + n \frac{\partial c_m}{\partial n_1} \right] \\
 & + \frac{n^4 \exp(-n^2 \gamma_m / v^2)}{T^2 v^5} \left[3c_m \gamma_m + n \gamma_m \frac{\partial c_m}{\partial n_1} \right] \\
 & - \frac{n^6 \exp(-n^2 \gamma_m / v^2)}{T^2 v^7} \left[2 c_m \gamma_m^2 + n c_m \gamma_m \frac{\partial \gamma_m}{\partial n_1} \right] \quad (5-58)
 \end{aligned}$$

Equation (5-58) is now used to evaluate the third integral in (5-1) with the result

$$\begin{aligned}
 \int_V \left[\left(\frac{\partial P}{\partial n_1} \right)_{T, V, n_2} - \frac{RT}{V} \right] dV &= \frac{2(B_{om} RT - A_{om} - C_{om} / T^2)}{v} \\
 & + \frac{3(b_m RT - a_m)}{2v^2} + \frac{6a_m \alpha_m}{5v^5} + \frac{1}{v} \left[RTn \frac{\partial B_{om}}{\partial n_1} - n \frac{\partial A_{om}}{\partial n_1} \right. \\
 & \left. - \frac{n}{T^2} \frac{\partial C_{om}}{\partial n_1} \right] + \frac{1}{2v^2} \left[RTn \frac{\partial b_m}{\partial n_1} - n \frac{\partial a_m}{\partial n_1} \right] + \frac{1}{5v^5} \\
 & \left[a_m n \frac{\partial \alpha_m}{\partial n_1} + \alpha_m n \frac{\partial a_m}{\partial n_1} \right] + \frac{1}{T^2 \gamma_m} \left[3c_m + n \frac{\partial c_m}{\partial n_1} \right] - \frac{c_m}{T^2 \gamma_m^2} \\
 & \left[2 \gamma_m + n \frac{\partial \gamma_m}{\partial n_1} \right]
 \end{aligned}$$

$$\begin{aligned}
 & - \frac{\exp(-\gamma_m/v^2)}{2T^2} \left[\frac{2c_m}{\gamma_m} + \frac{2n}{\gamma_m} \frac{\partial c_m}{\partial n_1} - \frac{c_m}{v^2} + \frac{n}{v^2} \frac{\partial c_m}{\partial n_1} \right. \\
 & \left. - \frac{2c_m \gamma_m}{v^4} - \frac{c_m n}{v^4} \frac{\partial \gamma_m}{\partial n_1} - \frac{2c_m n}{\gamma_m v^2} \frac{\partial \gamma_m}{\partial n_1} - \frac{2c_m n}{\gamma_m^2} \frac{\partial \gamma_m}{\partial n_1} \right]
 \end{aligned}
 \tag{5-59}$$

In order to differentiate A_{o_m} , B_{o_m} , C_{o_m} , etc. as required in (5-59) it is necessary to recall the set of combining rules presented in Chapter IV for determining these mixture coefficients from the pure substance constants, this set being

$$A_{o_m} = y_1^2 A_{o_{11}} + 2y_1 y_2 A_{o_{12}} + y_2^2 A_{o_{22}} \tag{5-60}$$

$$B_{o_m} = y_1^2 B_{o_{11}} + 2y_1 y_2 B_{o_{12}} + y_2^2 B_{o_{22}} \tag{5-61}$$

$$C_{o_m} = y_1^2 C_{o_{11}} + 2y_1 y_2 C_{o_{12}} + y_2^2 C_{o_{22}} \tag{5-62}$$

$$\gamma_m = y_1^2 \gamma_{11} + 2y_1 y_2 \gamma_{12} + y_2^2 \gamma_{22} \tag{5-63}$$

$$a_m = y_1^3 a_{111} + 3y_1^2 y_2 a_{112} + 3y_1 y_2^2 a_{122} + y_2^3 a_{222} \tag{5-64}$$

$$b_m = y_1^3 b_{111} + 3y_1^2 y_2 b_{112} + 3y_1 y_2^2 b_{122} + y_2^3 b_{222} \tag{5-65}$$

$$c_m = y_1^3 c_{111} + 3y_1^2 y_2 c_{112} + 3y_1 y_2^2 c_{122} + y_2^3 c_{222} \tag{5-66}$$

$$\alpha_m = y_1^3 \alpha_{111} + 3y_1^2 y_2 \alpha_{112} + 3y_1 y_2^2 \alpha_{122} + y_2^3 \alpha_{222} \quad (5-67)$$

where

$$A_{oij} = (A_{oi} A_{oj})^{\frac{1}{2}} \quad (5-68)$$

$$B_{oij} = \frac{1}{8} (B_{oi}^{1/3} + B_{oj}^{1/3})^3 \quad (5-69)$$

$$C_{oij} = (C_{oi} C_{oj})^{\frac{1}{2}} \quad (5-70)$$

$$\gamma_{ij} = (\gamma_i \gamma_j)^{\frac{1}{2}} \quad (5-71)$$

$$a_{ijk} = (a_i a_j a_k)^{1/3} \quad (5-72)$$

$$b_{ijk} = (b_i b_j b_k)^{1/3} \quad (5-73)$$

$$c_{ijk} = (c_i c_j c_k)^{1/3} \quad (5-74)$$

$$\alpha_{ijk} = (\alpha_i \alpha_j \alpha_k)^{1/3} \quad (5-75)$$

Using equation (5-34) and equations (5-60) thru (5-67) the partial derivatives required in (5-59) are given as

$$n \frac{\partial A_{om}}{\partial n_1} = 2y_2 \left[y_1 A_{o1} + (y_2 - y_1) A_{o12} - y_2 A_{o2} \right] \quad (5-76)$$

$$n \frac{\partial B_{om}}{\partial n_1} = 2y_2 \left[y_1 B_{o1} + (y_2 - y_1) B_{o12} - y_2 B_{o2} \right] \quad (5-77)$$

$$n \frac{\partial C_{om}}{\partial n_1} = 2y_2 \left[y_1 C_{o1} + (y_2 - y_1) C_{o12} - y_2 C_{o2} \right] \quad (5-78)$$

$$n \frac{\partial r_m}{\partial n_1} = 2y_2 \left[y_1^2 r_1 + (y_2 - y_1) r_{12} - y_2^2 r_2 \right] \quad (5-79)$$

$$n \frac{\partial a_m}{\partial n_1} = 3y_2 \left[y_1^2 a_1 + (2y_1 y_2 - y_1^2) a_{112} - (2y_1 y_2 - y_2^2) a_{122} - y_2^2 a_2 \right] \quad (5-80)$$

$$n \frac{\partial b_m}{\partial n_1} = 3y_2 \left[y_1^2 b_1 + (2y_1 y_2 - y_1^2) b_{112} - (2y_1 y_2 - y_2^2) b_{122} - y_2^2 b_2 \right] \quad (5-81)$$

$$n \frac{\partial c_m}{\partial n_1} = 3y_2 \left[y_1^2 c_1 + (2y_1 y_2 - y_1^2) c_{112} - (2y_1 y_2 - y_2^2) c_{122} - y_2^2 c_2 \right] \quad (5-82)$$

$$n \frac{\partial \alpha_m}{\partial n_1} = 3y_2 \left[y_1^2 \alpha_1 + (2y_1 y_2 - y_1^2) \alpha_{112} - (2y_1 y_2 - y_2^2) \alpha_{122} - y_2^2 \alpha_2 \right] \quad (5-83)$$

Equations (5-76) thru (5-83) are now substituted into (5-59) to give

$$\begin{aligned} \int_V \left[\left(\frac{\partial P}{\partial n_1} \right)_{T, V, n_2} - \frac{RT}{V} \right] dV &= \frac{2(y_1^2 A_{11} + y_2^2 A_{12})}{V} \\ &+ \frac{3(y_1^2 B_{111} + 2y_1 y_2 B_{112} + y_2^2 B_{122})}{2V^2} \\ &+ \frac{3(a_m \alpha_{1mm} + \alpha_m a_{1mm})}{5V^5} + \frac{3c_{1mm}}{T^2 \gamma_m} \end{aligned}$$

$$\begin{aligned}
 & - \frac{2c_m \gamma_{1m}}{T^2 \gamma_m^2} - \frac{\exp(-\gamma_m/v^2)}{T^2} \left[\frac{3c_{1mm}}{\gamma_m} \right. \\
 & \left. - \frac{2c_m \gamma_{1m}}{\gamma_m^2} + \frac{3c_{1mm}}{2v^2} - \frac{2c_m \gamma_{1m}}{\gamma_m v^2} - \frac{c_m \gamma_{1m}}{v^4} \right] \quad (5-84)
 \end{aligned}$$

where

$$A_{ij} = B_{oij} RT - A_{oij} - C_{oij}/T^2 \quad (5-85)$$

$$B_{ijk} = b_{ijk} RT - a_{ijk} \quad (5-86)$$

$$\alpha_{1mm} = (\alpha_1 \alpha_m \alpha_m)^{1/3} = y_1^2 \alpha_1 + 2y_1 y_2 \alpha_{112} + y_2^2 \alpha_{122} \quad (5-87)$$

$$a_{1mm} = (a_1 a_m a_m)^{1/3} = y_1^2 a_1 + 2y_1 y_2 a_{112} + y_2^2 a_{122} \quad (5-88)$$

$$c_{1mm} = (c_1 c_m c_m)^{1/3} = y_1^2 c_1 + 2y_1 y_2 c_{112} + y_2^2 c_{122} \quad (5-89)$$

$$\gamma_{1m} = (\gamma_1 \gamma_m)^{1/2} = y_1 \gamma_1 + y_2 \gamma_{12} \quad (5-90)$$

A final expression for the equilibrium equation is obtained by substituting equations (5-5), (5-41), (5-55) and (5-84) into (5-1), the result being

$$RT \ln \frac{y_1 P}{P_1^o} = RT \ln \frac{Pv}{P_1^o v_1^o} + v_1^s (P - P_1^o) + \frac{2A_{11}}{v_1^o} + \frac{3B_{111}}{2(v_1^o)^2}$$

$$\begin{aligned}
 & + \frac{6a_{111}\alpha_{111}}{5(v_1^0)^5} + \frac{c_{111}}{T^2\gamma_{11}} + \frac{c_{111}}{T^2} \exp \left[-\gamma_{11}/(v_1^0)^2 \right] \\
 & \left[\frac{\gamma_{11}}{(v_1^0)^4} + \frac{1}{2(v_1^0)^2} - \frac{1}{\gamma_{11}} \right] - \frac{2 \left[y_1(A_{11}-A_{12})+A_{12} \right]}{v} \\
 & - \frac{3 \left[y_1^2(B_{111}-2B_{112}+B_{122})+2y_1(B_{112}-B_{122})+B_{122} \right]}{2v^2} \\
 & - \frac{3(a_m\alpha_{1mm} + \alpha_m a_{1mm})}{5v^5} - \frac{3c_{1mm}}{T^2\gamma_m} + \frac{2c_m\gamma_{1m}}{T^2\gamma_m^2} \\
 & + \frac{\exp(-\gamma_m/v^2)}{T^2} \left[\frac{3c_{1mm}}{\gamma_m} - \frac{2c_m\gamma_{1m}}{\gamma_m^2} + \frac{3c_{1mm}}{2v^2} \right. \\
 & \left. - \frac{2c_m\gamma_{1m}}{\gamma_mv^2} - \frac{c_m\gamma_{1m}}{v^4} \right] \tag{5-91}
 \end{aligned}$$

Equations (5-56) and (5-91) form a set of two equations in two unknowns, specific volume and composition, for a given temperature and pressure and the set may be solved simultaneously, by the method indicated in section V-C, in order to determine the composition.

In the simplified case where terms involving y_1 , P_1^0 and $1/v_1^0$ are neglected but terms involving ratios of these quantities are retained equation (5-56) becomes

$$P = \frac{RT}{v} + \frac{B_{o2}RT - A_{o2} - C_{o2}/T^2}{v^2} + \frac{b_2RT - a_2}{v^3}$$

$$+ \frac{a_2 \alpha_2}{v^6} + \frac{c_2(1 + \gamma_2/v^2) \exp(-\gamma_2/v^2)}{T^2 v^3} \quad (5-92)$$

and (5-91) reduces to

$$\begin{aligned} RT \ln \frac{y_1^P}{P_1^O} &= RT \ln \frac{Pv}{P_1^O v_1^O} + v_1^S P - \frac{2A_{12}}{v} \\ &- \frac{3B_{122}}{2v^2} - \frac{3(a_2 \alpha_{122} + \alpha_2 a_{122})}{5v^5} - \frac{3c_{122}}{T^2 \gamma_2} \\ &+ \frac{2c_2 \gamma_{12}}{T^2 \gamma_2^2} + \frac{\exp(-\gamma_2/v^2)}{T^2} \left[\frac{3c_{122} \gamma_2^{-2c_2} \gamma_{12}}{\gamma_2^2} \right. \\ &\left. + \frac{3c_{122} \gamma_2^{-4c_2} \gamma_{12}}{2 \gamma_2 v^2} - \frac{c_2 \gamma_{12}}{v^4} \right] \quad (5-93) \end{aligned}$$

Equation (5-92) may be solved for the specific volume at a given temperature and pressure and this value in turn used in (5-93) to determine the composition.

3. Martin-Hou Equation of State

The Martin-Hou equation of state is given as

$$P = \frac{RT}{v-b} + \frac{F_2}{(v-b)^2} + \frac{F_3}{(v-b)^3} + \frac{F_4}{(v-b)^4} + \frac{F_5}{(v-b)^5} \quad (5-94)$$

where

$$F_i = A_i + B_i T + C_i \exp(-kT/T_c) \quad (i=2,3,4,5) \quad (5-95)$$

and

$$B_4 = C_4 = 0 \quad (5-96)$$

For pure component 1, (5-94) is written as

$$P = \frac{nRT}{V-nb_1} + \frac{n^2 F_{21}}{(V-nb_1)^2} + \frac{n^3 F_{31}}{(V-nb_1)^3} + \frac{n^4 F_{41}}{(V-nb_1)^4} + \frac{n^5 F_{51}}{(V-nb_1)^5} \quad (5-97)$$

Differentiation of (5-97) leads to

$$\left(\frac{\partial P}{\partial n}\right)_{T,V} = \frac{RTV}{(V-nb_1)^2} + \frac{2nF_{21}V}{(V-nb_1)^3} + \frac{3n^2F_{31}V}{(V-nb_1)^4} + \frac{4n^3F_{41}V}{(V-nb_1)^5} + \frac{5n^4F_{51}V}{(V-nb_1)^6} \quad (5-98)$$

so that the second integral in (5-1) becomes

$$\int_{v_1^0}^{\infty} \left[\left(\frac{\partial P}{\partial n}\right)_{T,V} - \frac{RT}{V} \right] dV = RT \ln \left(\frac{v_1^0}{v_1^0 - b_1} \right) + \frac{RTb_1 + 2F_{21}}{v_1^0 - b_1} + \frac{2F_{21}b_1 + 3F_{31}}{2(v_1^0 - b_1)^2} + \frac{3F_{31}b_1 + 4F_{41}}{3(v_1^0 - b_1)^3} + \frac{4F_{41}b_1 + 5F_{51}}{4(v_1^0 - b_1)^4} + \frac{F_{51}b_1}{(v_1^0 - b_1)^5} \quad (5-99)$$

Equation (5-94) may be evaluated at the sublimation point of component 1 as

$$P_1^o = \frac{RT}{v_1^o - b_1} + \frac{F_{21}}{(v_1^o - b_1)^2} + \frac{F_{31}}{(v_1^o - b_1)^3} + \frac{F_{41}}{(v_1^o - b_1)^4} + \frac{F_{51}}{(v_1^o - b_1)^5} \quad (5-100)$$

Now, (5-99) and (5-100) are combined to produce

$$\int_{v_1^o}^{\infty} \left[\left(\frac{\partial P}{\partial n} \right)_{T,V} - \frac{RT}{V} \right] dV = RT \ln \left(\frac{v_1^o}{v_1^o - b_1} \right) - RT + P_1^o v_1^o + \frac{F_{21}}{(v_1^o - b_1)} + \frac{F_{31}}{2(v_1^o - b_1)^2} + \frac{F_{41}}{3(v_1^o - b_1)^3} + \frac{F_{51}}{4(v_1^o - b_1)^4} \quad (5-101)$$

For a gas mixture (5-94) is written as

$$P = \frac{RT}{v - b_m} + \frac{F_{2m}}{(v - b_m)^2} + \frac{F_{3m}}{(v - b_m)^3} + \frac{F_{4m}}{(v - b_m)^4} + \frac{F_{5m}}{(v - b_m)^5} \quad (5-102)$$

or

$$P = \frac{nRT}{V - nb_m} + \frac{n^2 F_{2m}}{(V - nb_m)^2} + \frac{n^3 F_{3m}}{(V - nb_m)^3} + \frac{n^4 F_{4m}}{(V - nb_m)^4} + \frac{n^5 F_{5m}}{(V - nb_m)^5} \quad (5-103)$$

and the latter equation is differentiated and algebraically reduced to give

$$\begin{aligned}
 \left(\frac{\partial P}{\partial n_1}\right)_{T,V,n_2} = & (RTV + n^2 RT \frac{\partial b_m}{\partial n_1} + n^2 \frac{\partial F_{2m}}{\partial n_1}) / (V - nb_m)^2 \\
 & + (2nF_{2m}V + 2n^3 F_{2m} \frac{\partial b_m}{\partial n_1} + n^3 \frac{\partial F_{3m}}{\partial n_1}) / (V - nb_m)^3 \\
 & + (3n^2 F_{3m}V + 3n^4 F_{3m} \frac{\partial b_m}{\partial n_1} + n^4 \frac{\partial F_{4m}}{\partial n_1}) / (V - nb_m)^4 \\
 & + (4n^3 F_{4m}V + 4n^5 F_{4m} \frac{\partial b_m}{\partial n_1} + n^5 \frac{\partial F_{5m}}{\partial n_1}) / (V - nb_m)^5 \\
 & + (5n^4 F_{5m}V + 5n^6 F_{5m} \frac{\partial b_m}{\partial n_1}) / (V - nb_m)^6 \quad (5-104)
 \end{aligned}$$

Equation (5-104) is now used to evaluate the third integral in (5-1), the result being

$$\begin{aligned}
 \int_V^\infty \left[\left(\frac{\partial P}{\partial n_1}\right)_{T,V,n_2} - \frac{RT}{V} \right] dV = & RT \ln \left(\frac{V}{V - nb_m} \right) \\
 & + \left[RT (b_m + n \frac{\partial b_m}{\partial n_1}) + (2F_{2m} + n \frac{\partial F_{2m}}{\partial n_1}) \right] / (V - b_m) \\
 & + \left[F_{2m} (b_m + n \frac{\partial b_m}{\partial n_1}) + \frac{1}{2} (3F_{3m} + n \frac{\partial F_{3m}}{\partial n_1}) \right] / (V - b_m)^2 \\
 & + \left[F_{3m} (b_m + n \frac{\partial b_m}{\partial n_1}) + \frac{1}{3} (4F_{4m} + n \frac{\partial F_{4m}}{\partial n_1}) \right] / (V - b_m)^3 \\
 & + \left[F_{4m} (b_m + n \frac{\partial b_m}{\partial n_1}) + \frac{1}{4} (5F_{5m} + n \frac{\partial F_{5m}}{\partial n_1}) \right] / (V - b_m)^4
 \end{aligned}$$

$$+ \left[F_{5_m} (b_m + n \frac{\partial b_m}{\partial n_1}) \right] / (v-b_m)^5 \quad (5-105)$$

Equations (5-102) and (5-105) are combined to yield

$$\begin{aligned} \int_V^{\infty} \left[\left(\frac{\partial P}{\partial n_1} \right)_{T,V,n_2} - \frac{RT}{V} \right] dV &= RT \ln \left(\frac{v}{v-b_m} \right) - RT \\ &+ Pv + Pn \frac{\partial b_m}{\partial n_1} + (F_{2_m} + n \frac{\partial F_{2_m}}{\partial n_1}) / (v-b_m) \\ &+ \frac{1}{2} (F_{3_m} + n \frac{\partial F_{3_m}}{\partial n_1}) / (v-b_m)^2 + \frac{1}{3} (F_{4_m} + n \frac{\partial F_{4_m}}{\partial n_1}) / (v-b_m)^3 \\ &+ \frac{1}{4} (F_{5_m} + n \frac{\partial F_{5_m}}{\partial n_1}) / (v-b_m)^4 \end{aligned} \quad (5-106)$$

In order to evaluate the partial derivatives required in (5-106) it is recalled that

$$F_{i_m} = A_{i_m} + B_{i_m} T + C_{i_m} \exp(-kT/T_{c_m}) \quad (i=2,3,4,5) \quad (5-107)$$

where the constants A_{i_m} , B_{i_m} , C_{i_m} and T_{c_m} are given as, according to the set of combining rules discussed in Chapter IV

$$A_{i_m} = y_1^2 A_{i_{11}} + 2y_1 y_2 A_{i_{12}} + y_2^2 A_{i_{22}} \quad (i=2,3,4,5) \quad (5-108)$$

$$B_{i_m} = y_1^2 B_{i_{11}} + 2y_1 y_2 B_{i_{12}} + y_2^2 B_{i_{22}} \quad (i=2,3,5) \quad (5-109)$$

$$C_{i_m} = y_1^2 C_{i_{11}} + 2y_1 y_2 C_{i_{12}} + y_2^2 C_{i_{22}} \quad (i=2,3,5) \quad (5-110)$$

$$T_{c_m} = y_1^2 T_{c_{11}} + 2y_1 y_2 T_{c_{12}} + y_2^2 T_{c_{22}} \quad (5-111)$$

in which

$$A_{i_{jk}} = \frac{1}{8} (A_{i_j}^{1/3} + A_{i_k}^{1/3})^3 \quad (i=2,3,4,5) \quad (5-112)$$

$$B_{i_{jk}} = (B_{i_j} B_{i_k})^{1/2} \quad (i=2,3,5) \quad (5-113)$$

$$C_{i_{jk}} = \frac{1}{8} (C_{i_j}^{1/3} + C_{i_k}^{1/3})^3 \quad (i=2,3,5) \quad (5-114)$$

$$T_{c_{jk}} = (T_{c_j} T_{c_k})^{1/2} \quad (5-115)$$

In addition

$$b_m = y_1 b_1 + y_2 b_2 \quad (5-116)$$

Now, equation (5-34), equations (5-108) thru (5-111) and equation (5-116) may be used to evaluate the partial derivatives appearing in (5-106), the result being

$$n \frac{\partial b_m}{\partial n_1} = y_2 (b_1 - b_2) \quad (5-117)$$

$$n \frac{\partial F_{i_m}}{\partial n_1} = y_2 \left[\frac{\partial A_{i_m}}{\partial y_1} + \frac{\partial B_{i_m}}{\partial y_1} T + \frac{\partial C_{i_m}}{\partial y_1} \exp(-kT/T_{c_m}) \right. \\ \left. + \frac{\partial T_{c_m}}{\partial y_1} \frac{C_{i_m} kT}{T_{c_m}^2} \exp(-kT/T_{c_m}) \right] \quad (i=2,3,4,5) \quad (5-118)$$

where

$$\frac{\partial A_{i_m}}{\partial y_1} = 2y_2 \left[y_1 A_{i_1} + (y_2 - y_1) A_{i_{12}} - y_2 A_{i_2} \right] \quad (i=2,3,4,5) \quad (5-119)$$

$$\frac{\partial B_{i_m}}{\partial y_1} = 2y_2 \left[y_1 B_{i_1} + (y_2 - y_1) B_{i_{12}} - y_2 B_{i_2} \right] \quad (i=2,3,5) \quad (5-120)$$

$$\frac{\partial C_{i_m}}{\partial y_1} = 2y_2 \left[y_1 C_{i_1} + (y_2 - y_1) C_{i_{12}} - y_2 C_{i_2} \right] \quad (i=2,3,5) \quad (5-121)$$

$$\frac{\partial T_{c_m}}{\partial y_1} = 2y_2 \left[y_1 T_{c_1} + (y_2 - y_1) T_{c_{12}} - y_2 T_{c_2} \right] \quad (5-122)$$

Equations (5-117) thru (5-122) are now substituted into (5-106) to obtain

$$\begin{aligned} \int_V^{\infty} \left[\left(\frac{\partial P}{\partial n_1} \right)_{T,V,n_2} - \frac{RT}{V} \right] dV = RT \left[\ln \left(\frac{v}{v-b_m} \right) - 1 \right] \\ + P \left[v + y_2(b_1 - b_2) \right] + \frac{\bar{F}_{2_m}}{(v-b_m)} + \frac{\bar{F}_{3_m}}{2(v-b_m)^2} \\ + \frac{\bar{F}_{4_m}}{3(v-b_m)^3} + \frac{\bar{F}_{5_m}}{4(v-b_m)^4} \end{aligned} \quad (5-123)$$

where

$$\bar{F}_{i_m} = F_{i_m} + 2y_1 y_2 \left[A_{i_{11}} + B_{i_{11}} T + C_{i_{11}} \exp(-kT/T_{c_m}) \right]$$

$$\begin{aligned}
 & - 2y_2^2 \left[A_{i22} + B_{i22} T + C_{i22} \exp(-kT/T_{c_m}) \right] \\
 & + 2y_2 (y_2 - y_1) \left[A_{i12} + B_{i12} T + C_{i12} \exp(-kT/T_{c_m}) \right] \\
 & + 2y_2 \left[y_1 T_{c11} + (y_2 - y_1) T_{c12} - y_2 T_{c22} \right] \frac{C_{i_m} kT}{T_{c_m}^2} \exp(-kT/T_{c_m}) \\
 & (i=2,3,4,5) \tag{5-124}
 \end{aligned}$$

The final expression for the equilibrium equation is obtained by substituting equations (5-5), (5-41), (5-96), (5-101) and (5-123) into (5-1) with the result

$$\begin{aligned}
 RT \ln \frac{y_1 P}{P_1^0} &= RT \ln \frac{Pv}{P_1^0 v_1^0} + v_1^s (P - P_1^0) + RT \ln \frac{v_1^0}{v_1^0 - b_{11}} \\
 & - RT \ln \frac{v}{v - b_m} + P_1^0 v_1^0 + \frac{F_{211}}{(v_1^0 - b_{11})} + \frac{F_{311}}{2(v_1^0 - b_{11})^2} \\
 & + \frac{A_{411}}{3(v_1^0 - b_{11})^3} + \frac{F_{511}}{4(v_1^0 - b_{11})^4} - P \left[v + (1 - y_1)(b_{11} - b_{22}) \right] \\
 & - \frac{\bar{F}_{2m}}{(v - b_m)} - \frac{\bar{F}_{3m}}{2(v - b_m)^2} - \frac{y_1(2 - y_1)(A_{411} - 2A_{412} + A_{422})}{3(v - b_m)^3} \\
 & - \frac{2A_{412} - A_{422}}{3(v - b_m)^3} - \frac{\bar{F}_{5m}}{4(v - b_m)^4} \tag{5-125}
 \end{aligned}$$

where

$$F_{ijj} = A_{ijj} + B_{ijj} T + C_{ijj} \exp(-kT/T_{c_{jj}}) \quad (5-126)$$

$$\bar{F}_{i_m} = F_{i_m} + 2y_1(1-y_1) [A_{i_{11}} + B_{i_{11}} T + C_{i_{11}} \exp(-kT/T_{c_m})]$$

$$-2(1-2y_1+y_1^2) [A_{i_{22}} + B_{i_{22}} T + C_{i_{22}} \exp(-kT/T_{c_m})]$$

$$+2(1-3y_1+2y_1^2) [A_{i_{12}} + B_{i_{12}} T + C_{i_{12}} \exp(-kT/T_{c_m})]$$

$$+2(1-y_1) [T_{c_{12}} - T_{c_{22}} + y_1(T_{c_{11}} - 2T_{c_{12}} + T_{c_{22}})]$$

$$\frac{C_{i_m} kT}{T_{c_m}^2} \exp(-kT/T_{c_m}) \quad (i=2,3,5) \quad (5-127)$$

At a given temperature and pressure equations (5-102) and (5-125) form the desired set of two equations in two unknowns, v and y_1 , which may be solved simultaneously to determine the composition.

For the simplified case in which terms involving y_1 , P_1^0 and $1/v_1^0$ are neglected, with the exception of those involving ratios of these quantities, equation (5-102) reduces to

$$P = \frac{RT}{v-b_{22}} + \frac{F_{222}}{(v-b_{22})^2} + \frac{F_{322}}{(v-b_{22})^3} + \frac{A_{422}}{(v-b_{22})^4}$$

$$+ \frac{F_{522}}{(v-b_{22})^5} \quad (5-128)$$

and (5-125) becomes

$$\begin{aligned} RT \ln \frac{y_1^P}{p_1^0} &= RT \ln \frac{Pv}{p_1^0 v_1^0} + v_1^S P - RT \ln \left(\frac{v}{v-b_{22}} \right) \\ &+ p_1^0 v_1^0 - P(v+b_{11}-b_{22}) - \frac{D_2}{(v-b_{22})} - \frac{D_3}{2(v-b_{22})^2} \\ &- \frac{2A_{412} - A_{422}}{3(v-b_{22})^3} - \frac{D_5}{4(v-b_{22})^4} \end{aligned} \quad (5-129)$$

where

$$\begin{aligned} D_i &= 2 \left[A_{i12} + B_{i12} T + C_{i12} \exp(-kT/T_{c22}) \right] \\ &+ 2 (T_{c12} - T_{c22}) \frac{C_{i22} kT}{T_{c22}^2} \exp(-kT/T_{c22}) \\ &- F_{i22} \quad (i=2,3,5) \end{aligned} \quad (5-130)$$

At a given temperature and pressure, (5-128) may be solved directly for the specific volume and this value used in (5-129) to determine y_1 .

4. Virial Equation of State

The virial equation of state, equation (4-16), may be written as

$$P = \frac{RT}{V} + \frac{RTB}{V^2} + \frac{RTC}{V^3} \quad (5-131)$$

and for pure 1 is expressed as

$$P = \frac{nRT}{V} + \frac{n^2 RTB_1}{V^2} + \frac{n^3 RTC_1}{V^3} \quad (5-132)$$

Differentiation of (5-132) yields

$$\left(\frac{\partial P}{\partial n}\right)_{T,V} = \frac{RT}{V} + \frac{2nRTB_1}{V^2} + \frac{3n^2 RTC_1}{V^3} \quad (5-133)$$

so that the second integral in (5-1) becomes

$$\int_{V_1^0}^{\infty} \left[\left(\frac{\partial P}{\partial n}\right)_{T,V} - \frac{RT}{V} \right] dV = \frac{2RTB_1}{V_1^0} + \frac{3RTC_1}{2(V_1^0)^2} \quad (5-134)$$

The virial equation of state is written for the gas mixture as

$$P = \frac{RT}{V} + \frac{RTB_m}{V^2} + \frac{RTC_m}{V^3} \quad (5-135)$$

or

$$P = \frac{nRT}{V} + \frac{n^2 RTB_m}{V^2} + \frac{n^3 RTC_m}{V^3} \quad (5-136)$$

and, therefore,

$$\left(\frac{\partial P}{\partial n_1}\right)_{T,V,n_2} = \frac{RT}{V} + \frac{RT}{V^2} \frac{\partial}{\partial n_1} (n^2 B_m) + \frac{RT}{V^3} \frac{\partial}{\partial n_1} (n^3 C_m) \quad (5-137)$$

Thus, the third integral in (5-1) becomes

$$\int_V^{\infty} \left[\left(\frac{\partial P}{\partial n_1} \right)_{T,V,n_2} - \frac{RT}{V} \right] dV = \frac{RT}{V} \frac{\partial}{\partial n_1} (n^2 B_m) + \frac{RT}{2V^2} \frac{\partial}{\partial n_1} (n^3 C_m) \quad (5-138)$$

or

$$\int_V^{\infty} \left[\left(\frac{\partial P}{\partial n_1} \right)_{T,V,n_2} - \frac{RT}{V} \right] dV = \frac{RT}{V} \left[2B_m + n \frac{\partial B_m}{\partial n_1} \right] + \frac{RT}{2V^2} \left[3C_m + n \frac{\partial C_m}{\partial n_1} \right] \quad (5-139)$$

The mixture coefficients B_m and C_m are given by equations (4-65) and (4-66), respectively, these equations being repeated here as

$$B_m = y_1^2 B_{11} + 2y_1 y_2 B_{12} + y_2^2 B_{22} \quad (5-140)$$

$$C_m = y_1^3 C_{111} + 3y_1^2 y_2 C_{112} + 3y_1 y_2^2 C_{122} + y_2^3 C_{222} \quad (5-141)$$

Equations (5-34), and (5-140) and (5-141) may now be used to determine the partial derivatives required in (5-139), the result being

$$n \frac{\partial B_m}{\partial n_1} = 2y_2 \left[y_1 B_{11} + (y_2 - y_1) B_{12} - y_2 B_{22} \right] \quad (5-142)$$

$$n \frac{\partial C_m}{\partial n_1} = 3y_2 \left[y_1^2 C_{111} + (2y_1 y_2 - y_1^2) C_{112} - (2y_1 y_2 - y_2^2) C_{122} - y_2^2 C_{222} \right] \quad (5-143)$$

Equations (5-140) thru (5-143) are now substituted into (5-139) to yield

$$\int_V^\infty \left[\left(\frac{\partial P}{\partial n_1} \right)_{T,V,n_2} - \frac{RT}{V} \right] dV = \frac{2RT}{V} \left[y_1 B_{11} + y_2 B_{12} \right] + \frac{3RT}{2V} \left[y_1^2 C_{111} + 2y_1 y_2 C_{112} + y_2^2 C_{122} \right] \quad (5-144)$$

A final expression for the equilibrium equation is now obtained by substituting equations (5-5), (5-41), (5-134) and (5-144) into (5-1) to give

$$RT \ln \frac{y_1 P}{P_1^0} = RT \ln \frac{Pv}{P_1^0 v_1^0} + v_1^s (P - P_1^0) + \frac{2RT B_{11}}{v_1^0} + \frac{3RT C_{111}}{2(v_1^0)^2} - \frac{2RT}{v} \left[y_1 (B_{11} - B_{12}) + B_{12} \right] - \frac{3RT}{2v^2} \left[y_1^2 (C_{111} - 2C_{112} + C_{122}) + 2y_1 (C_{112} - C_{122}) + C_{122} \right] \quad (5-145)$$

Equations (5-135) and (5-145) form the desired set of two equations in two unknowns, y_1 and v , at a given temperature

and pressure. This set may be solved simultaneously to determine the composition.

In the simplified case where terms involving y_1 , P_1^0 and $1/v_1^0$ are assumed negligible, with the exception of terms involving ratios of these quantities, equation (5-135) becomes

$$P = \frac{RT}{v} + \frac{RTB_{22}}{v^2} + \frac{RTC_{222}}{v^3} \quad (5-146)$$

and (5-145) reduces to

$$RT \ln \frac{y_1 P}{P_1^0} = RT \ln \frac{Pv}{P_1^0 v_1^0} + v_1^s P - \frac{2RTB_{12}}{v} - \frac{3RTC_{122}}{2v^2} \quad (5-147)$$

As before, (5-146) may be solved for v at any given temperature and pressure and this value in turn used in (5-147) to determine the composition y_1 .

C. Method of Solution

The procedure for solving the equations presented in the preceding section is to first solve the set for the simplified case. For this special case the equation of state reduces to that of pure component 2 and is therefore a function of v only at a given temperature and pressure. Thus, this equation may be solved for v by using Newton's iterative method for one equation in one unknown, and this value in turn used in the simplified equilibrium equation to determine the mole fraction y_1 . Newton's method requires

an initial value for v in order to begin the iteration, and this value is obtained by multiplying the particular equation of state being employed by the quantity v/P $\left[(v-b)/P \right.$ for the Martin-Hou equation $\left. \right]$ and then substituting RT/P , from the ideal gas law, for all v 's appearing on the right hand side of the equation. The result of this operation is

$$v = \frac{RT}{P} + \frac{\beta}{RT} + \frac{\gamma_P}{(RT)^2} + \frac{\delta_P^2}{(RT)^3} \quad (5-148)$$

$$v = \frac{RT}{P} + \frac{B_0 RT - A_0 - C_0/T^2}{RT} + \frac{P(bRT - a)}{(RT)^2} + \frac{P^4 a \alpha}{(RT)^5} \\ + \frac{PC \left[1 + \gamma_P^2/(RT)^2 \right] \exp \left[-\gamma_P^2/(RT)^2 \right]}{(RT^2)^2} \quad (5-149)$$

$$v = b + \frac{RT}{P} + \frac{F_2}{RT - Pb} + \frac{PF_3}{(RT - Pb)^2} + \frac{P^2 A_4}{(RT - Pb)^3} \\ + \frac{P^3 F_5}{(RT - Pb)^4} \quad (5-150)$$

$$v = \frac{RT}{P} + B + \frac{PC}{RT} \quad (5-151)$$

for the Beattie-Bridgeman, Benedict-Webb-Rubin, Martin-Hou and virial equations of state, respectively. Equations (5-148) thru (5-151) are not considered to be accurate volume-explicit equations of state but they are better than the ideal gas law for determining initial values of v to ensure convergence in the Newton iterative method.

In order to complete the solution for the simplified case, the composition y_1 is determined by using the value of v obtained from the above method and solving equations (5-49), (5-93), (5-129) and (5-147), these equations resulting from the use of the Beattie-Bridgeman, Benedict-Webb-Rubin, Martin-Hou and virial equations of state, respectively. In order to solve the foregoing equations for y_1 , values of P_1^O , v_1^O and v_1^S are required at each temperature. The values of P_1^O and v_1^S used here have been taken from references 92 and 93 and are listed in Table I. Values of v_1^O at each temperature are calculated from equations (5-148) thru (5-151) using P_1^O . The values so calculated are considered to be sufficiently accurate since P_1^O is always quite small.

TABLE I
VALUES OF P_1^O , v_1^S
FOR CO_2

T, °K	P_1^O , atm	v_1^S , lit/mole
140	0.00184	0.0270
150	0.00841	0.0272
160	0.0314	0.0274
170	0.0993	0.0276
180	0.274	0.0278
190	0.679	0.0280

After solving the equations for the simplified case, the general solution is obtained by solving the sets of two equations in two unknowns presented in the preceding

section, this being accomplished by using the Newton-Raphson iterative method for the simultaneous solution of non-linear equations. This method requires reasonably close initial guesses for the two unknowns, y_1 and v , to ensure convergence during the iteration process. The values of y_1 and v obtained for the simplified case make good initial guesses and have been used as such in the present work. A discussion of the Newton and Newton-Raphson method is presented in Appendix B. In addition, the computer flow diagrams for the iterative solutions of the equations presented in Chapter IV are included in Appendix D.

The pure gas constants for the Beattie-Bridgeman equation are taken from reference 61, Benedict-Webb-Rubin from 67, 81 and 83, and Martin-Hou from 58 and 64 and these values are listed in Tables II, III and IV, respectively.

TABLE II
BEATTIE-BRIDGEMAN CONSTANTS
(atm, lit/mole, $^{\circ}\text{K}$)

A_0	5.0065	1.3445
B_0	0.10476	0.05046
a	0.07132	0.02617
b	0.07235	-0.00691
c	66.0×10^4	4.2×10^4

TABLE III
BENEDICT-WEBB-RUBIN CONSTANTS
(atm, lit/mole, °K)

	CO ₂	N ₂
A ₀	2.73726	1.19250
B ₀	0.0499086	0.0458000
C ₀	1.38558x10 ⁵	5.88907x10 ³
a	0.136802	0.0149000
b	0.00412368	0.00198154
c	1.49167x10 ⁴	5.48064x10 ²
α	0.846556x10 ⁻⁴	0.291545x10 ⁻³
γ	0.00539329	0.00750000

The critical temperatures listed in Table IV have been taken from reference 94.

TABLE IV
MARTIN-HOU CONSTANTS
(atm, lit/mole, °K)

	CO ₂	N ₂
b	0.020591713	0.020999999
A ₂	-4.5852792	-1.5296017
B ₂	0.0048652509	0.0028360157
C ₂	-77.544346	-25.727366
A ₃	0.25327630	0.076770659
B ₃	-0.14657252x10 ⁻³	-0.50760631x10 ⁻⁴
C ₃	6.6404483	2.0850164
A ₄	-0.0081897877	-0.0024514358
A ₅	0.74749516x10 ⁻⁴	0.32831952x10 ⁻⁴
B ₅	0.19623218x10 ⁻⁶	0.30582839x10 ⁻⁷
C ₅	-0.48510253x10 ⁻³	-0.0014670686
T _c	304.2	126.1

k= 5.475

Since two potential functions, that of Lennard-Jones and of Kihara, are employed here two different sets of pure gas second virial coefficients will be available for use in solving equations (5-135), (5-145) thru (5-147) and (5-151). In addition, each of these sets may or may not include the quadrupole correction discussed in sections IV - B - 4a and IV - B - 4b. The two sets of coefficients used here, both with and without quadrupole corrections, are listed in Tables V and VI.

TABLE V
PURE GAS SECOND VIRIAL COEFFICIENTS
CALCULATED FROM LENNARD-JONES (6-12) POTENTIAL
(atm, lit/mole, °K)

T	Without Quadrupole Correction		With Quadrupole Correction	
	B ₁	B ₂	B ₁	B ₂
140	-0.48605	-0.07944	-0.49887	-0.08156
150	-0.43118	-0.06905	-0.44175	-0.07086
160	-0.38571	-0.06019	-0.39453	-0.06176
170	-0.34740	-0.05256	-0.35485	-0.05394
180	-0.31469	-0.04592	-0.32107	-0.04713
190	-0.28667	-0.04009	-0.29197	-0.04117

The force constants used in the Lennard-Jones (6-12) potential for determining the values given in Table V were taken from reference 67 and are listed in Table VII while the Kihara parameters used in obtaining the values in Table VI were taken from reference 56 and are presented in Table VIII.

TABLE VI
 PURE GAS SECOND VIRIAL COEFFICIENTS
 CALCULATED FROM KIHARA POTENTIAL
 (atm, lit/mole, °K)

T	Without Quadrupole Correction		With Quadrupole Correction	
	B ₁	B ₂	B ₁	B ₂
140	-0.62990	-0.07995	-0.72083	-0.08318
150	-0.53193	-0.06898	-0.60016	-0.07169
160	-0.45667	-0.05974	-0.50938	-0.06205
170	-0.39733	-0.05186	-0.43906	-0.05385
180	-0.34952	-0.04506	-0.38324	-0.04680
190	-0.31029	-0.03914	-0.33802	-0.04067

TABLE VII
 FORCE CONSTANTS FOR THE LENNARD-JONES (6-12) POTENTIAL

	$\epsilon/k, ^\circ K$	$b_0, \text{lit/mole}$
CO ₂	189.0	0.1139
N ₂	95.05	0.06378

The values used for the quadrupole moments in determining the quadrupole corrections for CO₂ and N₂ are presented in reference 95, these values being 3.0×10^{-26} and 1.5×10^{-26} esu-cm², respectively.

TABLE VIII
PARAMETERS FOR THE KIHARA POTENTIAL

	$M_o, \text{\AA}$	$S_o, \text{\AA}^2$	$V_o, \text{\AA}^3$	$\rho_o, \text{\AA}$	$U_o/k, ^\circ K$
CO ₂	9.173	2.469	0.1781	2.790	380.20
N ₂	4.694	0.8130	0.03856	3.148	136.67

The pure gas third virial coefficients used here are determined from equations (4-28) and (4-29), these equations being derived from the Lennard-Jones (6-12) potential. Since two sets of c_j (references 67 and 72) have been proposed for use in (4-28) two third virial coefficients may be determined at a given temperature. A tabulation of the two coefficients as a function of temperature is presented in Table IX.

TABLE IX
PURE GAS THIRD VIRIAL COEFFICIENTS
CALCULATED FROM LENNARD-JONES (6-12) POTENTIAL
(atm, lit/mole, $^\circ K$)

T	Using c_j of Reference 67 in Equation (4-28)		Using c_j of Reference 72 in Equation (4-29)	
	C_1	C_2	C_1	C_2
140	-0.02631	+0.00224	-0.02839	+0.00219
150	-0.01224	+0.00213	-0.01384	+0.00209
160	-0.00399	+0.00203	-0.00521	+0.00199
170	+0.00096	+0.00193	-0.00000	+0.00190
180	+0.00393	+0.00185	+0.00317	+0.00183
190	+0.00578	+0.00178	+0.00509	+0.00176

The only remaining constants necessary for the solution of the equilibrium equation based on the virial equation of state are the second and third virial interaction coefficients. The values for B_{12} based on the Lennard-Jones (6-12) potential, both with and without quadrupole corrections, have been determined by the methods discussed in section IV - C - 2a and are presented in Table X while those based on the Kihara potential and determined by the methods proposed in section IV - C - 2b are given in Table XI.

TABLE X
SECOND VIRIAL INTERACTION COEFFICIENTS
CALCULATED FROM THE LENNARD-JONES (6-12) POTENTIAL

T	Without Quadrupole Correction	With Quadrupole Correction
	B_{12}	B_{12}
140	-0.20327	-0.20805
150	-0.17989	-0.18393
160	-0.15960	-0.16363
170	-0.14369	-0.14631
180	-0.12877	-0.13138
190	-0.11606	-0.11835

The values for C_{112} and C_{122} have been determined by the method of Ewald as discussed in section IV - C - 2a using the c_j of both references 67 and 72 in equation (4-28), and the results are listed in Table XII.

TABLE XI
SECOND VIRIAL INTERACTION COEFFICIENTS
CALCULATED FROM THE KIHARA POTENTIAL

Without Quadrupole Correction		With Quadrupole Correction
T	B_{12}	B_{12}
140	-0.20979	-0.22355
150	-0.18235	-0.19343
160	-0.15995	-0.16905
170	-0.14134	-0.14894
180	-0.12563	-0.13208
190	-0.11222	-0.11776

TABLE XII
THIRD VIRIAL INTERACTION COEFFICIENTS
CALCULATED FROM THE LENNARD-JONES (6-12) POTENTIAL
(atm, lit/mole, °K)

Using c_j of Reference 67 in Equation (4-28)			Using c_j of Reference 72 in Equation (4-28)	
T	C_{112}	C_{122}	C_{112}	C_{122}
140	+0.00204	+0.00361	+0.00146	+0.00344
150	+0.00384	+0.00364	+0.00340	+0.00351
160	+0.00476	+0.00357	+0.00441	+0.00346
170	+0.00519	+0.00346	+0.00491	+0.00337
180	+0.00535	+0.00333	+0.00512	+0.00325
190	+0.00535	+0.00320	+0.00516	+0.00313

VI. EXPERIMENTAL EQUIPMENT

A. General Description

A continuous flow system for attaining equilibrium was employed in this investigation rather than the static or recirculating systems which have been utilized in similar equilibrium studies. This type of system has also been employed in recent investigations by Sonntag⁽²⁵⁾ and Hiza and Herring⁽⁹⁶⁾ with highly satisfactory results. The flow method is particularly advantageous in that it permits a continuous analysis of the gas phase which is of great importance in assuring the attainment of equilibrium and in determining the true equilibrium composition.

The schematic diagram of the experimental setup used in this investigation is presented in Figure 1 and the control panel associated with this equipment may be seen in Figures 2 and 3. The experimental setup as diagrammed in Figure 1 may be considered in three basic sections; (1) the cryostat, (2) test gas flow path and (3) heat exchanger coolant flow path. Each of these divisions are discussed in the following sections.

1. Cryostat

The cryostat is a vacuum jacketed vessel containing the bath liquid, liquid-level indicator, equilibrium coil, equilibrium cell, heater, circulating pump, cooling coil, platinum resistance thermometer, thermocouples and temperature controller sensing bulb. These items, with the

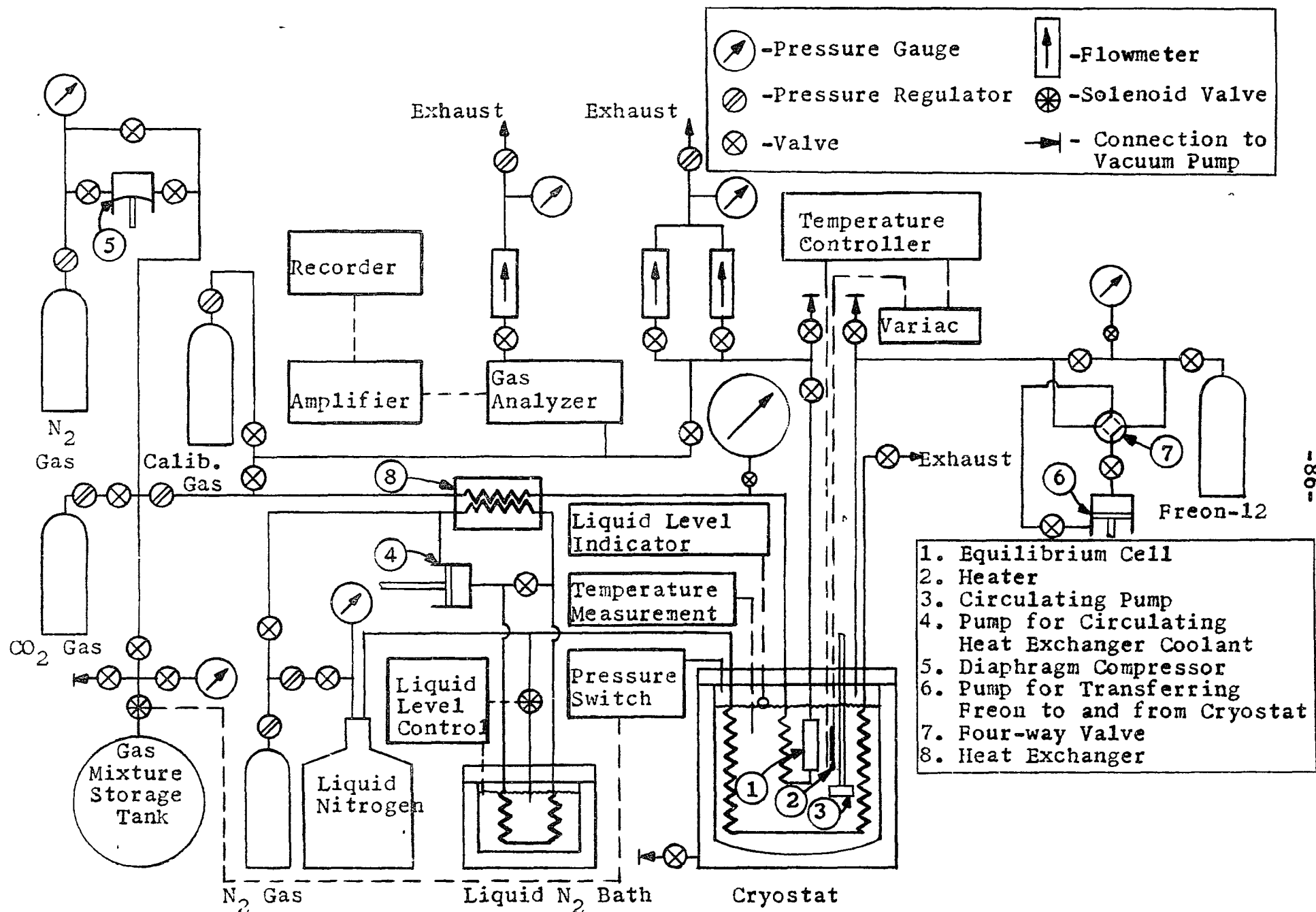


Figure 1. Schematic Diagram of Experimental Equipment

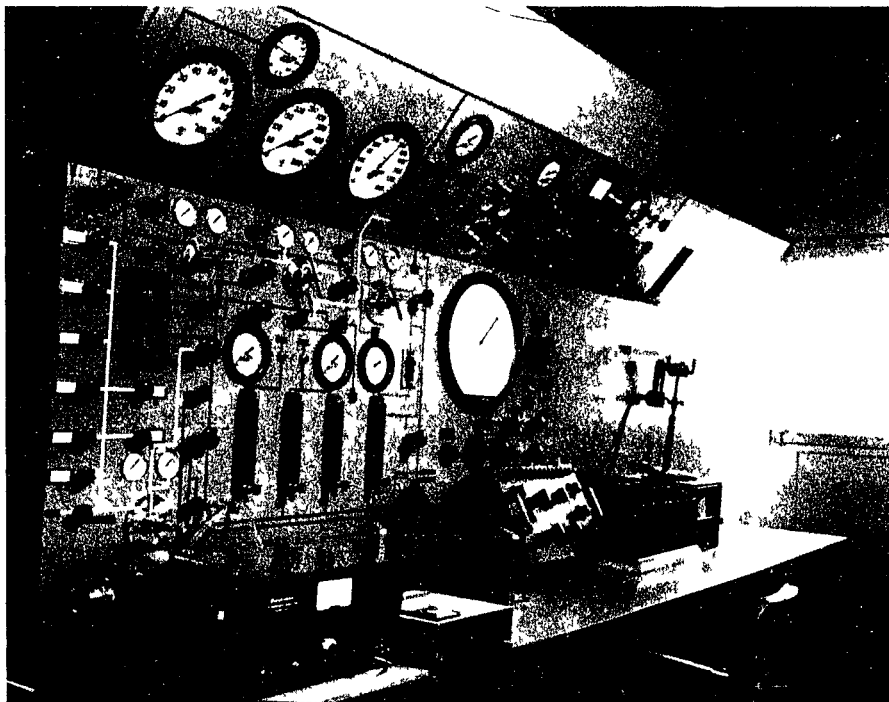


Figure 2. View of Control Panel

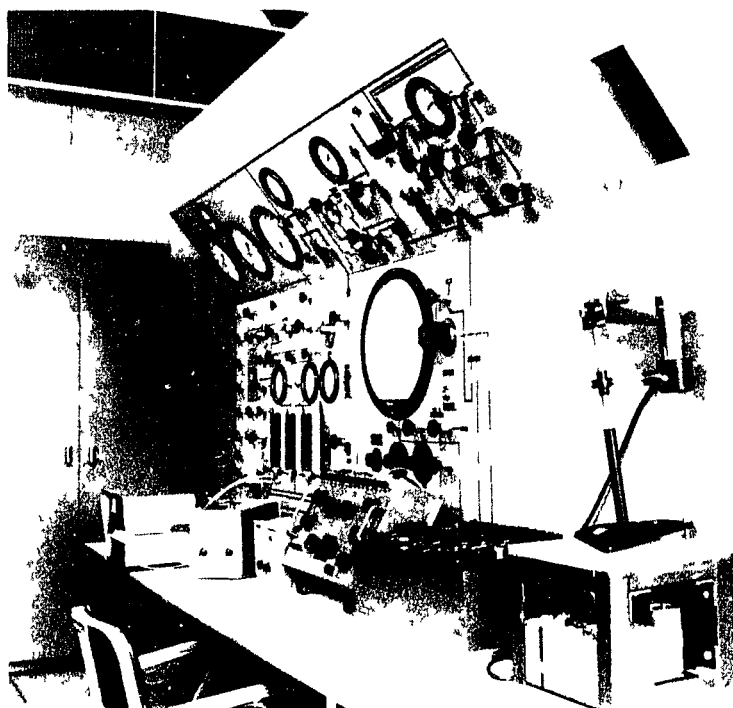


Figure 3. View of Control Panel

exception of the bath liquid, may be seen in Figures 4 thru 10.

The bath liquid used in the cryostat is Freon-12, this liquid being advantageous over other bath liquids in that it is nonflammable, nonexplosive and nontoxic. However, because its vapor pressure is less than atmospheric over the temperature range covered in the present study the cryostat is sealed by bolting the cryostat to a suspended top plate with a double O-ring seal between the two. Figure 10 shows the cryostat bolted in place, Figure 11 shows the O-ring grooves and Figure 12 shows the insulation in place around the cryostat. In addition, all tubes and wires which pass into the cryostat are sealed by passing all such elements through Conax packing glands located in the top plate as shown in Figure 13. The only remaining element to be sealed is the drive shaft of the circulating pump which is accomplished with a standard rotary seal. However, so that the rotary seal will not have to withstand the low temperatures encountered at the top plate, it is placed on a brass plate located approximately 12 inches above the top plate. The drive shaft extends through the rotary seal and is enclosed in a stainless steel pipe which is sealed to the top plate and the brass plate, thus sealing the cryostat. This pipe is visible in Figure 13.

During operation liquid nitrogen is passed through the cooling coil, Figure 9, at a rate sufficient to cool the bath liquid slightly below the desired temperature.

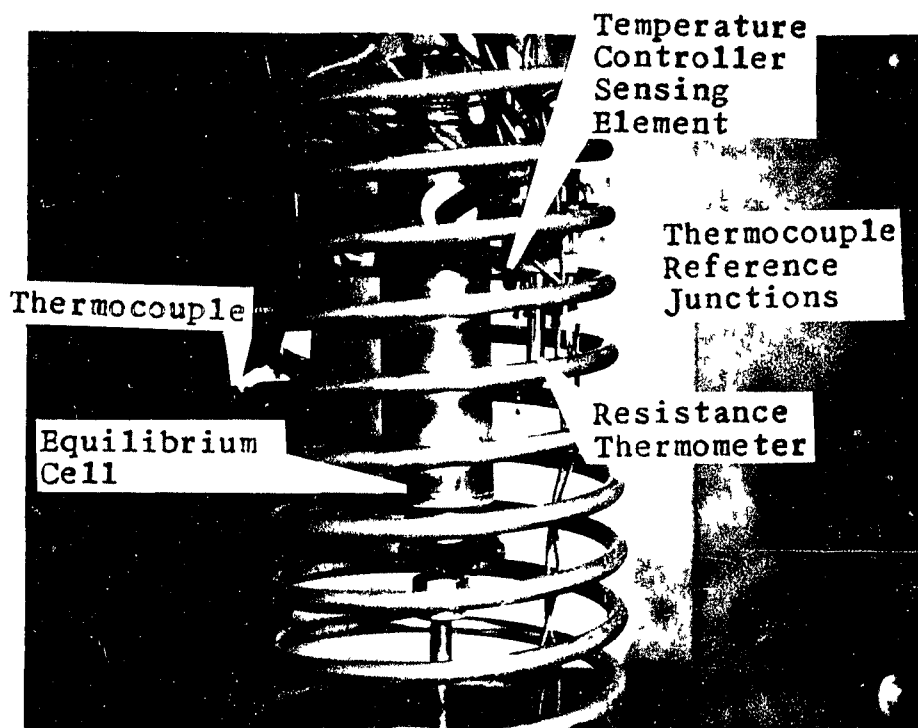


Figure 4. View Showing Equilibrium Cell, Resistance Thermometer, Temperature Controller Sensing Element and Thermocouple Reference Junctions

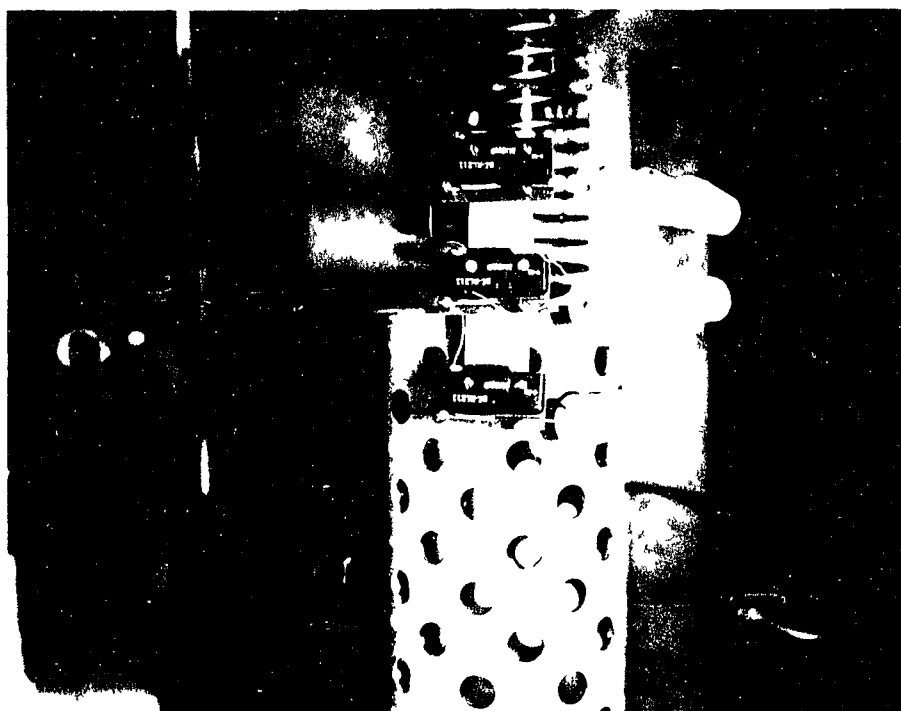


Figure 5. Heater and Float Actuated Switches for Liquid Level Indicator

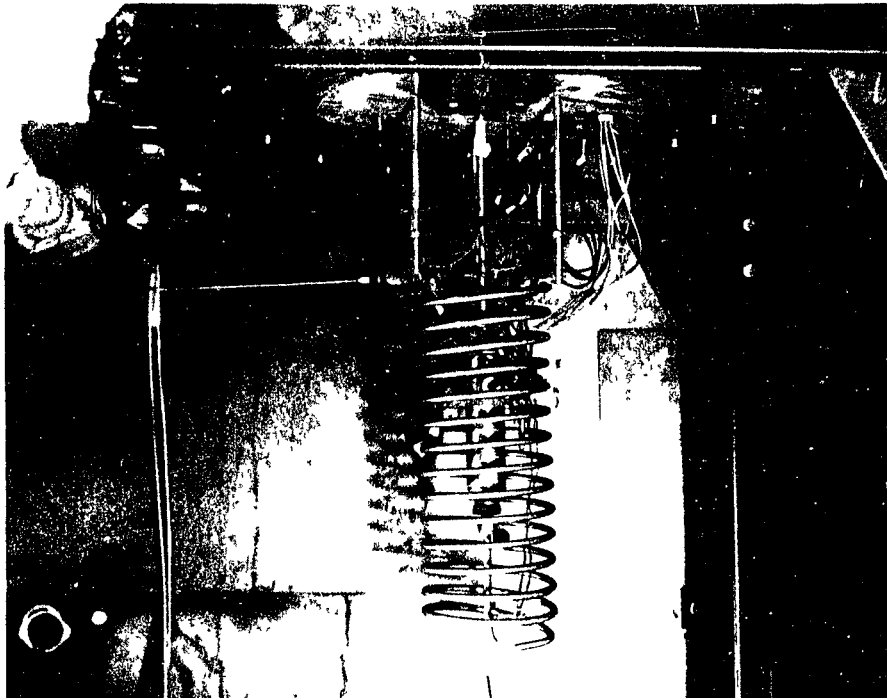


Figure 6. First Step in Cryostat Assembly: Equilibrium Coil and Cell, Resistance Thermometer, Temperature Controller Sensing Element and Thermocouples in Place

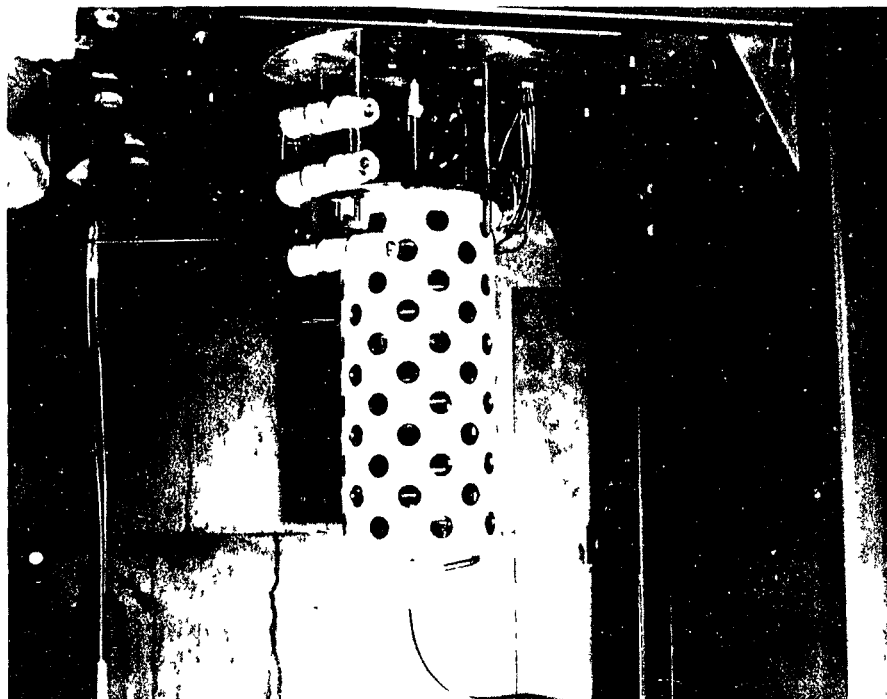


Figure 7. Second Step in Cryostat Assembly: Heater and Float Actuated Switches for Liquid Level Indicator Added

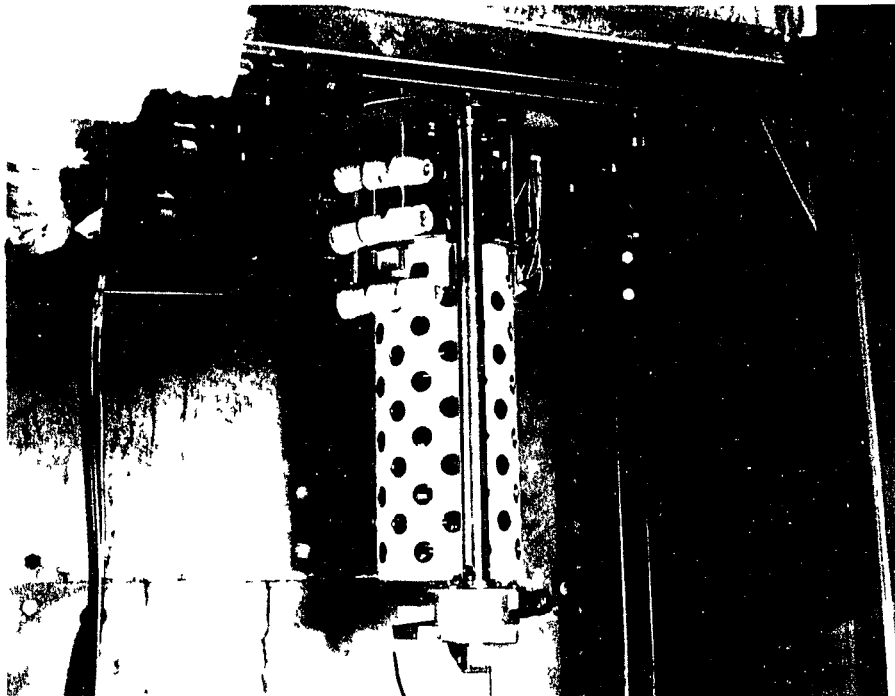


Figure 8. Third Step in Cryostat Assembly:
Centrifugal Circulating Pump Added

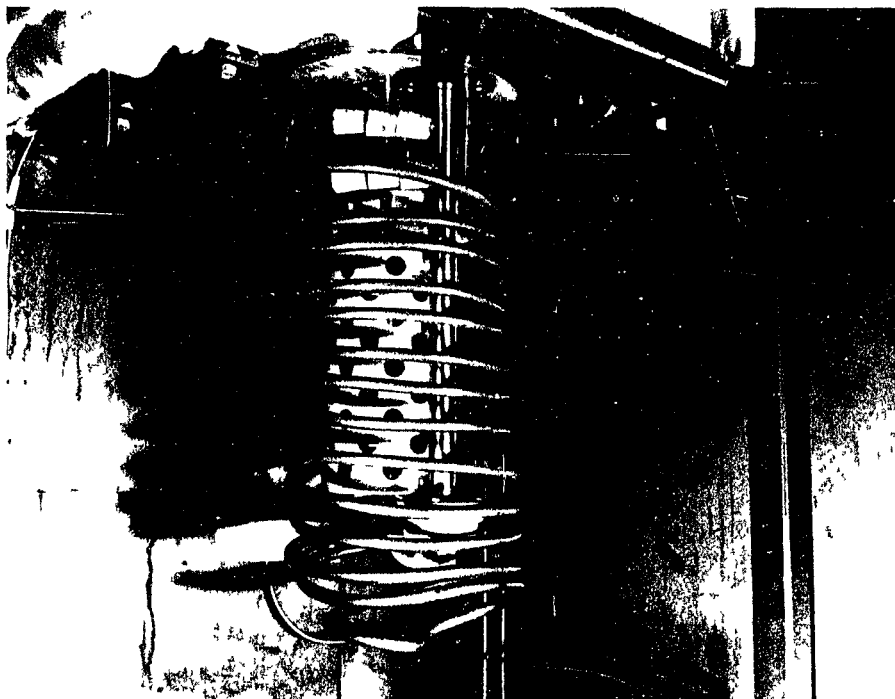


Figure 9. Fourth Step in Cryostat Assembly:
Cooling Coil Added

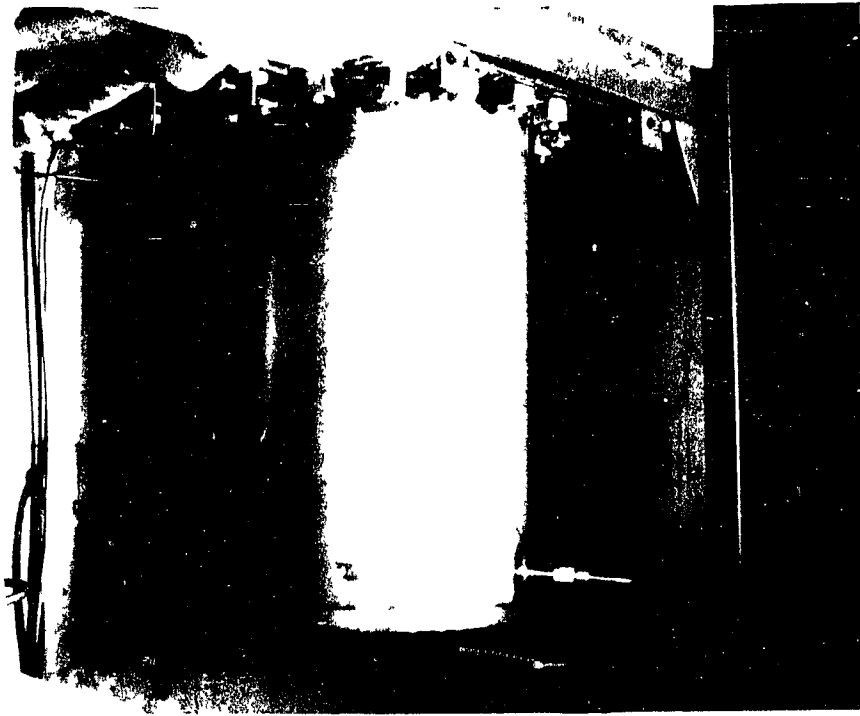


Figure 10. Fifth Step in Cryostat Assembly:
Cryostat Bolted to Top Plate

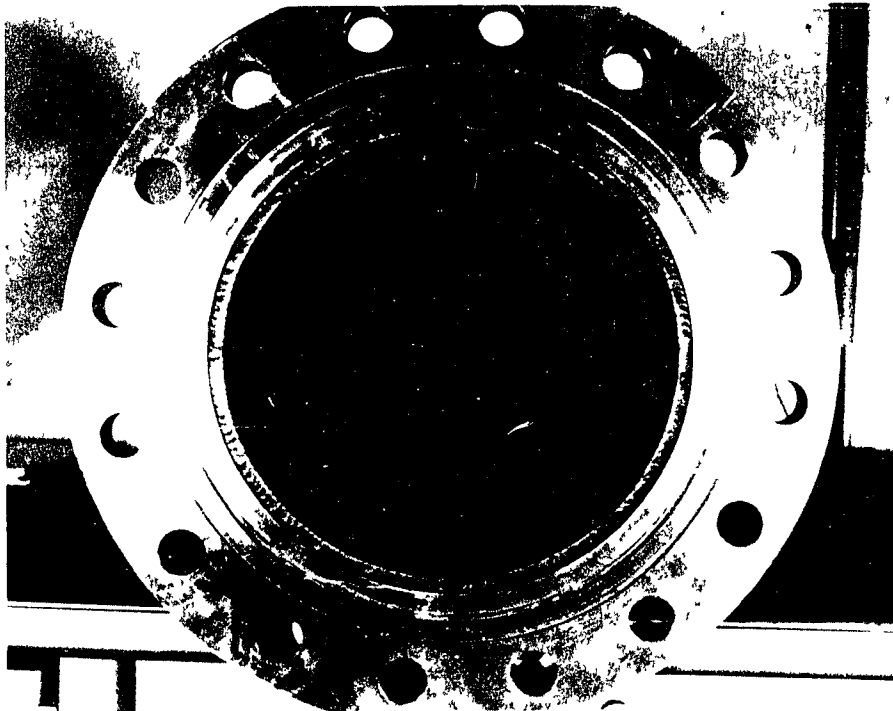


Figure 11. View Showing O-ring Grooves in
Cryostat Flange

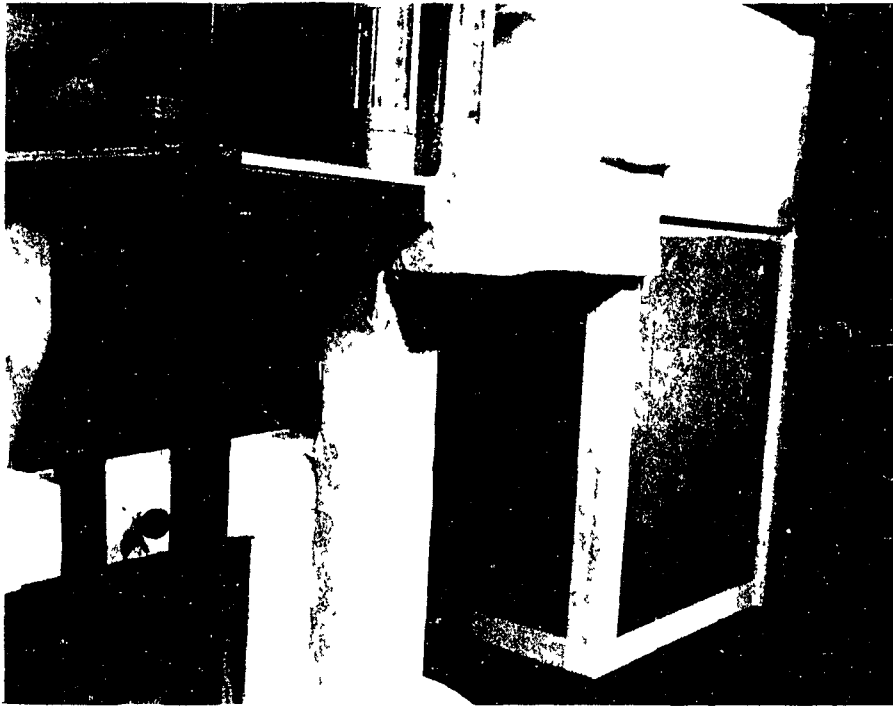


Figure 12. Final Step in Cryostat Assembly:
Insulation Placed Around Cryostat and
Top Plate

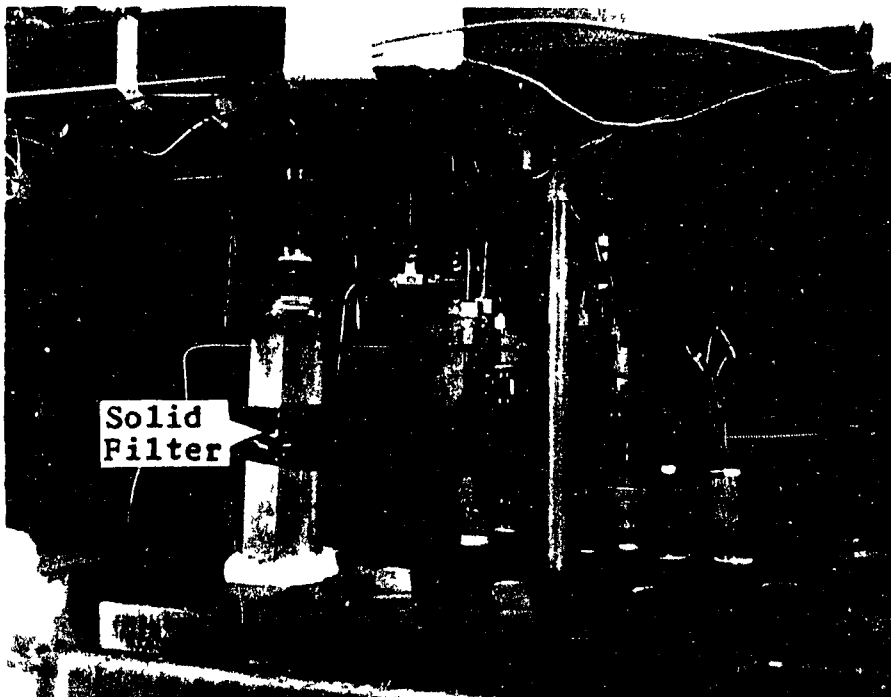


Figure 13. Cryostat Top Plate and Solid Filter

This subcooling is counterbalanced by the heater, Figures 5 and 7, which is actuated by an automatic temperature controller whose sensing element is partially visible in Figure 4. In order to reduce temperature gradients in the bath to a minimum, a centrifugal circulating pump, Figure 8, is used to agitate the liquid. The inlet of this pump is located just below and the outlet just above the equilibrium coil so that the liquid surrounding the equilibrium cell and coil will be well circulated.

All thermocouples located within the cryostat, with the exception of one located next to the platinum resistance thermometer, are referenced to the platinum resistance thermometer. Thus, these thermocouples indicate differences in temperature between that at the location of the thermocouple and that of the resistance thermometer. Since the accuracy of a potentiometer used in determining thermocouple emf's is inversely proportional to the magnitude of the reading, this method will minimize the errors in determining the temperatures within the cryostat. The reference junctions for the above thermocouples may be seen encircling the platinum resistance thermometer in Figure 4. The one thermocouple not referenced to the platinum resistance thermometer is referenced to the ice point and is used for preliminary temperature measurements until a desired equilibrium temperature is attained, at which time the resistance thermometer is used.

2. Test Gas Flow Path

The test gas, which is pressurized with the diaphragm compressor shown in Figure 14, flows from the storage tank, Figure 14, through a solenoid valve to the high pressure side of the pressure regulator. The low pressure side of the regulator separates into two branches, one going directly to the gas analyzer to permit sampling of the test gas. The other branch passes through a heat exchanger which serves to precool the test gas before entering the cryostat. It is probable that some carbon dioxide will condense from the test gas during this cooling process, a portion of which will be deposited on the heat exchanger walls with the remainder carried through the heat exchanger. It is desirable to filter out as much solid carbon dioxide as possible from the test gas before it enters the cryostat since this will delay the eventual plugging of the equilibrium cell. Therefore, a solid filter, seen in Figure 13, is located on the outlet of the heat exchanger, its purpose being to trap any solid carbon dioxide formed but not deposited in the heat exchanger.

From the solid filter the test gas enters the cryostat and flows through the equilibrium coil during which time it reaches the equilibrium temperature. Additional carbon dioxide will condense in the equilibrium coil, part of it depositing on the tube walls and the remainder carried into the equilibrium cell. The equilibrium cell contains a filter whose purpose is to trap all solid carbon dioxide. It is imperative that

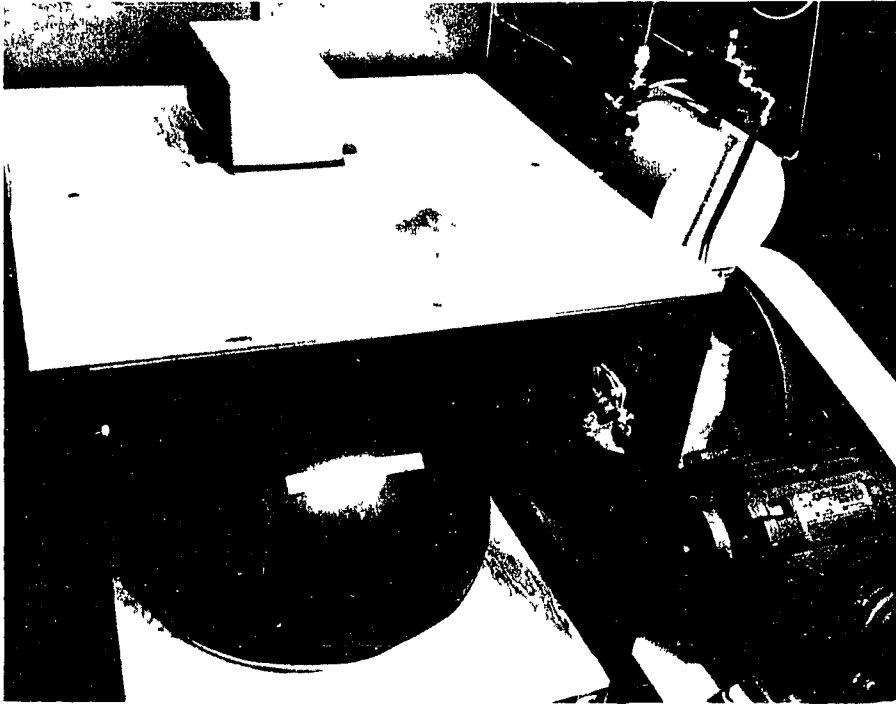


Figure 14. Storage Tank and Diaphragm Compressor

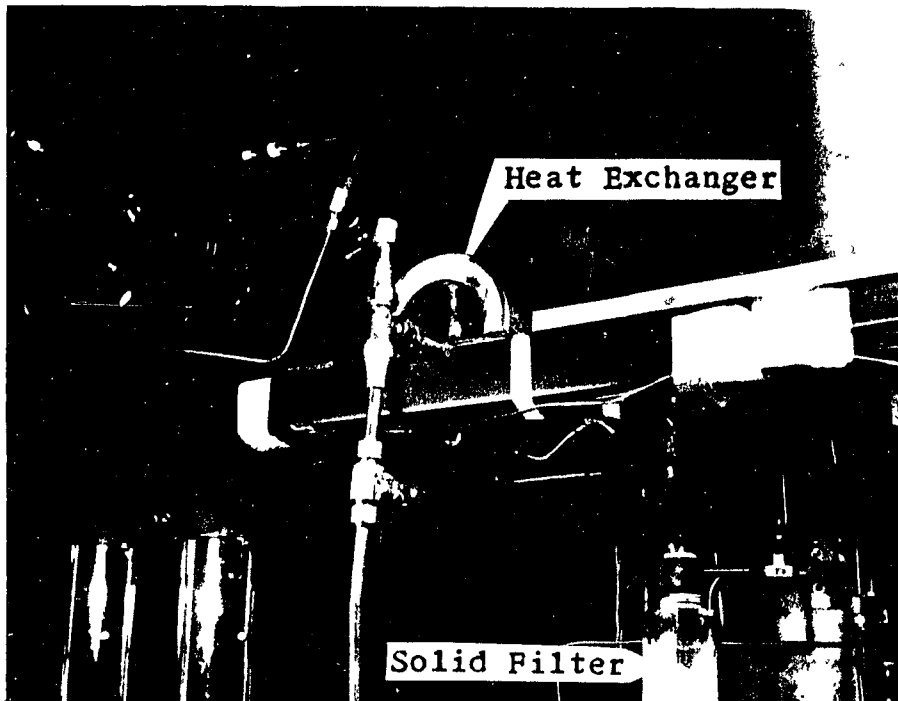


Figure 15. Heat Exchanger and Solid Filter

absolutely no solid pass through this filter since it would sublime upon leaving the cryostat and thus give an erroneously high equilibrium composition. After leaving the equilibrium cell the test gas passes out of the cryostat, is warmed to ambient temperature, and passes through the gas analyzer where the equilibrium composition is determined.

3. Heat Exchanger Coolant Flow Path

The heat exchanger coolant flow path, as seen in Figure 1, consists of a closed loop in which the coolant is circulated by a small compressor. The coolant used is nitrogen gas which passes from the outlet of the compressor through a cooling coil immersed in a liquid nitrogen bath, this bath being maintained at a constant level by an automatic controller so that the amount of the cooling coil in contact with liquid nitrogen will remain constant. The nitrogen gas liquifies in the cooling coil and then passes through the counterflow heat exchanger where it is evaporated by the heat absorbed from the test gas. Finally, the nitrogen gas leaves the heat exchanger and returns to the compressor, thus completing the loop.

A bypass valve is connected between the compressor outlet and the inlet to the heat exchanger in order to regulate the amount of coolant which passes through the liquid nitrogen bath. The purpose for such regulation is to control the temperature of the coolant as it enters the heat exchanger, this in turn fixing the temperature of the test

gas leaving the heat exchanger. Thus it is possible to control the temperature of the test gas as it enters the cryostat at a value slightly above the equilibrium temperature.

B. Construction

1. Cryostat

The cryostat, Figures 10 and 11, consists of a stainless steel, vacuum jacketed, flanged vessel with an inside diameter of $10\frac{1}{4}$ inches and a maximum depth of 22 inches. The outside diameter of the cryostat is $12\frac{1}{2}$ inches and the overall height is 24 inches. The flange is 1 inch thick, $17\frac{1}{2}$ inches in diameter, and contains 16- $1\frac{1}{8}$ inch bolt holes located on a $15\frac{1}{4}$ inch diameter bolt circle. In addition, the flange contains two grooves for a 12 and 13 inch O-ring, these O-rings being made of silicone rubber.

The top plate of the cryostat, Figure 13, consists of a 1 inch stainless steel plate to which the cryostat is fastened with 16-1"x4" stainless steel bolts and stainless steel nuts, flat washers and lock washers. The top plate is stationary and all equipment contained within the cryostat is suspended from it, as seen in Figure 9.

The cryostat was originally designed to operate with no insulation other than the vacuum jacket. However, during operation this did not prove adequate and the cryostat was then insulated with a 3 inch layer of fiberglass and enclosed in a styrofoam box as seen in Figure 12.

2. Heater

The heater used in the temperature regulation was constructed as follows: $\frac{3}{4}$ inch holes spaced 1 inch apart were drilled in a $5\frac{1}{4}$ inch OD by 10 inches long brass cylinder with a $\frac{1}{8}$ inch wall, the holes permitting liquid to flow through the cylinder. A $6\frac{3}{4}$ inch by $5\frac{1}{4}$ inch brass ring $\frac{1}{8}$ inch thick was then silver soldered to this cylinder at one end. Next, two layers of a thermal setting insulating tape were wrapped on the outside of the brass cylinder. Thereafter, 40 feet of 28 gage Chromel A resistance wire giving a resistance of approximately 100 ohms was wrapped on the insulating tape in such a manner that none of the wire crossed over the holes drilled previously. Two additional layers of the thermal setting insulating tape were wrapped over the resistance wire and the entire assembly then placed in a baking oven to bond the insulating tape. Finally, upon completion of the bonding process, the insulating tape located over the holes in the cylinder wall was removed. The completed heater may be seen in Figure 5 where the outline of the heater resistance wire is also visible.

The heater is suspended from the top plate by four thin stainless steel tubes connected to the top plate and fastened to the ring at the top of the heater. The heater may be seen suspended in place in Figure 7.

3. Liquid Level Indicator

The schematic diagram for the liquid level indicating device may be seen in Figure 16. Three micro-switches were used in this device, these switches being actuated by float operated levers with the floats made from corks. The micro-switches with levers and floats may be seen in Figures 5 and 7. The micro-switches were spaced 2 inches apart. As seen from the schematic diagram, Figure 16, as each micro-switch is closed a resistor is added in parallel with a resulting increase in the current flow through the circuit. Thus the current reading may be used to indicate the liquid level.

This particular arrangement indicates only an incremental change in liquid level rather than giving a continuous level indication as more complex arrangements might. However, this is adequate for the present investigation since it is important only to know that the level is above a given point rather than knowing the exact level. If desired, the present device could be made to approach a continuous reading device by placing several micro-switches close together. The present arrangement has one advantage over others reported in the literature in that the current reading is independent of temperature since the resistors are located externally to the cryostat. The float operated switches also eliminate the possibility of sticking associated with float operated slide wire devices.

Note: Switches enclosed by dotted line are float-actuated micro-switches located within the cryostat.

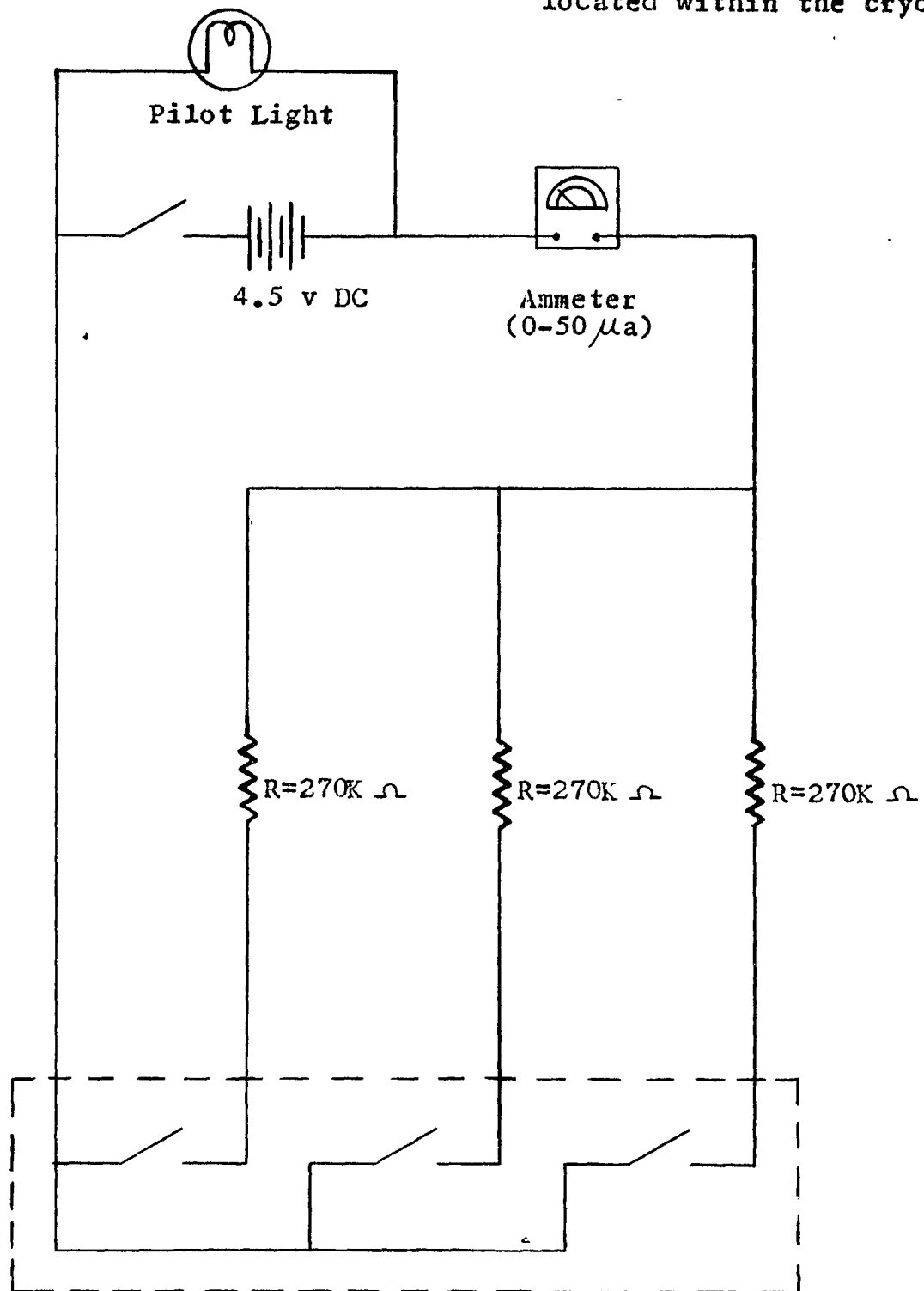


Figure 16. Schematic Diagram of Liquid Level Indicator

4. Heat Exchanger

A cross-sectional view of the heat exchanger may be seen in Figure 17. The heat exchanger is constructed as follows: a semicircular groove $\frac{1}{4}$ inch in diameter is turned in a helical pattern on the outside of a $3\frac{1}{2}$ inch OD, 3 inch ID, 12 inch long brass cylinder with a $\frac{1}{8}$ inch space between the grooves, these grooves being stopped before reaching the ends of the cylinder. Two $\frac{3}{8}$ inch holes are then drilled radially through the grooves at each end of the cylinder at the point where the grooves stop. Two $\frac{3}{8}$ inch OD copper tubes are then placed in these holes from the inside of the cylinder and silver soldered flush with the bottom of the groove. Next, a semicircular groove $\frac{1}{4}$ inch in diameter is turned in a helical pattern on the outside of a 4 inch OD, $3\frac{1}{2}$ ID, 12 inch long brass cylinder with a $\frac{1}{8}$ inch space between the grooves. This cylinder is then shrunk fit over the preceding cylinder. Thereafter, a $\frac{1}{4}$ inch OD stainless steel tube with a 0.049 inch wall is wrapped in the groove of the outer cylinder, the total length of tube being approximately 22 feet. After the stainless steel tube is in place, a $4\frac{1}{2}$ inch OD, $4\frac{1}{4}$ inch ID, 12 inch long stainless steel sleeve is pressed over the entire assembly. As the final step, the void space around the stainless steel tube located between the outer brass cylinder and stainless steel sleeve is filled with Wood's metal, a low melting alloy.

Originally, the heat exchanger was not filled with the Wood's metal inasmuch as it was assumed the stainless steel sleeve would force the stainless steel tube into good

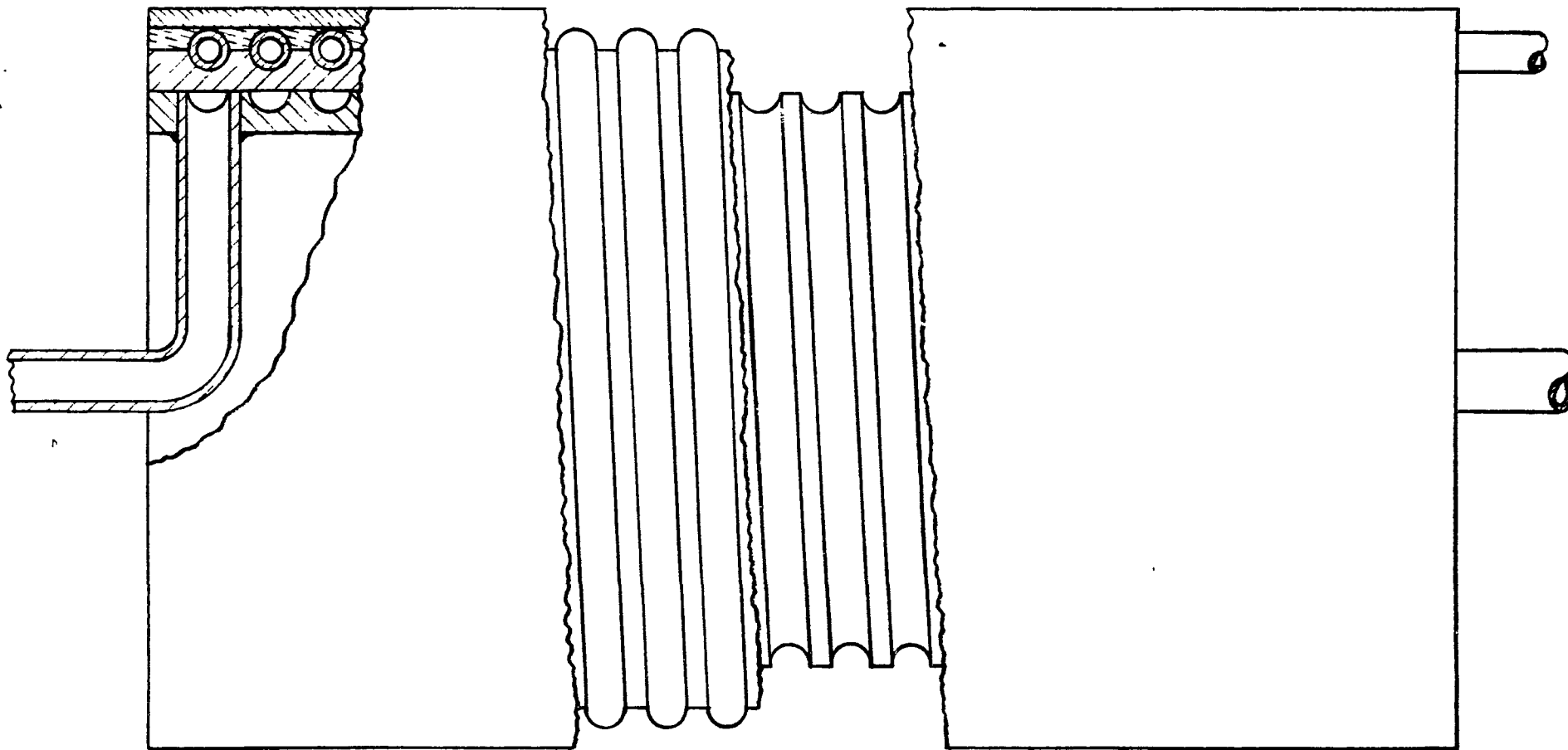


Figure 17. Heat Exchanger Construction

contact with the brass cylinder. However, the heat exchanger performance did not prove satisfactory under these conditions and the Wood's metal was added to improve the performance. The heat exchanger is partially visible in Figure 15.

5. Equilibrium Cell and Coil

The equilibrium cell consists of a $1\frac{1}{4}$ inch OD solid stainless steel rod 6 inches long with a $\frac{45}{64}$ inch hole bored through lengthwise to produce a cylinder. Each end of this cylinder is tapped for a $\frac{1}{2}$ inch pipe thread. The cylinder is then packed to within an inch of one end with copper wool and the remaining space is filled with fine spun glass. The function of the copper wool and spun glass is to trap all solid carbon dioxide particles carried into the equilibrium cell since it is absolutely necessary that no solid particles be permitted to pass beyond this point.

After placing the copper wool and spun glass in the equilibrium cell, high pressure stainless steel tube fittings are turned into the pipe threads located on each end. These fittings are then silver soldered in place to eliminate possible leakage of the high pressure test gas which might build up damagingly high pressures in the cryostat. In case such leakage should occur, a pressure switch is connected to the cryostat which will close the solenoid valve on the outlet of the test gas storage tank if the pressure becomes too high.

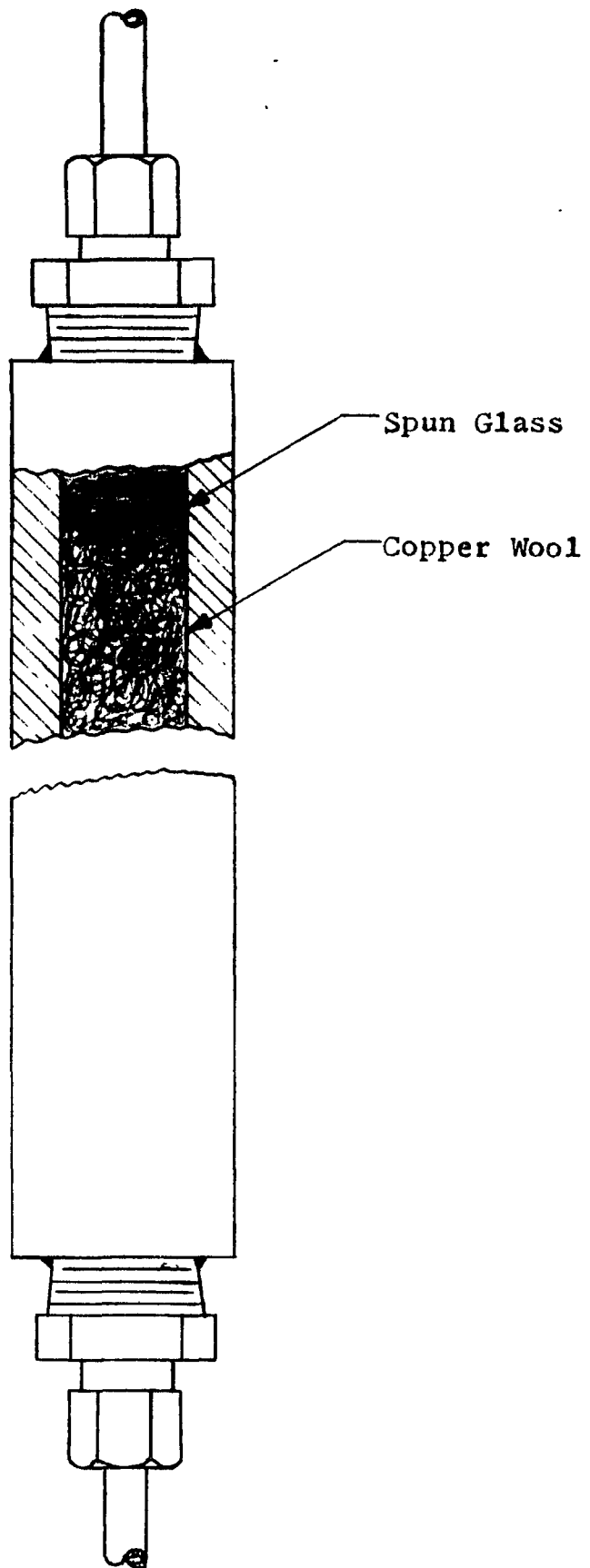


Figure 18. Equilibrium Cell Construction

The equilibrium coil consists of approximately 12 feet of $\frac{1}{4}$ inch OD stainless steel tube with a 0.049 inch wall which is formed into a 4 inch diameter coil. This coil is immersed in the bath liquid and its function is to cool the test gas to the equilibrium temperature. The equilibrium coil is connected to the equilibrium cell where all solid carbon dioxide is trapped. The equilibrium cell is so placed that the end containing the spun glass is the exit side. A cross-sectional view of the equilibrium cell is presented in Figure 18 while the equilibrium coil and cell may be seen in Figures 4 and 6.

6. Solid Filter

The solid filter, located on the outlet side of the heat exchanger, is made from a 6 inch length of $1 \frac{3}{4}$ inch hexagonal stainless steel bar. A $\frac{29}{32}$ inch hole is drilled axially to a depth of 5 inches. The open end of the bar is then tapped for a $\frac{3}{4}$ inch pipe thread and the bar is then filled with copper wool. A high pressure stainless steel tube fitting is then turned into the pipe thread, this joint being sealed with Teflon tape. It was not deemed necessary to silver solder this joint, as was done in the equilibrium cell, since a small leak at this point is undesirable but tolerable. A cross-sectional view of the solid filter may be seen in Figure 19 while the filter connected in place is visible in Figures 13 and 15.

As was indicated earlier, the purpose of this filter is to trap any solid carbon dioxide which forms but is not

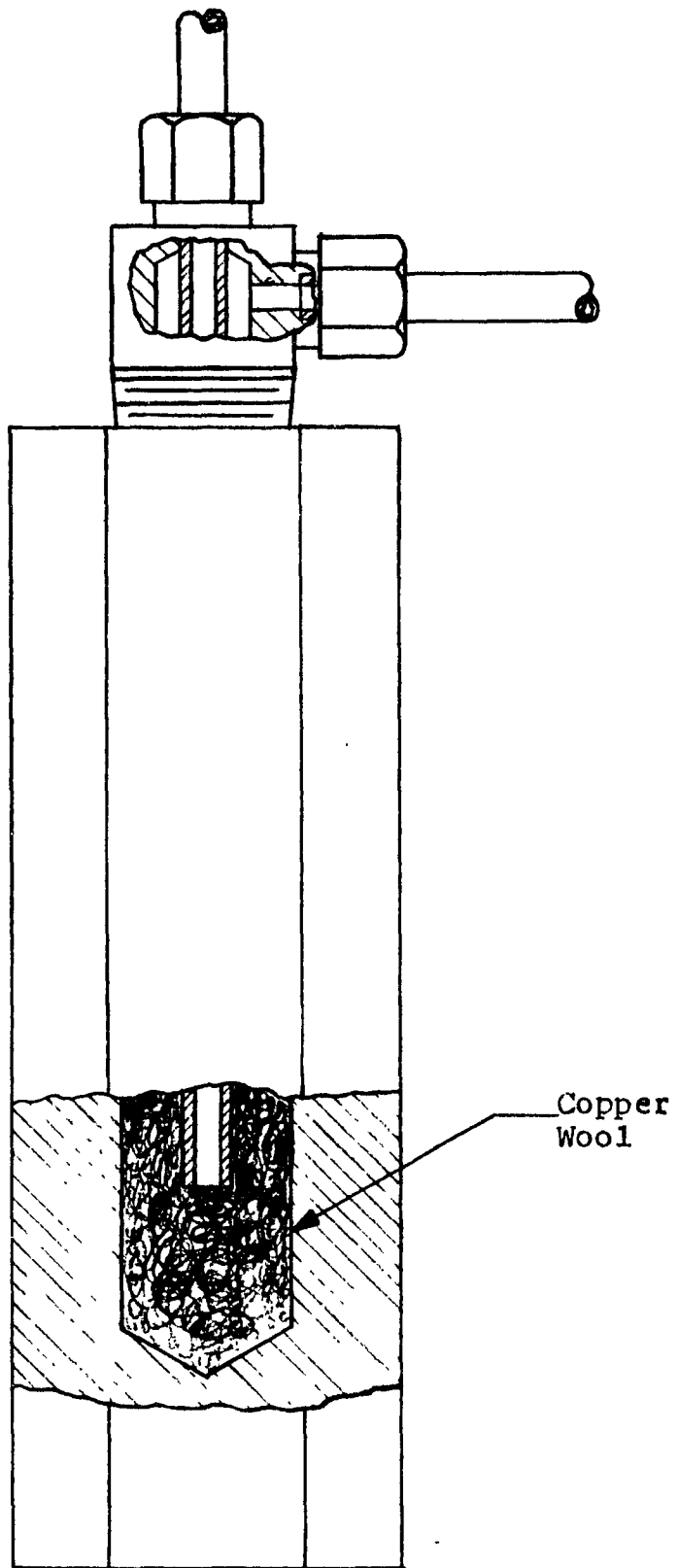


Figure 19. Solid Filter Construction

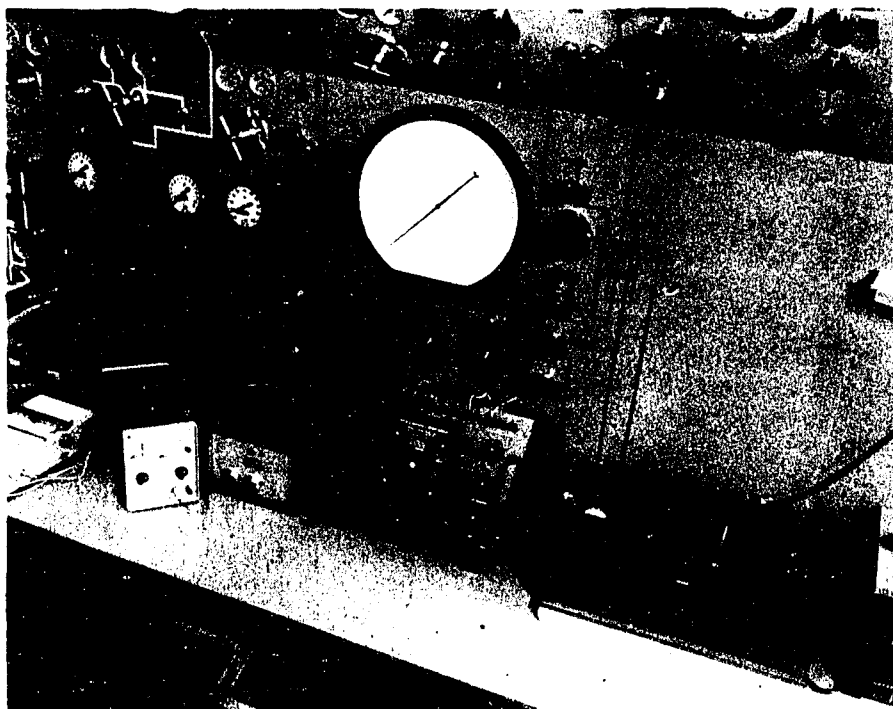


Figure 20. Potentiometer-Null Detector,
Temperature Controller and Mueller Bridge

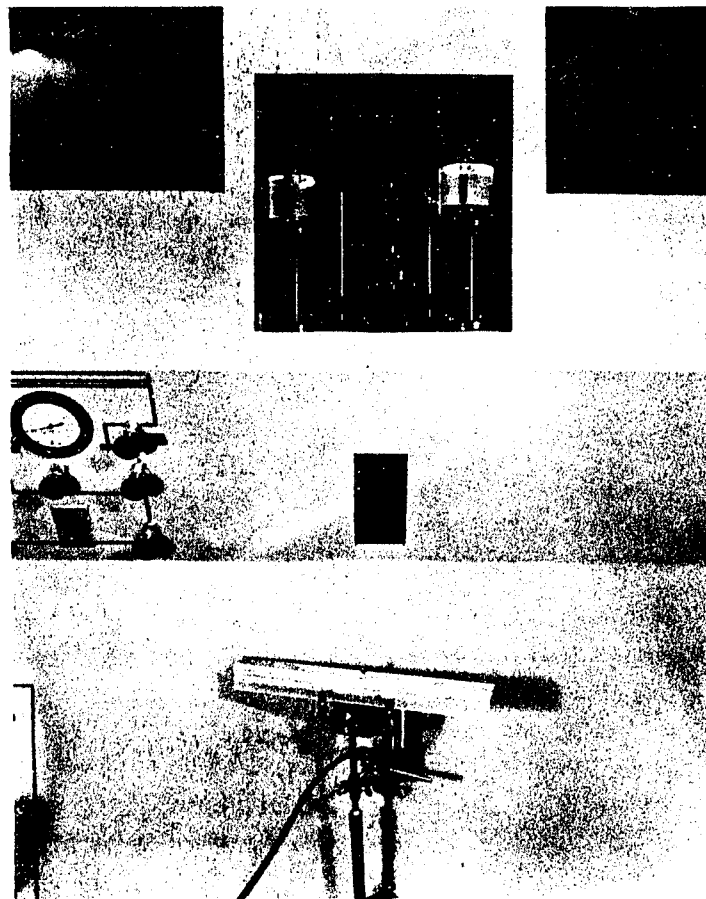


Figure 21. Galvanometer, Julius Suspension and Lamp and Scale

deposited in the heat exchanger thus delaying the plugging of the equilibrium cell.

C. Instrumentation

1. Temperature Measurement

The equilibrium temperature is measured with a Leeds and Northrup type 8163 platinum resistance thermometer in conjunction with a Leeds & Northrup model G-2 Mueller temperature bridge, the latter being shown in Figure 20. The null detecting equipment associated with this bridge consists of a Leeds and Northrup model 2284-d galvanometer supported in a Julius suspension in addition to a model 2100 lamp and scale. In the present installation the galvanometer is suspended above the lamp and scale thus necessitating use of the Leeds and Northrup model 2178 prism and 2104 reflecting mirror. The null detecting equipment may be seen in Figure 21. It was found necessary to enclose the Julius suspension as shown in Figure 21 in order to eliminate motion induced by forced air circulation within the room.

All other temperatures are measured with 20 gage copper-constantan thermocouples in conjunction with a Leeds and Northrup type K-3 potentiometer. The null detecting device used in this case is a model 9834 electronic D-C null detector. The null detector and potentiometer may be seen in Figure 20. All thermocouples, except those discussed in section VI-A-1, are referenced to the ice point. The one

thermocouple located in the cryostat and referenced to the ice point was calibrated against the platinum resistance thermometer. This calibration was then used for all other thermocouples which seems permissible inasmuch as all the thermocouples used were made from the same spools of wire.

2. Temperature Control

The temperature was automatically controlled by a Hallikainen model 1053 Thermotrol control unit, this unit being visible in Figure 20. The sensing element for this controller is a resistance bulb located adjacent to the platinum resistance thermometer, as seen in Figure 4. The resistance bulb used is a Hallikainen model 1358A designed for the range -100° to -400°F .

During operation the coolant flow rate through the cryostat is such that the bath liquid is cooled slightly below the desired equilibrium temperature and the temperature control unit then actuates a heater to maintain the equilibrium temperature.

3. Pressure Measurement

The equilibrium pressures are read on a calibrated 16" Heise bourdon tube gage with a 0-3000 psig range, this gage being visible in Figure 20. All other measured pressures are read on standard bourdon tube gauges.

4. Pressure Control

The one pressure requiring precise control is the

equilibrium pressure and this is accomplished with a Grove model 15KX high pressure regulator. This regulator is located immediately to the right of the Heise gage as seen in Figure 20. The two pressure regulators located on the discharge of the analyzer and main exhaust lines are standard pancake regulators. These regulators serve to eliminate surging in the flowmeters at low flow rates as well as to maintain a constant calibrating pressure in the gas analyzer. All other regulators used in the system are standard Matheson gas regulators.

5. Gas Analysis

The equilibrium mixture is analyzed in a Beckman L/B model 15A non-dispersive infrared analyzer. This instrument consists of an analyzer and amplifier unit, the former being sensitized to detect for carbon dioxide. During operation, the analyzer unit produces a signal proportional to the percentage of carbon dioxide present. This signal goes to the amplifier unit and is expressed as an electrical voltage on a panel meter located on the amplifier. In addition, so that a permanent record of this voltage may be obtained, the amplifier output is fed to a Beckman model 93500 potentiometric recorder. This ensemble of analyzing equipment may be seen in Figure 2.

6. Liquid Nitrogen Level Control

The liquid nitrogen bath used to cool the heat exchanger coolant is maintained at a constant level by a

Johns and Frame model B level controller. This controller includes a gas cross-charged sensing element placed in the liquid nitrogen bath which closes a relay when the level falls below that desired. This relay in turn actuates a normally-closed solenoid valve located in the liquid nitrogen transfer line thus permitting liquid nitrogen to flow until the desired level is reached at which time the sensing element opens the relay.

7. Flowmeters

A Matheson model 622PBV-602 flowmeter with a 0-2 SCFH capacity is located on the analyzer exhaust line. A model 622PBV-603, 0-10 SCFH, and a model 622PBV-604, 0-35 SCFH, are placed in parallel on the main exhaust line, one or the other being used depending upon the particular flow rate desired.

VII. EXPERIMENTAL ERRORS

A. Temperature Measurement

1. Resistance Thermometer-Bridge-Galvanometer

The Leeds and Northrup model 8163 platinum resistance thermometer used in the present investigation was calibrated at the National Bureau of Standards to an accuracy of ± 0.0001 ohm. The sensitivity of the resistance thermometer in the temperature range encountered here is no less than ± 0.1 ohm/ $^{\circ}$ C. Therefore, the sensitivity of the resistance thermometer may be considered to be at least $\pm 0.001^{\circ}$ C.

The Leeds and Northrup model 2284-d reflecting galvanometer has a sensitivity of $0.1 \mu\text{v}/\text{mm}$ at a distance of 1 meter. The galvanometer is placed on a Julius suspension which is designed to damp out vertical as well as rotary motions of the galvanometer thus reducing the possibility of galvanometer deflections due to vibrations. Using this galvanometer and the G-2 Mueller bridge it was found that a change in the thermometer resistance of 0.0001 ohm could be detected, this change giving a galvanometer scale deflection of just over 1 mm. Thus it is possible to detect a change in temperature of $\pm 0.001^{\circ}$ C. However, in as much as the International Temperature Scale is considered to be accurate to only $\pm 0.01^{\circ}$ C, the same value shall be taken as the accuracy of the temperature measurement with the resistance thermometer-bridge-galvanometer network used here.

2. Thermocouple-Potentiometer-Null Detector

All thermocouples were constructed of 20 gage copper-constantan wire taken from the same spools. One typical thermocouple, that located adjacent to the resistance thermometer in the cryostat, was calibrated against the resistance thermometer at 10°C intervals over the temperature range to be encountered. A curve of $\text{EMF}_{\text{observed}}$ versus $\text{EMF}_{\text{standard}}$ was then made and all points fell within 1 microvolt of a smooth curve drawn through the calibration points. Inasmuch as the sensitivity of the thermocouples used is no less than $25\mu\text{v}/^{\circ}\text{C}$ in the temperature range encountered this would indicate that the thermocouple accuracy is at least $\pm 0.04^{\circ}\text{C}$.

The maximum error for the K-3 potentiometer is $\pm (0.5\mu\text{v} + 0.015\%$ of the reading) in the range used here. Since thermocouples No. 2 thru 6 (see Figure 22) located in the cryostat are referenced to the resistance thermometer their readings are very small and the maximum error in the potentiometer reading may be taken as $\pm 0.5\mu\text{v}$ (0.02°C) for these thermocouples. The sensitivity of the Leeds and Northrup model 9834 D-C null detector is $0.2\mu\text{v}$ per scale division or approximately 0.01°C per division.

Inasmuch as thermocouples Nos. 2 thru 6 are differential thermocouples which produce small emfs, their accuracy in temperature measurement is not limited by a calibration curve but instead by the error limits of the potentiometer. Thus the accuracy of temperature measurement with the

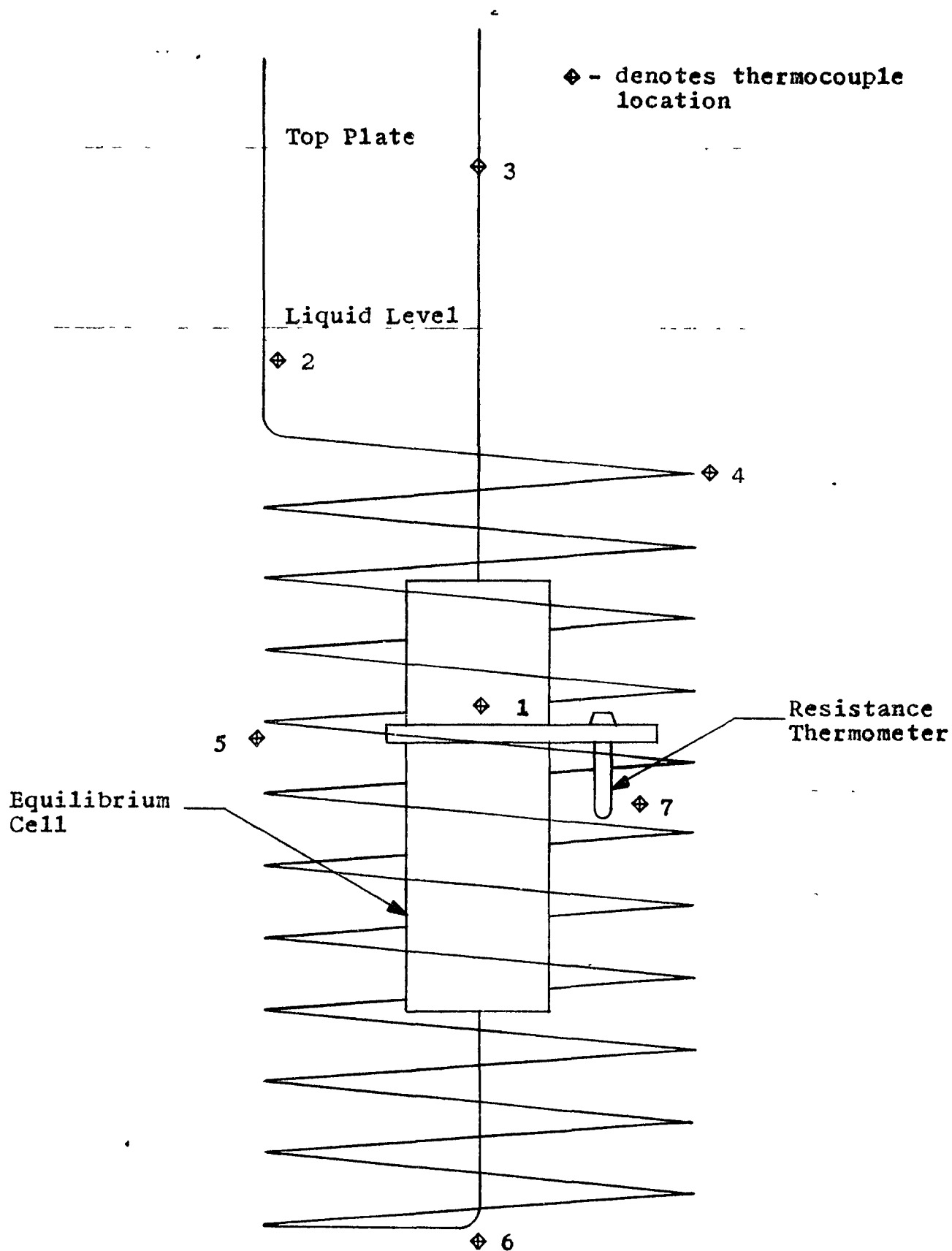


Figure 22. Location of Thermocouples in Cryostat

thermocouple-potentiometer-null detector set up is considered to be $\pm 0.02^{\circ}\text{C}$ for thermocouples Nos. 2 thru 6 and $\pm 0.04^{\circ}\text{C}$ for all others.

B. Temperature Control

The Hallikainen model 1053 Thermotrol temperature controller may be used as an on-off, proportional, or proportional with reset controller with the maximum sensitivities in each case being ± 0.001 , ± 0.023 and $\pm 0.00023^{\circ}\text{C}$, respectively. These sensitivities are meaningless, however, as far as the accuracy of temperature control is concerned since this is affected by several factors such as time constants of the sensing element and heater, configuration of bath components and bath agitation.

As used in the present installation, the maximum temperature variation from a set point was $\pm 0.05^{\circ}\text{C}$ as indicated by the platinum resistance thermometer with the variation in some cases being as low as $\pm 0.02^{\circ}\text{C}$. It is felt that this might have been improved if the bath liquid in the vicinity of the cooling coils had been agitated more completely than was accomplished by the centrifugal circulating pump used in the present setup. Nevertheless, it can be conservatively stated that the temperature control during this investigation was at least $\pm 0.05^{\circ}\text{C}$.

C. Cryostat Temperature Distribution

The temperature distribution within the cryostat was determined by five thermocouples which were referenced to

the platinum resistance thermometer (see Figure 22). During operation none of these thermocouples, with the exception of Nos. 2 and 3, differed by more than $\pm 3 \mu v$ or approximately $\pm 0.1^{\circ}C$ from the temperature of the resistance thermometer and in many cases the difference was from 0 to $\pm 2 \mu v$ (0 to $\pm 0.08^{\circ}C$). Thermocouple No. 3, located just below the top plate and well above the bath liquid, read as high as 500-750 μv ($20-30^{\circ}C$) warmer while thermocouple No. 2, located near the bath liquid and approximately $1\frac{1}{2}$ inches below No. 3, read as high as 0-55 μv ($0-2.25^{\circ}C$) warmer than the resistance thermometer. This is not unreasonable since the conduction through the top plate could account for such differences.

Thermocouple No. 1 (see Figure 22) which is located inside the equilibrium cell was found to give highly irregular and unrealistic readings during operation. A check indicated that this thermocouple had shorted against the copper wool located in the equilibrium cell. However, after considering (1) the temperature distribution within the cryostat as indicated by thermocouples Nos. 2-6, (2) that the temperature of the test gas leaving the heat exchanger was never more than $10^{\circ}C$ above the desired equilibrium temperature, and (3) the dwell time of the test gas in the equilibrium coil it was decided that the test gas was at the equilibrium temperature in the equilibrium cell. If in fact the test gas was not reaching the equilibrium temperature but remaining at some higher value, the magnitude of the difference should increase with flow rate through the equilibrium cell with a corresponding increase in equilibrium composition. However, a check of several

test points at different flow rates as discussed in section G of this chapter showed no change in equilibrium composition. Thus it is concluded that the test gas was actually reaching the equilibrium temperature.

D. Pressure

The equilibrium pressure is read on a 0-3000 psig 16" Heise bourdon tube gage with 3 psi divisions which has been thermally compensated and calibrated to an accuracy of $\pm 0.1\%$ of full scale. Thus the pressure readings may be considered accurate within ± 0.2 atm (± 3 psi). The Grove pressure regulator used maintained the equilibrium pressure within ± 0.2 atm (± 3 psi) until the equilibrium cell began to plug at which time the pressure would fluctuate widely.

All other pressures were measured on standard bourdon tube gages with accuracies of $\pm \frac{1}{2}\%$ of full scale reading, this being deemed sufficient since pressures other than the equilibrium pressure are not extremely important.

E. Gas Analysis

The Beckman L/B model 15A non-dispersive infrared analyzer used for gas analysis has a maximum error of $\pm 1\%$ of full scale with a sensitivity of half that value. Three sets of cells, $3/16$, $3/4$ and $5\frac{1}{4}$ inches in length were used in conjunction with this analyzer, each set being used to detect carbon dioxide concentrations, expressed as mole fractions, in the range 0.02-0.10, 0.004-0.02 and less

than 0.004, respectively.

Calibrating cylinders consisting of varying percentages of carbon dioxide in nitrogen were obtained from the Matheson Company for purposes of calibrating the analyzer. The analyzer was calibrated by first setting the zero or reference point of both the analyzer and recorder with pre-purified nitrogen flowing through the analyzer. Then a calibrating gas with a carbon dioxide concentration just above that to be expected during a particular run was passed through the analyzer and the gain set to produce a 90% scale deflection. A 90% scale deflection was used rather than full scale deflection inasmuch as the amplifier response becomes very flat in the upper 10% of the scale reading. With the amplifier output at 90%, the span control of the recorder was also set to give a 90% scale deflection. Thereafter at least two calibrating gases with carbon dioxide concentrations intermediate between zero and that used for the 90% scale deflection were passed through the analyzer and the resulting amplifier output recorded. Thus a minimum of four points were used for each calibration and in every case a smooth curve was drawn which passed through all calibration points.

It was noted that with the zero and 90% points of the recorder and amplifier set to read the same, the amplifier meter reading was slightly higher at intermediate points than that indicated by the recorder with this difference reaching a maximum of $1\frac{1}{2}\%$ at approximately mid-scale and tending toward zero as the zero and 90% readings were

approached. However, several calibration curves were made from both the amplifier meter readings and recorder values. Preliminary tests were then made and the meter reading and recorder value produced by the gas being analyzed were used with the appropriate calibration curve and in every case the analysis was the same. Therefore, all calibration curves used in the final analysis were made from the recorder values inasmuch as this provided a permanent record which could be rechecked if necessary.

The accuracy of the Beckman 93500 recorder is specified to be $\pm 0.5\%$ of full scale. Prior to collecting any experimental data, one calibration curve per day was made and rechecked over a period of eight hours for four consecutive days and in no case was a change in calibration evident on the recorder. Thus, the accuracy of the gas analysis made here is considered to be $\pm 0.5\%$ of the full scale reading.

During experimental runs the pressure in the analyzer was maintained at 3 psig and the analyzer flow rate held at 0.5 SCFH for the 3/16 inch cells and 1.0 SCFH for the other two sets of cells. The results of several tests indicated that a variation in pressure of ± 0.1 psig from that given above had no apparent effect on the analyzer calibration. Likewise, several tests indicated that the flow rate could vary from 0.3 to 1.5 SCFH in all three sets of cells with no variation in calibration. Nevertheless, during experimental runs the analyzer pressure and flow rate were maintained at the values specified above with all analyzer calibrations being made under the same conditions.

Several test points were rechecked using cells of different lengths to ascertain if different cells would give the same analysis. In every case the check points were within the $\pm 0.5\%$ of full scale considered to be the accuracy of the gas analyzer.

F. Flow Rate

The Matheson Company quotes the following accuracies on the flowmeters used here: 622 PBV-602, $\pm 4\%$; 622 PBV-603 and 622 PBV-604, $\pm 2\%$. The first meter is used on the analyzer exhaust line and the other two on the main exhaust line.

The average flow rate used in most of the experimental runs was such as to give a gas velocity through the equilibrium coil of 0.04 ft/sec with a resulting dwell time in the cryostat of 5-6 minutes. Approximately half of the test points were rechecked at various flow rates with no detectable change in equilibrium composition. This would indicate that a precise flow measurement is not required and that the flow meter accuracies quoted above are more than adequate.

Inasmuch as some variation in flow rate through the analyzer is tolerable as indicated in section VII-E the accuracy specified previously for the analyzer flow meter is also sufficient.

G. Equilibrium Cell

The equilibrium cell with the solid filter is essentially

the heart of the system and must perform the vital task of retaining all the solid carbon dioxide since any solid particles escaping the equilibrium cell would sublime and give an erroneously high gas phase analysis. If all solid particles were not being retained in the equilibrium cell the percentage of carbon dioxide in the exhaust stream would increase with flow rate. In order to check the ability of the equilibrium cell to retain the solid, a total of 38, or slightly over half, of the experimental points were checked at various flow rates. Of these check points, the same equilibrium composition was obtained 19 times, a smaller value 12 times and a larger value 7 times for the low flow rates as compared to the higher flow rates. The 18 points which were not the same were within the limits of error of the gas analyzer. Thus it is concluded that the equilibrium cell functioned highly satisfactorily as a solid trap.

H. Throttling Valves

There is a possibility of a composition change when a gas mixture is severely throttled across a small orifice. If such a change does occur it should be a function of flow rate through the orifice. In order to test for any such throttling effects across the valves in the experimental equipment, various mixtures flowing at different rates were analyzed and in no case could a change in composition be detected. In these tests the mixtures were throttled in such a manner that the pressure was reduced by a factor

of 1500 which is far greater than that experienced in the actual experimental runs. Thus it would seem that no composition change due to throttling was present in this investigation.

I. Gas Purity

The test gas mixtures were made from Liquid Carbonic high-purity nitrogen with a quoted minimum purity of 99.98% and Matheson Coleman grade carbon dioxide with a quoted minimum purity of 99.99%. The nitrogen used for setting the zero on the gas analyzer was Matheson pre-purified nitrogen with a quoted minimum purity of 99.996%. The gas mixtures used for calibrating the analyzer were prepared by the Matheson Company from 99.7% minimum purity nitrogen and 99.99% minimum purity carbon dioxide with the analysis of the mixture being made on a mass spectrograph.

J. Attainment of Equilibrium

Equilibrium is attained by two alternate methods in this investigation. Initially a carbon dioxide-nitrogen gas mixture in which the carbon dioxide concentration is greater than that present at equilibrium is passed thru the equilibrium cell at which point the excess carbon dioxide is condensed and trapped. The vapor in equilibrium with this solid then passes through the gas analyzer at which point the carbon dioxide concentration is determined. Eventually the system will plug due to the solid carbon dioxide being deposited. When plugging becomes evident as indicated by

the inability to maintain a constant flow rate, pure nitrogen gas at the desired equilibrium pressure is passed through the system. In this case carbon dioxide is evaporated rather than deposited, the amount evaporated being such as to produce a vapor mixture which is in equilibrium with the solid. The majority of the check points referred to in section VII-G were made by this latter method. Inasmuch as the original and check points, which were run by the two methods described here, gave results always equal to or within experimental accuracy of each other, it is concluded that true equilibrium was attained.

The continuous analysis provided by the analyzer as well as the permanent record produced by the recorder were invaluable in determining the attainment of equilibrium. With the analyzer-recorder setup it is possible to observe the composition of the gas stream from the equilibrium cell approach the equilibrium value at which time the recorder levels off and gives a constant reading for as long as a test point is run. The experimental data were collected by maintaining a constant temperature and varying the pressure, and since the recorder chart moves at a constant speed it was possible to note the time required to reach equilibrium with a change in pressure which in most cases was 5 minutes or less. This time, of course, would vary with the flow rate through the system inasmuch as the lines between the analyzer and equilibrium cell must be purged after reaching each new equilibrium point.

As mentioned previously the flow rates used for the majority of the test points was such as to give a dwell time of 5-6 minutes in the cryostat. This would seem to be more than adequate in view of the fact that those points which were rechecked were done so at faster flow rates with no change in equilibrium composition.

VIII EXPERIMENTAL PROCEDURES

A. Pressure and Temperature Range

Equilibrium compositions were determined for six isotherms from 140° to 190°K in 10°K intervals, these being the same temperatures covered by Sonntag^(25,26). The maximum equilibrium pressure for each isotherm was 200 atmospheres while the minimum pressure at each temperature was that at which the experimental data agreed with that reported previously. These minimum pressures were 70, 70, 100, 100, 100 and 50 atmospheres for the 140° , 150° , 160° , 170° , 180° and 190°K isotherms, respectively.

B. Preparation of Gas Mixtures

The gas mixtures used must have a carbon dioxide concentration greater than the expected equilibrium composition if any solid CO_2 is to form. At the same time, the concentration should not greatly exceed the expected equilibrium value since this would cause an excessive amount of solid to form, resulting in premature plugging of the system. In addition, the maximum equilibrium composition expected for each isotherm will decrease with temperature. With these facts in mind an initial gas mixture of 0.06 carbon dioxide mole fraction in nitrogen was prepared by filling the evacuated storage tank with carbon dioxide to a predetermined pressure such that the desired composition was obtained after adding nitrogen until a tank pressure of 5000 psig was reached. This mixture was then used to obtain

the experimental data along the 190°K isotherm. Upon completion of this isotherm the tank pressure was reduced to a predetermined level such that a 0.04 CO₂ mole fraction mixture was obtained after adding nitrogen until a tank pressure of 5000 psig was again reached. This mixture was then used to obtain data along the 180°K isotherm. In a similar manner, the carbon dioxide concentration in the gas mixture was reduced after each isotherm such that the mole fraction of CO₂ in the mixture was 0.02 for the 170°K isotherm, 0.01 for 160°K, 0.005 for 150°K and 0.0025 for 140°K. With this method of operation carbon dioxide was used to obtain the initial mixture of 0.06 CO₂ mole fraction and thereafter it was necessary to use only nitrogen. The carbon dioxide concentration was changed at the completion of rather than during the running of an isotherm since it was convenient to do so while the equipment was being prepared for the next isotherm.

The nitrogen gas was pumped into the storage tank with the diaphragm compressor seen in Figure 14 of Chapter VI, a diaphragm compressor being used to prevent contamination of the test gas. Carbon dioxide could also be added to the storage tank through this compressor in the event that the CO₂ concentration was not sufficient to form a solid phase. However, this was not found necessary during testing since the mixtures indicated previously were adequate.

C. Run Procedure

The first step in making an experimental run was to

cool the cryostat from room temperature to the desired equilibrium temperature, this requiring approximately three hours due to the large mass of liquid. At the same time, the heat exchanger was cooled by circulating the coolant with the bypass valve completely closed. During this cooling time the gas analyzer was calibrated at at least four points and the calibration periodically rechecked. When the cryostat was cooled to within approximately 30°C of the equilibrium temperature, nitrogen gas was passed through the system at the pressure and flow rate to be maintained during the running of a test point. With the nitrogen flowing under these conditions the bypass valve on the heat exchanger coolant line was regulated so that the temperature of the nitrogen leaving the heat exchanger was approximately 1°C above the equilibrium temperature. This bypass valve setting resulted in an exit temperature of 5° - 10°C above the equilibrium value when the CO_2 - N_2 mixture was passed through the system. This rise in exit temperature is due to an increased load on the heat exchanger caused by the heat of sublimation liberated from the freezing carbon dioxide.

When the cryostat reached the desired temperature, the liquid nitrogen flow rate through the cryostat was reduced to that necessary to slightly subcool the bath liquid. At this time the automatic temperature regulator was turned on and adjusted to maintain the equilibrium temperature within as small a limit as possible, this being accomplished by using the regulator as a proportional with reset

controller. With the equilibrium temperature stabilized, the $\text{CO}_2\text{-N}_2$ gas mixture was passed through the system and regulated at the desired pressure and flow rate. As indicated previously, the heat exchanger exit temperature of this mixture was set at $5^\circ\text{-}10^\circ\text{C}$ above the equilibrium value. Care was exercised so that this exit temperature never fell below and was always less than 10°C above the equilibrium temperature, the latter restriction being made to keep the cooling required in the cryostat to a minimum.

After the temperature, pressure and flow rate were set, a portion of the gas stream from the equilibrium cell was passed thru the analyzer at the same flow rate and pressure at which the instrument had been calibrated. The gas mixture was allowed to pass through the system until equilibrium was attained as indicated by a leveling off in the recorded output of the analyzer. Thereafter the gas mixture was passed through the system for at least an additional twenty-five minutes during which time the following conditions were to be satisfied: (1) the recorder must indicate no further change in composition, (2) the gas mixture flow rate and pressure must remain constant and (3) the test gas must never reach a temperature below the equilibrium value. When these conditions were satisfied for the minimum period of twenty-five minutes the experimental readings were taken.

Since the gas phase was analyzed and recorded continuously it was possible to observe the composition approach the equilibrium value at which point it remained constant

for as long as the test gas was flowing through the system. This was felt to be a reliable indication of the attainment of equilibrium. Furthermore, it was not deemed necessary to run each experimental point several times and take an average of the resulting equilibrium compositions as done by others who performed similar investigations without a continuous gas phase analysis. Nevertheless, approximately half of the experimental points were rerun at different flow rates in order to (1) check the reproducibility of the experimental data, (2) test the ability of the equilibrium cell to retain all solid particles of carbon dioxide and (3) determine the ability of analyzer cells of different lengths to give the same analysis.

The experimental data were collected at constant temperature beginning with the highest pressure and successively decreasing the pressure. This order resulted in constantly decreasing equilibrium compositions over the pressure range encountered here. Since the accuracy of the gas analyzer is considered to be a fixed percentage of the full scale reading, it is desirable to have the experimental readings fall on the upper portion of the calibration curve in order to minimize the percent error in the reading itself. For this reason the analyzer was periodically recalibrated using a lower percentage CO_2 calibrating mixture each time to give the maximum scale deflection. Each calibration was rechecked between test points to check the analyzer for drift and reproducibility.

After a period of operation the system plugged due to the accumulation of solid carbon dioxide. When this

occurred, pure nitrogen gas was passed through the system and equilibrium was attained by picking up rather than depositing carbon dioxide. Several test points were rerun at different flow rates with this method of operation in order to provide the checks indicated previously. During a few such check points the temperature of the nitrogen leaving the heat exchanger was permitted to fall to 10° - 15°C below the equilibrium value with the test gas then being warmed rather than cooled in the cryostat. This would not be permissible if equilibrium is being attained by freezing out carbon dioxide from a CO_2 - N_2 mixture since the possibility would arise of equilibrium occurring at the lower temperature in the heat exchanger. This would result in a carbon dioxide concentration in the test gas which is insufficient to form a solid phase at the comparatively warmer temperature in the equilibrium cell. However, solid carbon dioxide is already present in the equilibrium cell for those check points discussed above and the difficulty just mentioned does not arise. It was desirable to run some check points in which the test gas was warmed rather than cooled in the cryostat so as to ensure that the gas was in fact reaching the equilibrium temperature even though the thermocouple located within the equilibrium cell was not functioning, as discussed in section VII-C.

D. Operational Difficulties

The only major difficulty encountered was experienced during preliminary testing of the equipment. It was noted

that the equilibrium composition was increasing with the gas flow rate through the system. Therefore, it was concluded that solid carbon dioxide particles were passing through the equilibrium cell. However, it was soon discovered that the problem did not exist with the equilibrium cell but rather with the method of throttling the gas flowing through the analyzer. During calibration the throttling necessary to maintain a constant flow rate through the analyzer was done on the inlet side of the instrument. However, during the preliminary test runs the gas coming from the equilibrium cell and flowing through the analyzer was throttled on the exhaust side of the unit, thus producing a pressure in the analyzer in excess of that at which the calibration had been made. As the flow rate through the system was increased the amount of throttling necessary to maintain a constant flow rate through the analyzer was also increased. This in turn produced an even higher pressure in the analyzer with a corresponding increase in the analyzer output. Once this was realized, all throttling of the gas through the analyzer was performed on the inlet side. Thereafter, no change in composition could be detected by changing the gas flow rate through the system thus indicating that the equilibrium cell was retaining all solid carbon dioxide particles.

IX. EXPERIMENTAL RESULTS

A. General

The experimental results reported here are those obtained in the following pressure and temperature range: 140° and 150°K, 70-200 atm; 160°, 170° and 180°K, 100-200 atm; and 190°K, 50-200 atm. However, inasmuch as this investigation is an extension of that initiated by Sonntag, references 25 and 26, the data from this latter work is also included here in order that the equilibrium compositions over the entire range 140° - 190°K and 5-200 atm (except 190°K, 5 atm) may be presented. As indicated before, the data collected previously at the high pressure end of the 140°, 150° and 190°K isotherms differed somewhat from that obtained here. Possible explanations for these differences are offered later in this chapter.

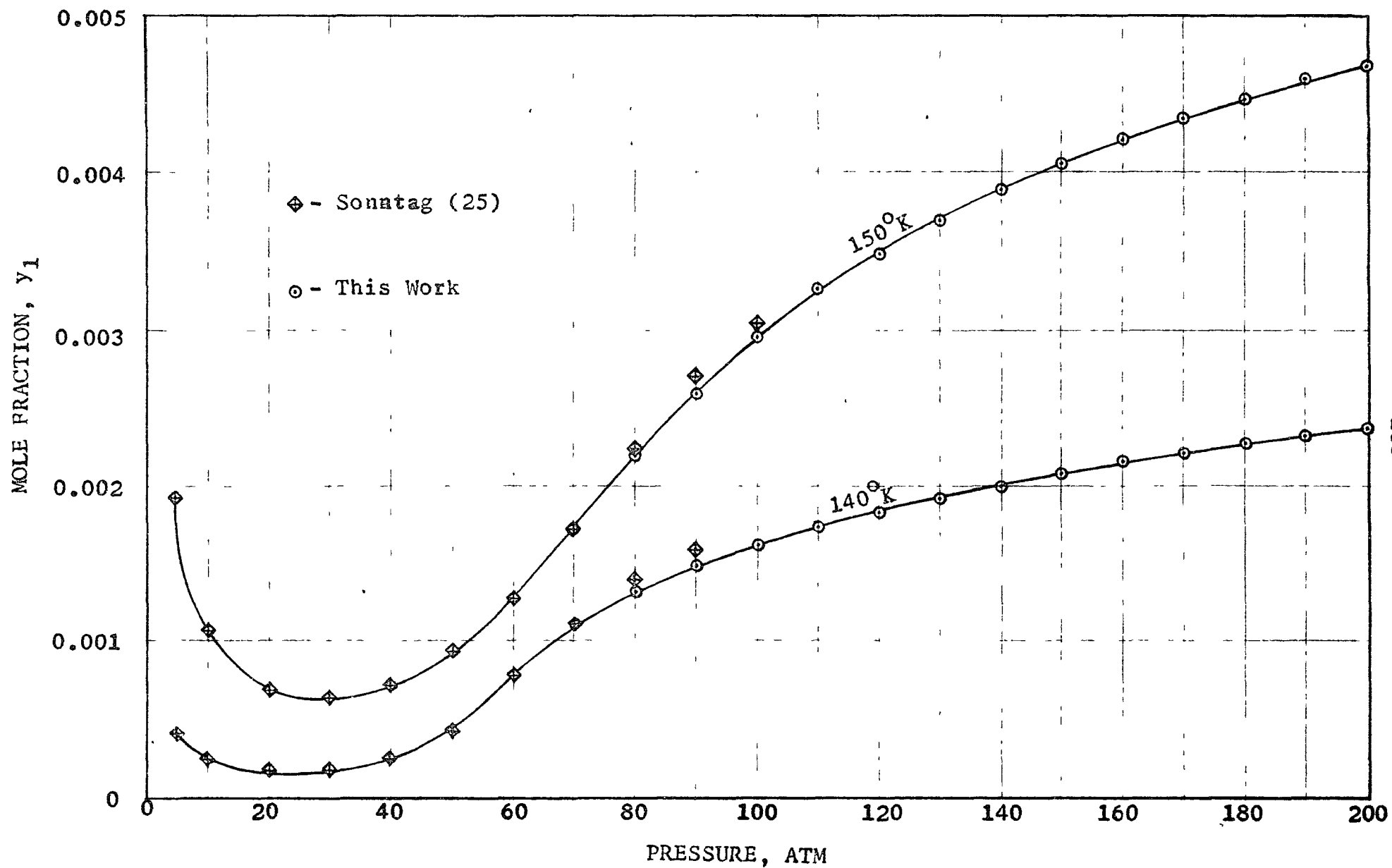
The experimental readings made during this investigation are presented in Appendix A where the letters A, B, and C designate those runs made with the 3/16, 3/4, and 5 $\frac{1}{4}$ inch gas analyzer cells, respectively. The numbers immediately following these letters are used to designate a particular run with each run having been numbered consecutively as it was made. Appendix A also includes the considered accuracy of the gas phase analysis for each test point, this accuracy being approximately $\pm 1\%$ of the analyzer reading in most cases. The variation in these accuracies is due to the use of different analyzer calibrations in an attempt to keep the experimental readings

on the upper portion of the calibration curve. As indicated previously, this was necessary in order to minimize analysis errors since the analyzer accuracy is a fixed percentage of the full scale deflection.

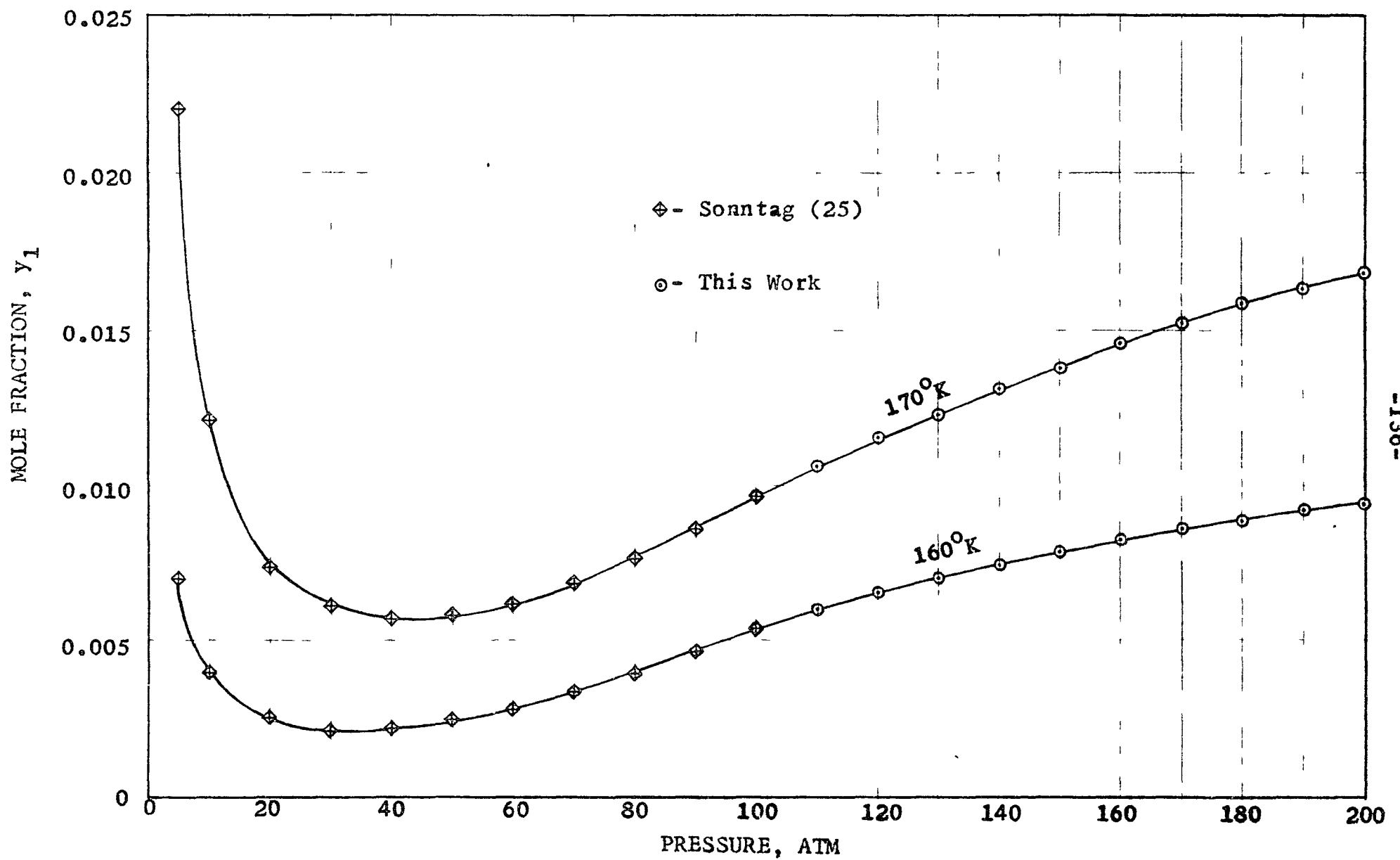
B. Results and Discussion

The experimental results are presented in Figures 23-25 as carbon dioxide mole fraction versus pressure at constant temperature for the six isotherms studied here, these results being presented on three separate graphs due to the wide variance in mole fraction encountered during the investigation. Inasmuch as Sonntag has discussed the minimum points as well as the behavior below 100 atmospheres for each of the six isotherms, no such discussion will be included here. However, the behavior of each isotherm in the pressure range covered in this study will be investigated.

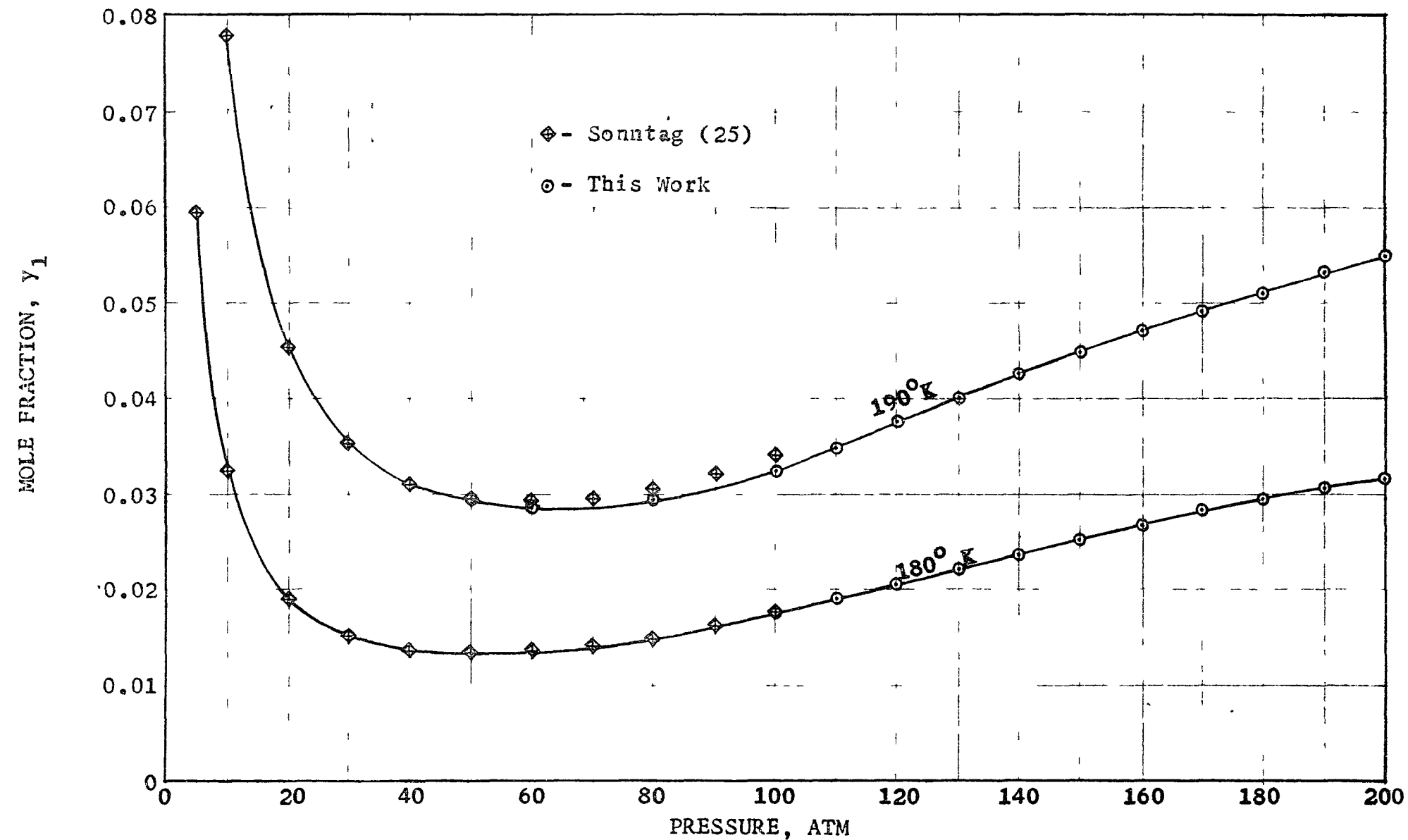
Sonntag's investigation revealed an inflection in the 140° and 150°K isotherms at which points the slope of the curves began to decrease with increasing pressure, these points occurring at approximately 60 and 75 atmospheres, respectively, for each of the isotherms. It would seem, therefore, that such behavior would occur at the higher temperature isotherms if the pressure is sufficiently increased and such is seen to be the case in Figures 23-25. This phenomenon has also been observed by Webster⁽¹⁶⁾ with the air-CO₂ system.



Mole Fraction y_1 vs. Pressure at 140° and 150°



Mole Fraction y_1 vs. Pressure at 160° and 170°r



Mole Fraction y_1 vs. Pressure at 180° and 190°K

Figure 25

At 140°K, the equilibrium compositions determined here at 80 and 90 atmospheres are slightly below those obtained by Sonntag. In view of the more accurate temperature measurement and control as well as the more effective heat exchanger employed here as compared to that used in the previous investigation, it would seem that the data obtained here would be the more valid. It should also be noted that the data of Sonntag at 70 atmospheres and below and that obtained here form a smooth and continuous curve which also indicates that the data collected here at 80 and 90 atmospheres is more valid. The accuracy of the gas analysis made at this temperature is ± 0.00001 mole fraction from 70-140 atmospheres and ± 0.00002 for all other points.

A possible reason for the differences between the two sets of data might lie in the respective amounts of carbon dioxide being frozen out in the equilibrium cell. In both investigations, the velocity through the equilibrium coil was held constant but Sonntag used a larger velocity, 0.1 compared to 0.04 ft/sec. In addition, the diameter of the equilibrium coil used in the previous study was larger, 0.277 compared to 0.152 inches. Finally, the gas mixture used in the former work contained a larger concentration of carbon dioxide, 0.05 compared to 0.025 mole fraction CO_2 . The net effect of all this is that a larger mass of carbon dioxide was being frozen out during the previous investigation than in the present one. Since the heat of sublimation must be absorbed when the solid forms, this would produce a

comparatively larger cooling load on the equilibrium vessel used in the prior investigation and this could in turn result in a temperature slightly higher than the desired equilibrium value with a corresponding increase in the equilibrium composition. This would seem to be a plausible explanation when it is realized that an increase in temperature of approximately 1°C , for this particular isotherm, could account for the differences between the two sets of data. Assuming this explanation to be valid, the two sets of data should agree at some lower pressure since the mass flow rate through the system, and therefore the mass of solid carbon dioxide being deposited, will decrease with pressure if the velocity is held constant. This is seen to be the case as evidenced by the agreement at 70 atmospheres indicated in Figure 23.

At 150°K , the equilibrium compositions determined here at 80, 90 and 100 atmospheres again fall below that of Sonntag. The reasoning given previously for the differences between the data is also applicable in this case with a temperature of less than 0.8°C above the equilibrium value accounting for the disagreement at this temperature. The two sets of data agree at 70 atmospheres and the data of Sonntag below that pressure and the data collected here form a smooth and continuous curve as seen in Figure 23. The gas phase analysis for this isotherm is accurate to ± 0.00002 mole fraction from 70-110 atmospheres and ± 0.00003 for all other points.

At 160° , 170° and 180°K the experimental data agrees with that obtained in the previous study at 100 atmospheres for each temperature and the two sets of data generate a smooth curve for each of the three isotherms as seen in Figures 24 and 25. The disagreement between the two sets of data observed at 140° and 150°K is not present here. As the temperature is increased the density of the gas mixture, and therefore the mass flow rate through the system, is reduced. At the same time the equilibrium composition increases with temperature. Therefore the amount of solid carbon dioxide being deposited in the equilibrium cell at 160° , 170° and 180°K will be considerably less than that at 140° and 150°K . Thus the reasons presented previously for the disagreement between the two sets of data at 140° and 150°K could still be valid but not be applicable at the three temperatures discussed here inasmuch as the mass of carbon dioxide freezing out of the mixture is reduced.

The accuracy of the mole fraction determination for the three isotherms discussed in the preceding paragraph are as follows: 160°K , ± 0.00003 from 100-110 atmospheres and ± 0.00005 for all other points; 170°K , ± 0.00005 for 100 atmospheres and ± 0.0001 for all other points; 180°K , ± 0.0001 from 100-110 atmospheres and ± 0.0002 for all other points.

The final isotherm studied was that at 190°K and the experimental results for this temperature may be seen in Figure 25. The mole fractions for this isotherm are considered

to be accurate to ± 0.0002 from 50 - 100 atmospheres and ± 0.0003 for all the other points. It will be noted that the equilibrium compositions determined here in the range 60-100 atmospheres again fell slightly below those reported previously while both sets of data agreed at 50 atmospheres. It is possible that the larger mass flow rates used by Sonntag as compared to those used here might account for some of this difference due to the warming tendency of the freezing carbon dioxide as discussed previously. However, it is felt that the major cause for this disagreement lies in the heat exchanger. A counterflow heat exchanger was used in both investigations to cool the test gas as it entered the cryostat. However, in the previous work the test gas leaving the cryostat was used as the coolant and, therefore, the temperature of the test gas entering the cryostat could not be regulated at a selected value as was possible in this investigation. The heat exchanger exit temperature was never more than 10°C above the equilibrium value in the present study while the same temperature in the previous investigation was approximately 50°C above the equilibrium value for the 190° isotherm. Thus it is felt that the heat exchanger performance plus the warming tendency of the freezing carbon dioxide could have produced a temperature approximately 1°C above the desired 190°K during Sonntag's investigation, this temperature difference being sufficient to account for the disagreement between the two sets of data.

The equilibrium compositions presented in Figures 23-25

are also presented in tabular form in Table XIII. The values presented there with the exception of those at 180°K , 190 atm and 190°K , 80 atm, are those obtained in either the original or check runs, depending upon which value seemed to be the best as indicated from a large scale plot of the data. The values listed at the two excepted points are the average of the original and check points inasmuch as these values seemed to best fit the large scale plot of the data. The data obtained by Sonntag, with the exception of those points in disagreement with the data collected here, are also included in Table XIII.

As mentioned previously, approximately 50 percent of the experimental points were rerun in order to provide checks on various components in the system. The results of the original and check runs are presented in Table XIV along with their respective gas analysis error limits and the gas velocity in the equilibrium coil, the latter being important in evaluating the capacity of the equilibrium cell to retain solid carbon dioxide particles. Examination of this table reveals that the results of the original and check runs are within the gas analysis error limits in every case. This provides a favorable check on the accuracy of the gas analysis as well as the reproducibility of the data. Table XIV also indicates that analyzer cells of different lengths will give the same analysis as evidenced by the checks made at 150°K ; 100 and 110 atmospheres and 180°K ; 100, 110 and 120 atmospheres. Finally, this table reveals that of the 38 points rerun, a lower flow rate

TABLE XIII

MOLE FRACTION OF CARBON DIOXIDE AS A
FUNCTION OF TEMPERATURE AND PRESSURE

P atm	T, °K					
	140	150	160	170	180	190
5	0.00042	0.00192	0.00699	0.0220	0.0595	-
10	0.00024	0.00107	0.00395	0.0121	0.0325	0.0778
20	0.00018	0.00069	0.00251	0.00735	0.0190	0.0455
30	0.00019	0.00064	0.00217	0.00608	0.0151	0.0354
40	0.00025	0.00071	0.00217	0.00572	0.0137	0.0311
50	0.00043	0.00094	0.00243	0.00580	0.0133	0.0293*
60	0.00078	0.00128	0.00280	0.00618	0.0135	0.0285
70	0.00112*	0.00172*	0.00332	0.00685	0.0141	0.0285 ⁽¹⁾
80	0.00132	0.00219	0.00393	0.00760	0.0150	0.0293 ⁽²⁾
90	0.00148	0.00259	0.00461	0.00860	0.0161	0.0305 ⁽¹⁾
100	0.00162	0.00295	0.00534*	0.00964*	0.0175*	0.0324
110	0.00173	0.00325	0.00595	0.0106	0.0190	0.0347
120	0.00183	0.00348	0.00650	0.0115	0.0205	0.0375
130	0.00192	0.00370	0.00698	0.0123	0.0221	0.0400
140	0.00200	0.00388	0.00741	0.0131	0.0236	0.0425
150	0.00208	0.00405	0.00784	0.0138	0.0251	0.0448
160	0.00215	0.00420	0.00821	0.0146	0.0267	0.0470
170	0.00221	0.00434	0.00856	0.0152	0.0282	0.0490
180	0.00227	0.00447	0.00885	0.0158	0.0295	0.0510
190	0.00232	0.00458	0.00912	0.0163	0.0306 ⁽³⁾	0.0530
200	0.00236	0.00468	0.00937	0.0168	0.0315	0.0548

* Points at which the data collected by Sonntag^(25, 26) and that obtained here agree. All values at pressures lower than these points are those reported by Sonntag.

(1) Values read from plot of data and not determined experimentally

(2) Average of 0.0292 ± 0.0002 and 0.0294 ± 0.0002

(3) Average of 0.0305 ± 0.0002 and 0.0307 ± 0.0002

TABLE XIV
COMPARISON OF ORIGINAL AND CHECK POINTS

$T, ^\circ K$	P, atm	$\bar{V}, ft/sec$	y_1
140	200	0.0269	0.00236 + 0.00002
	200	0.0394	0.00237 $\pm$ 0.00002
	180	0.0275	0.00227 + 0.00002
	180	0.0381	0.00227 $\pm$ 0.00002
	160	0.0284	0.00215 + 0.00002
	160	0.0372	0.00215 $\pm$ 0.00002
	140	0.0269	0.00200 + 0.00001
	140	0.0359	0.00200 $\pm$ 0.00002
	120	0.0283	0.00184 + 0.00001
	120	0.0353	0.00183 $\pm$ 0.00002
150	200	0.0340	0.00468 + 0.00003
	200	0.0455	0.00469 $\pm$ 0.00003
	180	0.0351	0.00447 + 0.00003
	180	0.0468	0.00446 $\pm$ 0.00003
	160	0.0362	0.00419 + 0.00003
	160	0.0435	0.00420 $\pm$ 0.00003
	140	0.0359	0.00388 + 0.00003
	140	0.0436	0.00388 $\pm$ 0.00003
	120	0.0385	0.00349 + 0.00003
	120	0.0467	0.00348 $\pm$ 0.00003
	110	0.0430	0.00324 + 0.00003
	110*	0.0515	0.00325 $\pm$ 0.00002
	100	0.0364	0.00295 + 0.00003
	100	0.0455	0.00295 $\pm$ 0.00003
	100*	0.0455	0.00295 $\pm$ 0.00002

TABLE XIV (CONT'D)

$T, ^\circ K$	P, atm	$V, ft/sec$	y_1
160	200	0.0291	0.00937 ± 0.00005
	200	0.0411	0.00937 ± 0.00005
	180	0.0355	0.00883 ± 0.00005
	180	0.0456	0.00885 ± 0.00005
	160	0.0320	0.00821 ± 0.00005
	160	0.0479	0.00823 ± 0.00005
	140	0.0401	0.00740 ± 0.00005
	140	0.0459	0.00741 ± 0.00005
	120	0.0413	0.00650 ± 0.00005
	120	0.0510	0.00650 ± 0.00005
170	200	0.0266	0.0168 ± 0.0002
	200	0.0480	0.0168 ± 0.0002
	180	0.0278	0.0158 ± 0.0002
	180	0.0475	0.0158 ± 0.0002
	160	0.0235	0.0146 ± 0.0002
	160	0.0499	0.0145 ± 0.0002
	140	0.0321	0.0131 ± 0.0001
	140	0.0498	0.0131 ± 0.0001
	120	0.0366	0.0115 ± 0.0001
	120	0.0494	0.0116 ± 0.0001
	100	0.0346	0.00965 ± 0.00005
	100	0.0497	0.00964 ± 0.0001

TABLE XIV (CONT'D)

T, °K	P, atm	V, ft/sec	y_1
180	200	0.0290	0.0315 + 0.0002
	200	0.0493	0.0315 $\pm$ 0.0002
	190	0.0298	0.0305 + 0.0002
	190	0.0447	0.0307 $\pm$ 0.0002
	160	0.0266	0.0267 + 0.0002
	160	0.0496	0.0267 $\pm$ 0.0002
	140	0.0366	0.0235 + 0.0002
	140	0.0475	0.0236 $\pm$ 0.0002
	120	0.0337	0.0204 + 0.0002
	120	0.0464	0.0205 $\pm$ 0.0002
	120*	0.0464	0.0205 $\pm$ 0.0001
	110	0.0451	0.0190 + 0.0002
	110*	0.0451	0.0190 $\pm$ 0.0001
	100	0.0498	0.0175 + 0.0002
	100*	0.0398	0.0175 $\pm$ 0.0001
190	200	0.0201	0.0548 + 0.0003
	200	0.0470	0.0548 $\pm$ 0.0003
	190	0.0225	0.0530 + 0.0003
	190	0.0290	0.0530 $\pm$ 0.0003
	180	0.0236	0.0510 + 0.0003
	180	0.0269	0.0510 $\pm$ 0.0003
	180	0.0303	0.0508 $\pm$ 0.0003
	160	0.0288	0.0470 + 0.0003
	160	0.0504	0.0470 $\pm$ 0.0003
	140	0.0281	0.0425 + 0.0003
	140	0.0481	0.0425 $\pm$ 0.0003
	110	0.0252	0.0347 + 0.0002
	110	0.0504	0.0347 $\pm$ 0.0002
	100	0.0222	0.0322 + 0.0002
	100	0.0500	0.0324 $\pm$ 0.0002
	80	0.0355	0.0292 + 0.0002
	80	0.0496	0.0294 $\pm$ 0.0002

* denotes runs made with different length analyzer cells

resulted in a lower equilibrium composition 12 times, the same value 19 times and a higher value 7 times. Such results indicate that no solid carbon dioxide particles were passing through the equilibrium cell and subliming as the gas was warmed to room temperature.

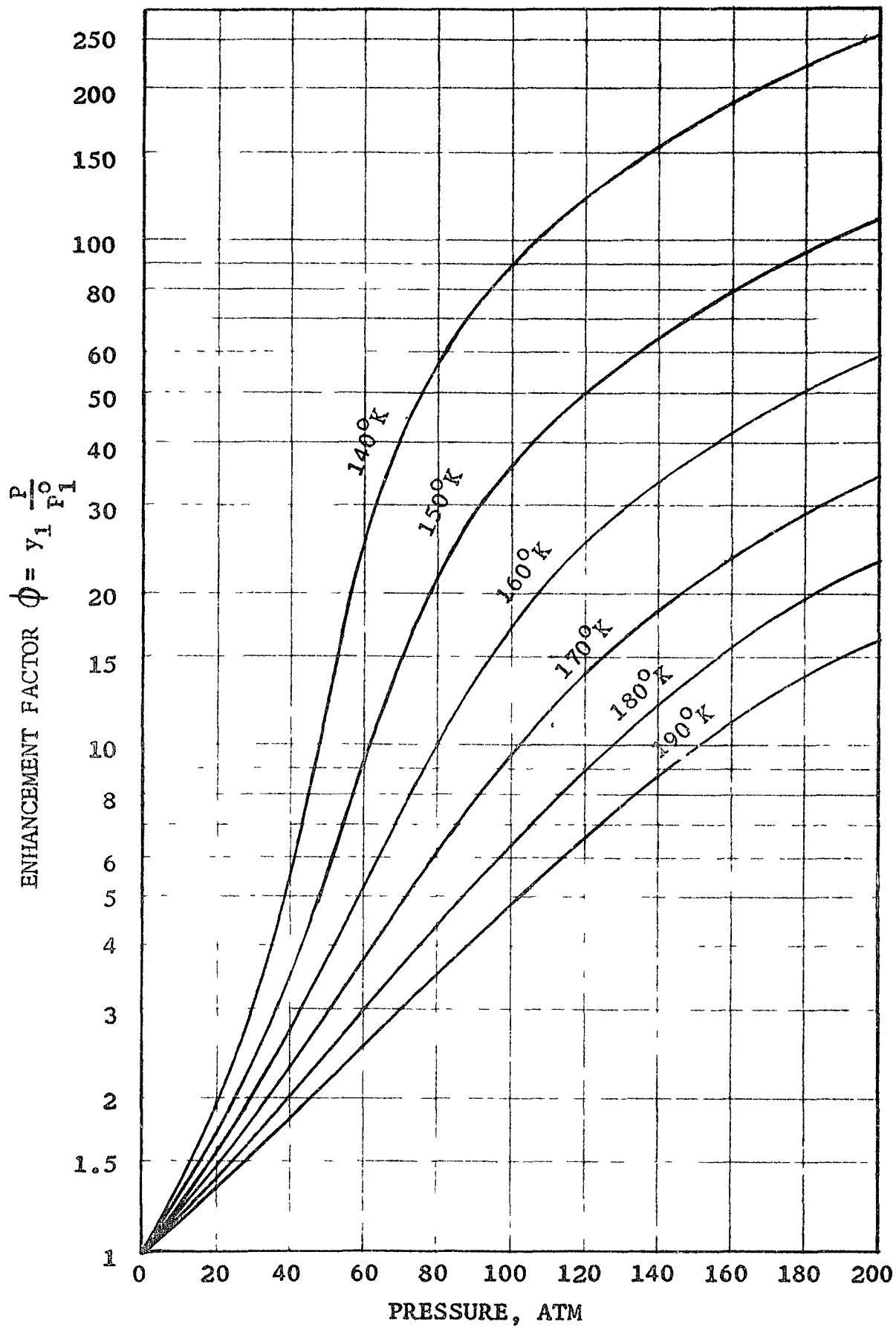
An alternate form of presenting the experimental data is in terms of the enhancement factor ϕ , this dimensionless parameter being defined in section III-G as the ratio of the actual carbon dioxide mole fraction in the gas phase to that predicted for ideal behavior of the system. The values for the enhancement factor are presented in Table XV, these values being computed from the mole fractions listed in Table XIII and the vapor pressures given in Table I. The enhancement factors are also plotted in Figure 26 as a function of pressure at constant temperature.

The enhancement factor, as defined above, will equal unity for ideal behavior and will steadily increase from this value as the system becomes more and more non-ideal. Thus the magnitude of this parameter will give an indication of the departure from ideal behavior. The enhancement factor would be expected to increase with increasing pressure at a given temperature or decreasing temperature at a given pressure, this being the direction in which the density of the gas phase increases. Figure 26 shows this to be the case as evidenced by an increase in the enhancement factor in the direction expected with a maximum value for this parameter of approximately 256 occurring at 140° K, 200 atmospheres.

TABLE XV
ENHANCEMENT FACTOR AS A FUNCTION
OF TEMPERATURE AND PRESSURE

$$\phi = y_1 \frac{P}{P_1^0}$$

P atm	T, °K					
	140	150	160	170	180	190
5	1.1414	1.1415	1.1131	1.1078	1.0858	-
10	1.3043	1.2723	1.2580	1.2185	1.1861	1.1458
20	1.9565	1.6409	1.5987	1.4804	1.3869	1.3402
30	3.0978	2.2830	2.0732	1.8369	1.6533	1.5641
40	5.4348	3.3769	2.7643	2.3041	2.0000	1.8321
50	11.685	5.5886	3.8694	2.9204	2.4270	2.1576
60	25.435	9.1320	5.3503	3.7341	2.9562	2.5184
70	42.609	14.316	7.4013	4.8288	3.6022	2.9249
80	57.391	20.832	10.013	6.1229	4.3796	3.4521
90	72.391	27.717	13.213	7.7946	5.2883	4.0427
100	88.043	35.077	17.006	9.7080	6.3869	4.7717
110	103.42	42.509	20.844	11.742	7.6277	5.6215
120	119.34	49.655	24.841	13.897	8.9781	6.6274
130	135.65	57.194	28.898	16.103	10.485	7.6583
140	152.17	64.590	33.038	18.469	12.058	8.7629
150	169.46	72.235	37.452	20.846	13.741	9.8969
160	186.96	79.905	41.834	23.525	15.591	11.075
170	204.73	87.729	46.344	26.022	17.496	12.268
180	222.06	95.672	50.732	28.640	19.380	13.520
190	239.56	103.47	55.185	31.188	21.219	14.831
200	256.52	111.30	59.682	33.837	22.993	16.141



Enhancement Factor Φ vs. Pressure at Constant Temperature
Figure 26

Some mention should also be made here concerning a possible phase transformation from the solid-vapor equilibrium system to that in which a liquid phase is introduced. It was expected that such a transformation might occur at the lower temperatures and higher pressures covered in the present investigation. However, this transformation did not occur in the present study due to the low initial concentrations of carbon dioxide which resulted in mixture critical temperatures below the range covered here. The fact that no phase transformation occurred is evidenced by the absence of discontinuities in the equilibrium compositions as functions of pressure, as might be expected due to a change in solubility occurring with a transformation. In addition, there was no indication in the operating behavior of the system which would indicate the presence of a liquid phase in the equilibrium cell. A further investigation would be worthwhile in order to determine a more complete phase diagram for the components used here. However, such an investigation is beyond the scope of the experimental equipment used in the present study.

X. COMPARISON OF ANALYTICAL AND EXPERIMENTAL RESULTS

A. General

In this final chapter a comparison will be made between the experimental results presented in the preceding chapter and those obtained with the five analytical methods discussed in Chapter V. These five methods are based on three empirical equations of state, Beattie-Bridgeman, Benedict-Webb-Rubin and Martin-Hou, as well as the statistical mechanics virial equation of state using two sets of virial coefficients through the third. Regarding the last two methods, the third virial coefficients used in both cases are based on the Lennard-Jones (6-12) potential while the second virial coefficients are based on the same potential in one case and on the Kihara potential in the other. In addition, there will be two sets of solutions for each method, one based on the simplifying assumption that all terms involving y_1 , P_1^0 and $1/v_1^0$ except those involving ratios of these quantities, are negligible, this solution being referred to as the simplified solution. The other solution arises from solving the general equations in which no simplifications have been made, this solution being termed the general solution. It should be remembered, however, that both solutions assume that the solid phase is pure and incompressible carbon dioxide.

It will be noted that experimental and analytical results for the pressure and temperature range covered by Sonntag (25, 26) are also included here even though the

region above 100 atmospheres is the primary area of interest in this investigation. This is done in order that the experimental and analytical results over the entire range covered in both investigations might be observed.

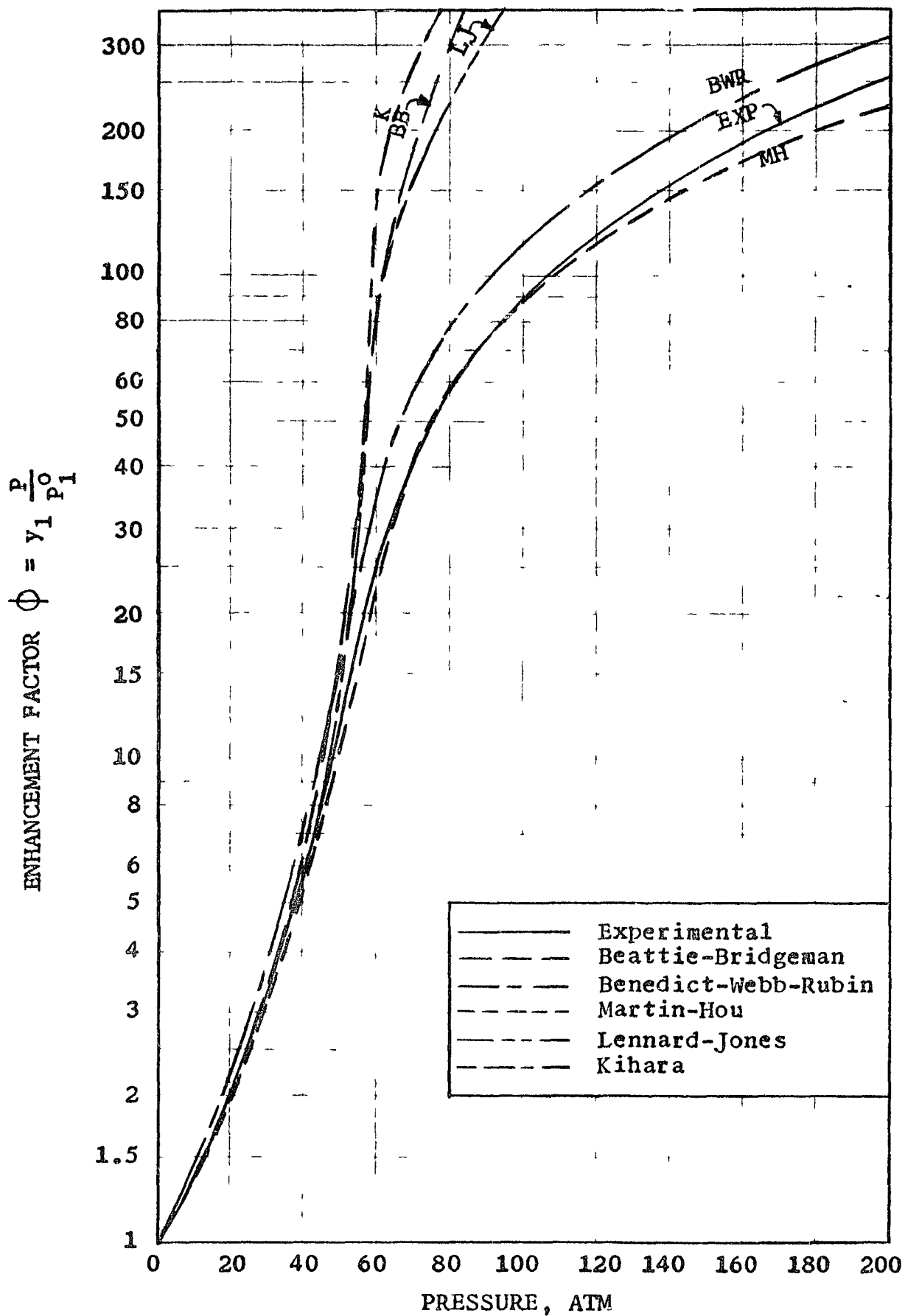
B. Discussion

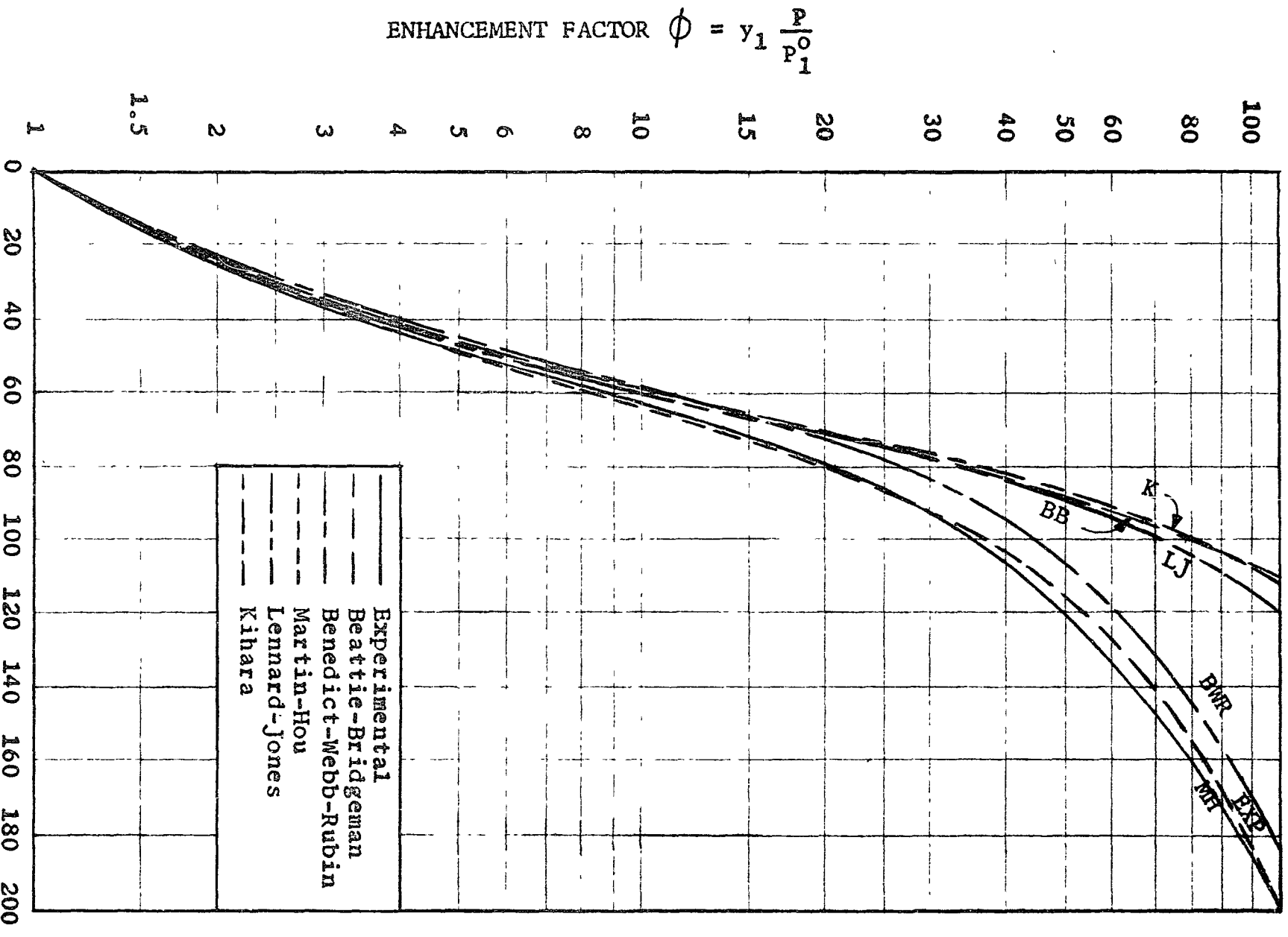
The ten analytical solutions obtained here are denoted as the General and Simplified Beattie-Bridgeman, General and Simplified Benedict-Webb-Rubin, General and Simplified Martin-Hou, General and Simplified Lennard-Jones and General and Simplified Kihara. The notation used for the first six solutions is self-explanatory while that of the remaining four indicates the results for the general and simplified cases using the virial equation of state with the second virial coefficients calculated from the Lennard-Jones potential in one case and the Kihara potential in the other. The results of the ten solutions as well as the experimental data are presented in graphical (Figures 27-32) and tabular (Table XVI) form as enhancement factor versus pressure at constant temperatures. Enhancement factors have been used since this parameter shows a greater numerical contrast than is offered by the mole fractions. Inasmuch as the simplified and general solutions are fairly close over a major portion of the temperature and pressure range, the simplified solutions have been omitted from Figures 27-32 in order to avoid unnecessary confusion. However, these solutions are included in Table XVI for purposes of comparison with the corresponding general solutions. The

analytical results presented here have been performed on the IBM 709 and IBM 7090 computers.

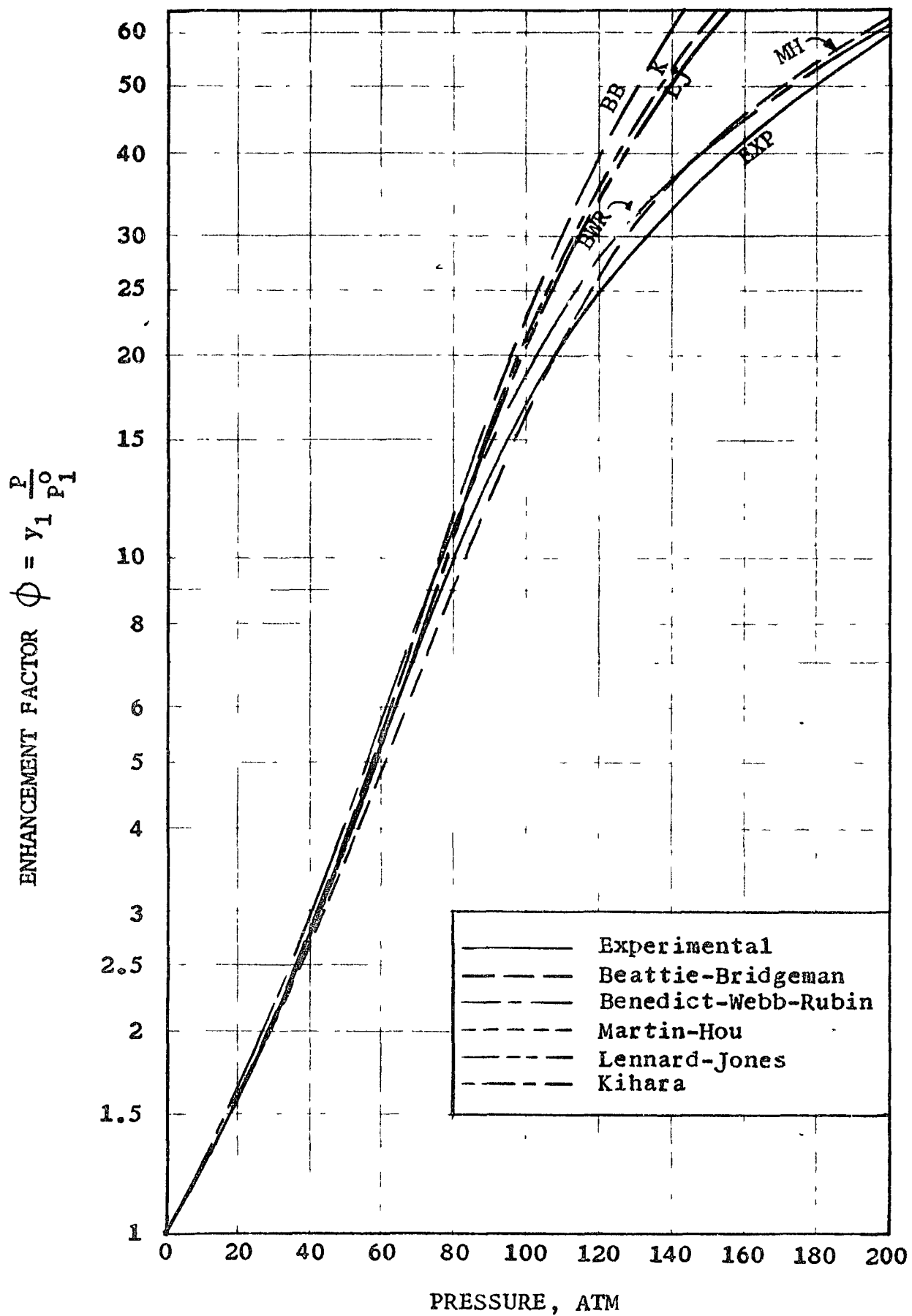
Before proceeding to any further discussion of the analytical results it should be noted that the General and Simplified Lennard-Jones and the General and Simplified Kihara solutions do not include the quadrupole correction on the second virial coefficient as discussed in Chapter IV. The effect of this correction is to increase the enhancement factor and therefore the mole fraction. This would improve the comparison between the experimental and analytical results for the Kihara solutions at 190° K only, while making the comparisons of the four solutions less favorable at the other temperatures. For this reason the quadrupole corrections were omitted from the results presented in Table XVI and Figures 27-32. In order that the effect of the quadrupole correction on the analytical results might be seen, the solutions at a few selected temperatures and pressures, both with and without quadrupole corrections, are listed in Tables XVII and XVIII.

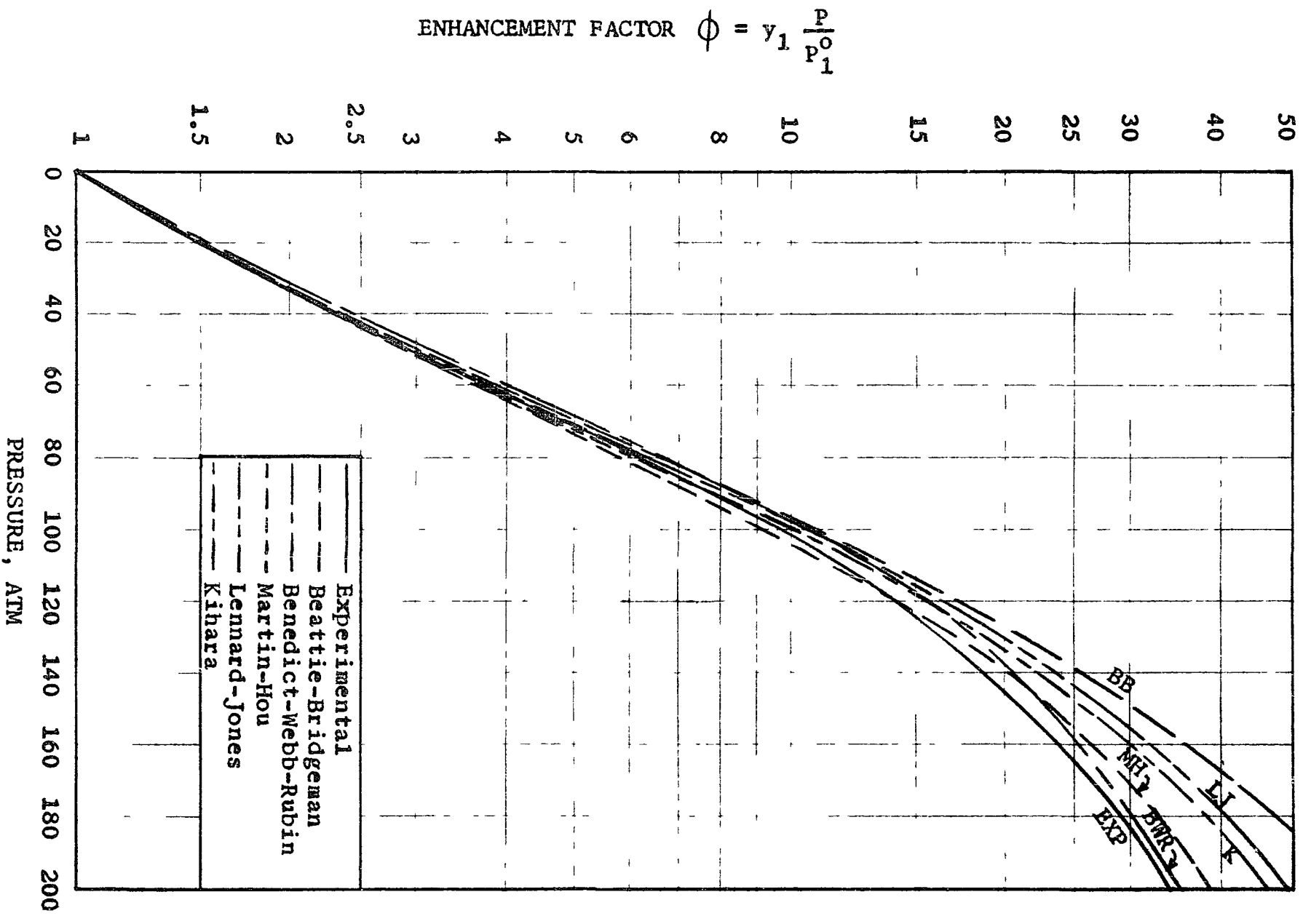
An additional comment should be made concerning the two sets of third virial coefficients presented in Tables IX and XII in Chapter V. It was discovered that the analytical enhancement factors based on the set of third virial coefficients evaluated with the c_j of Rowlinson, et al.⁽⁷²⁾ were consistently larger than those obtained with the second set. These larger enhancement factors were found to compare less favorably with those determined experimentally. Therefore, the third virial coefficients



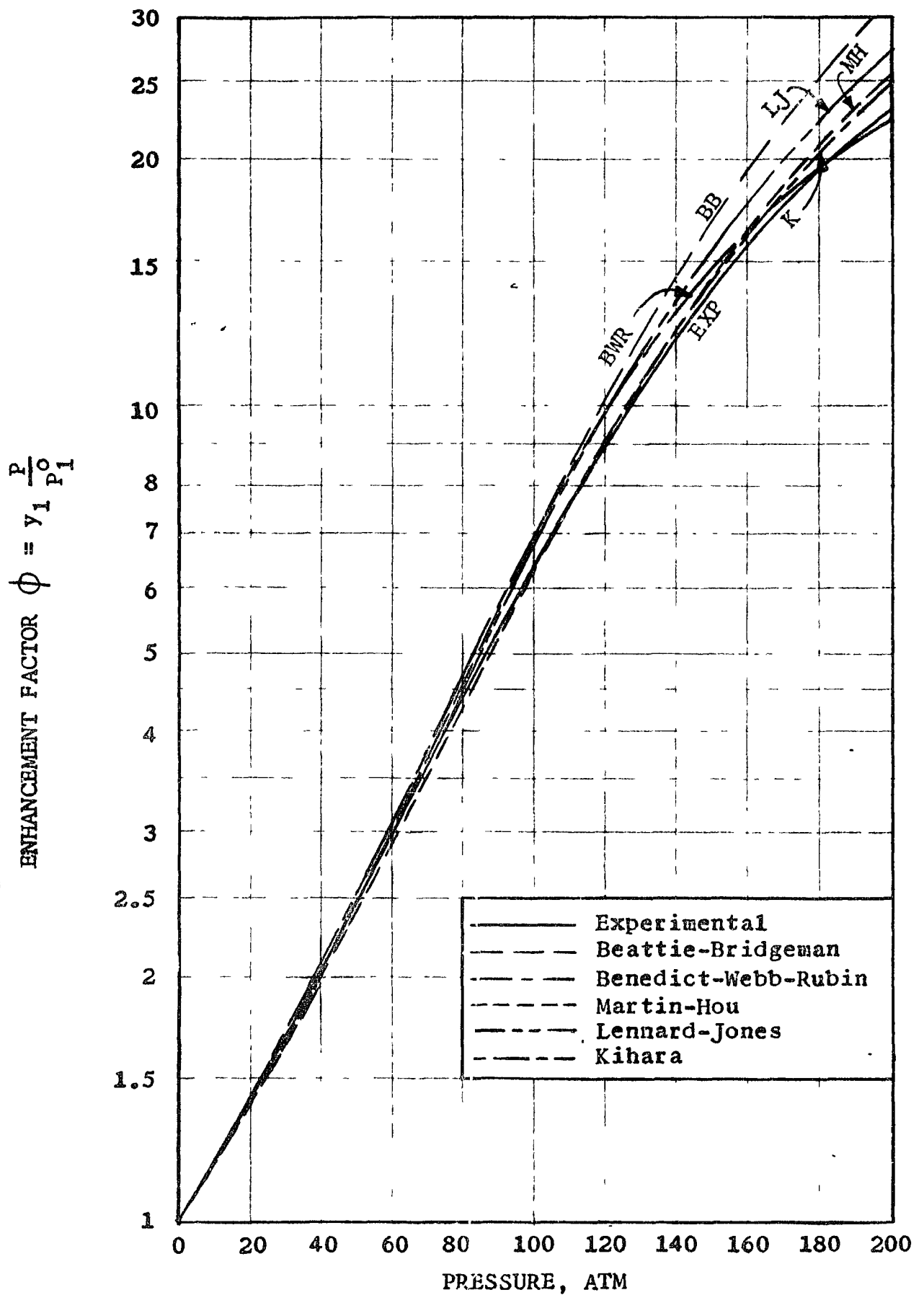


General Solutions at 150°K
Figure 28

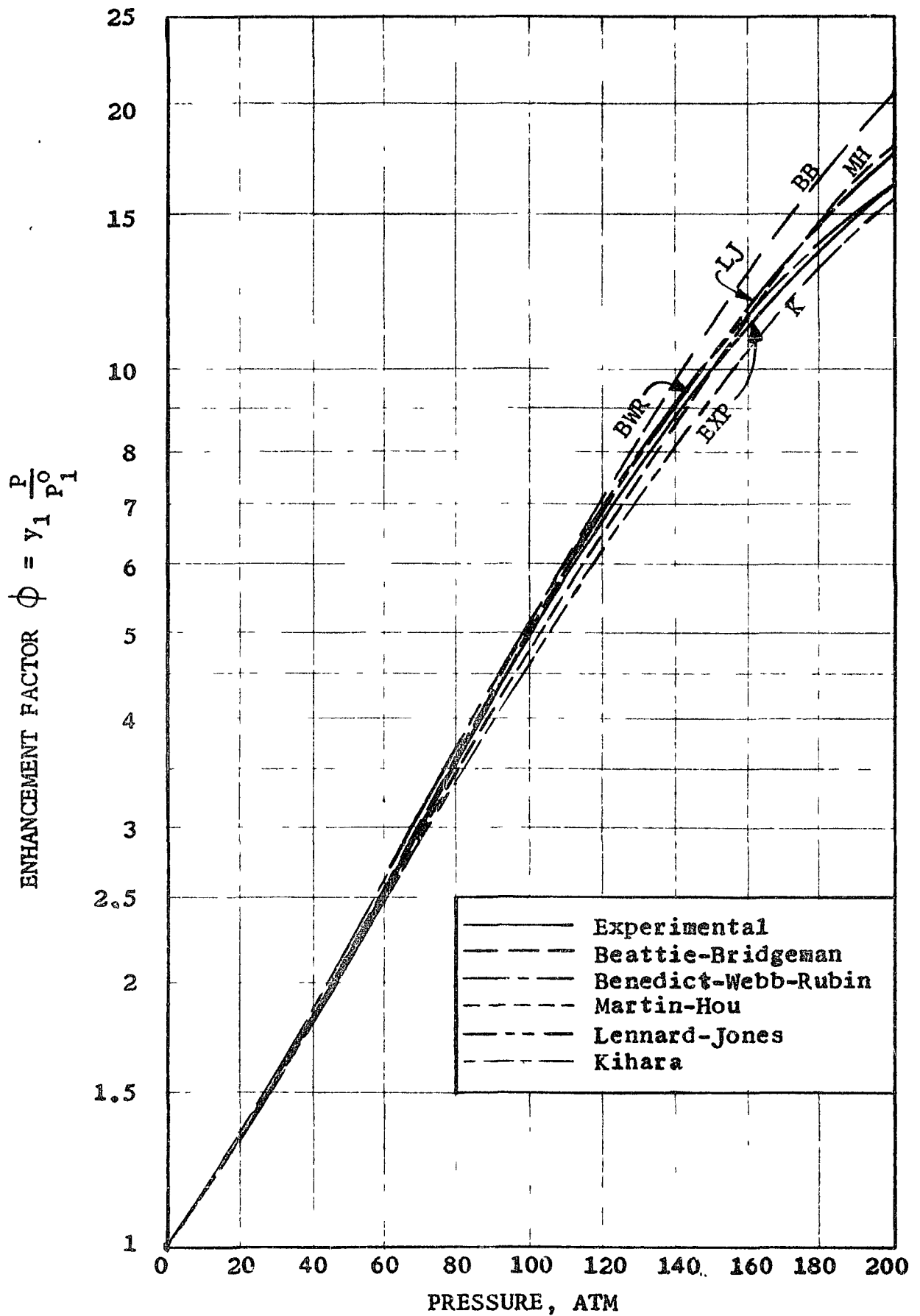




General Solutions at 170°K
Figure 30



General Solutions at 180°K
Figure 31



General Solutions at 190°K

Figure 32

TABLE XVI

COMPARISON OF ANALYTICAL AND EXPERIMENTAL ENHANCEMENT FACTORS

Notation:			Experimental Results				EXP
Simplified Beattie-Bridgeman	SBB		General Beattie-Bridgeman				GBB
Simplified Benedict-Webb-Rubin	SBWR		General Benedict-Webb-Rubin				GBWR
Simplified Martin-Hou	SMH		General Martin-Hou				GMH
Simplified Lennard-Jones	SLJ		General Lennard-Jones				GLJ
Simplified Kihara	SK		General Kihara				GK
A. Enhancement Factors at 1400K ($P_0 = 0.00184$ atm)							
$P, \text{ atm}$	EXP	SBB	SBWR	SMH	SLJ	SK	
5	1.1414	1.1798	1.2032	1.1694	1.1713	1.1780	
10	1.3043	1.4060	1.4619	1.3793	1.3840	1.4006	
20	1.9565	2.0777	2.2388	1.9848	2.0007	2.0546	
30	3.0978	3.3284	3.6779	3.0491	3.1032	3.2509	
40	5.4348	6.2615	6.7958	5.2396	5.5577	6.0268	
50	11.685	19.223	15.291	10.809	19.244	28.063	
60	25.435	76.621	34.873	24.017	82.602	113.99	
70	42.609	151.72	55.960	42.088	137.93	190.09	
80	57.391	240.78	75.633	57.753	197.14	273.39	
90	72.391	345.41	94.761	72.007	261.80	365.68	
100	88.043	466.58	113.68	85.761	332.20	467.42	
110	103.42	605.00	132.54	99.335	408.56	578.79	
120	119.34	761.31	151.34	112.86	490.77	699.64	
130	135.65	936.10	170.07	126.36	578.77	830.22	
140	152.17	1130.0	188.72	139.91	672.61	970.38	
150	169.56	1343.4	207.27	153.47	772.24	- *	
160	186.96	1577.5	225.69	167.05	877.48	-	
170	204.73	1832.0	243.95	180.66	988.43	-	
180	222.06	2107.9	262.02	194.28	1105.1	-	
190	239.56	2405.6	279.91	207.90	1227.1	-	
200	256.52	2728.5	297.55	221.52	1354.8	-	
$P, \text{ atm}$	EXP	GBB	GBWR	GMH	GLJ	GK	
5	1.1414	1.1798	1.2034	1.1692	1.1712	1.1779	
10	1.3043	1.4060	1.4622	1.3791	1.3840	1.4006	
20	1.9565	2.0777	2.2398	1.9846	2.0008	2.0548	
30	3.0978	3.3284	3.6815	3.0498	3.1042	3.2525	
40	5.4348	6.2615	6.8150	5.2450	5.5665	6.0408	
50	11.685	19.223	15.462	10.864	20.422	33.739	
60	25.435	88.208	35.780	24.379	92.737	140.60	
70	42.609	184.73	57.538	42.789	157.15	246.52	
80	57.391	310.02	77.829	58.545	228.76	377.03	
90	72.391	472.83	97.590	72.872	309.85	543.41	
100	88.043	684.94	117.18	86.699	401.59	765.21	
110	103.42	965.52	136.70	100.35	504.99	1099.2	
120	119.34	1352.0	156.18	113.94	621.15	- *	
130	135.65	1948.9	175.62	127.52	751.40	-	
140	152.17	10031	194.98	141.12	897.39	-	
150	169.56	- *	214.24	154.74	1061.2	-	
160	186.96	-	233.37	168.39	1245.7	-	
170	204.73	-	252.34	182.05	1454.7	-	
180	222.06	-	271.12	195.71	1693.9	-	
190	239.56	-	289.70	209.38	1972.2	-	
200	256.52	-	308.05	223.05	2305.0	-	

* Solution failed to converge

TABLE XVI (CONT'D)

B. Enhancement Factors at 150°K ($P_1^0 = 0.00841$ atm)

$P, \text{ atm}$	EXP	SBB	SBWR	SMH	SLJ	SK
5	1.1415	1.1459	1.1595	1.1373	1.1410	1.1435
10	1.2723	1.3208	1.3528	1.2998	1.3085	1.3142
20	1.6409	1.7940	1.8779	1.7303	1.7545	1.7708
30	2.2830	2.5310	2.6935	2.3778	2.4328	2.4691
40	3.3769	3.7643	4.0288	3.4062	3.5388	3.6158
50	5.5886	6.0254	6.3326	5.1287	5.5379	5.7050
60	9.1320	10.512	10.357	8.0561	9.6743	10.069
70	14.316	18.985	16.690	12.722	18.379	19.370
80	20.832	31.940	24.648	19.400	31.580	33.730
90	27.717	48.541	33.035	27.648	46.808	50.535
100	35.077	68.186	41.362	36.145	63.195	68.797
110	42.509	90.642	49.559	44.278	80.588	88.321
120	49.655	115.82	57.635	52.088	98.945	109.05
130	57.194	143.71	65.606	59.700	118.23	130.94
140	64.590	174.29	73.476	67.165	138.43	153.97
150	72.235	207.57	81.246	74.528	159.49	178.07
160	79.905	243.57	88.916	81.835	181.41	203.25
170	87.729	282.30	96.482	89.118	204.12	229.43
180	95.672	323.79	103.94	96.299	227.64	256.63
190	103.47	368.04	111.29	103.48	251.92	284.79
200	111.30	415.08	118.52	110.64	276.94	313.90

$P, \text{ atm}$	EXP	GBB	GBWR	GMH	GLJ	GK
5	1.1415	1.1455	1.1602	1.1364	1.1406	1.1431
10	1.2723	1.3204	1.3537	1.2989	1.3081	1.3139
20	1.6409	1.7940	1.8800	1.7294	1.7545	1.7709
30	2.2830	2.5310	2.6989	2.3776	2.4338	2.4709
40	3.3769	3.7643	4.0440	3.4088	3.5442	3.6235
50	5.5886	6.0628	6.3812	5.1430	5.5640	5.7397
60	9.1320	10.703	10.522	8.1127	9.8270	10.265
70	14.316	19.831	17.156	12.900	19.211	20.461
80	20.832	34.447	25.567	19.849	33.914	37.007
90	27.717	53.967	34.420	28.447	51.120	57.011
100	35.077	78.009	43.206	37.189	69.924	79.494
110	42.509	106.63	51.860	45.479	90.213	104.40
120	49.655	140.08	60.394	53.411	111.99	131.83
130	57.194	178.75	68.822	61.119	135.26	161.93
140	64.590	223.12	77.151	68.684	160.05	194.87
150	72.235	273.75	85.380	76.151	186.36	230.88
160	79.905	331.32	93.506	83.546	214.22	270.20
170	87.729	396.93	101.52	90.884	243.64	313.15
180	95.672	470.66	109.43	98.171	274.64	360.13
190	103.47	554.59	117.22	105.41	307.25	411.68
200	111.30	649.89	124.88	112.61	341.50	468.49

TABLE XVI (CONT'D)

C. Enhancement Factors at 160°K ($P_1^0 = 0.0314$ atm)

P, atm	EXP	SBB	SBWR	SMH	SLJ	SK
5	1.1131	1.1213	1.1274	1.1151	1.1180	1.1187
10	1.2580	1.2607	1.2773	1.2458	1.2529	1.2542
20	1.5987	1.6148	1.6579	1.5731	1.5914	1.5942
30	2.0732	2.1102	2.1890	2.0213	2.0570	2.0620
40	2.7643	2.8235	2.9457	2.6491	2.7160	2.7232
50	3.8694	3.8787	4.0411	3.5436	3.6780	3.6864
60	5.3503	5.4647	5.6280	4.8205	5.1226	5.1266
70	7.4013	7.8216	7.8605	6.6031	7.3107	7.2955
80	10.013	11.156	10.800	8.9896	10.504	10.447
90	13.213	15.536	14.328	12.048	14.755	14.647
100	17.006	20.902	18.207	15.806	19.870	19.728
110	20.844	27.153	22.226	20.168	25.594	25.450
120	24.841	34.206	26.270	24.865	31.755	31.643
130	28.898	42.000	30.282	29.625	38.257	38.203
140	33.038	50.497	34.242	34.319	45.033	45.064
150	37.452	59.669	38.143	38.925	52.056	52.194
160	41.834	69.497	41.980	43.427	59.305	59.567
170	46.344	79.967	45.752	47.837	66.741	67.146
180	50.732	91.067	49.457	52.203	74.368	74.933
190	55.185	102.79	53.096	56.496	82.169	82.908
200	59.682	115.13	56.665	60.749	90.131	91.060

P, atm	EXP	GBB	GBWR	GMH	GLJ	GK
5	1.1131	1.1199	1.1292	1.1125	1.1167	1.1174
10	1.2580	1.2595	1.2797	1.2431	1.2517	1.2531
20	1.5987	1.6142	1.6626	1.5704	1.5906	1.5938
30	2.0732	2.1118	2.1982	2.0191	2.0576	2.0636
40	2.7643	2.8311	2.9651	2.6493	2.7209	2.7304
50	3.8694	3.9032	4.0840	3.5512	3.6950	3.7084
60	5.3503	5.5362	5.7246	4.8479	5.1742	5.1892
70	7.4013	8.0162	8.0724	6.6760	7.4587	7.4680
80	10.013	11.629	11.220	9.1543	10.884	10.883
90	13.213	16.535	15.052	12.378	15.576	15.593
100	17.006	22.744	19.289	16.403	21.333	21.458
110	20.844	30.207	23.685	21.108	27.867	28.226
120	24.841	38.888	28.104	26.131	34.982	35.713
130	28.898	48.773	32.489	31.149	42.568	43.813
140	33.038	59.877	36.819	36.041	50.565	52.471
150	37.452	72.227	41.085	40.799	58.939	61.656
160	41.834	85.864	45.283	45.444	67.666	71.353
170	46.344	100.84	49.409	49.995	76.729	81.553
180	50.732	117.21	53.464	54.470	86.115	92.252
190	55.185	135.03	57.444	58.878	95.813	103.45
200	59.682	154.39	61.348	63.228	105.81	115.15

TABLE XVI (CONT'D)

D. Enhancement Factors at 170°K ($P_1^0 = 0.0993$ atm)

P_1 atm	EXP	SBB	SBWR	SMH	SLJ	SK
5	1.1078	1.1037	1.1020	1.1001	1.1025	1.1012
10	1.2185	1.2177	1.2215	1.2085	1.2149	1.2117
20	1.4804	1.4939	1.5107	1.4689	1.4855	1.4764
30	1.8369	1.8536	1.8862	1.8038	1.8344	1.8156
40	2.3041	2.3275	2.3776	2.2385	2.2896	2.2553
50	2.9204	2.9569	3.0234	2.8054	2.8899	2.8303
60	3.7341	3.7938	3.8689	3.5423	3.6867	3.5858
70	4.8288	4.8956	4.9583	4.4872	4.7422	4.5754
80	6.1229	6.3134	6.3209	5.6711	6.1191	5.8508
90	7.7946	8.0778	7.9549	7.1166	7.8594	7.4457
100	9.7080	10.193	9.8224	8.8403	9.9657	9.3600
110	11.742	12.642	11.860	10.856	12.401	11.561
120	13.897	15.400	13.999	13.166	15.107	14.000
130	16.103	18.442	16.185	15.741	18.029	16.626
140	18.469	21.744	18.382	18.512	21.122	19.402
150	20.846	25.290	20.565	21.391	24.352	22.295
160	23.525	29.064	22.724	24.306	27.694	25.284
170	26.022	33.057	24.849	27.198	31.131	28.353
180	28.640	37.257	26.938	30.059	34.650	31.489
190	31.188	41.659	28.987	32.887	38.234	34.680
200	33.837	46.255	30.996	35.655	41.884	37.921

P_1 atm	EXP	GBB	GBWR	GMH	GLJ	GK
5	1.1078	1.1001	1.1063	1.0935	1.0988	1.0973
10	1.2185	1.2143	1.2270	1.2016	1.2114	1.2034
20	1.4804	1.4916	1.5200	1.4616	1.4827	1.4742
30	1.8369	1.8541	1.9022	1.7968	1.8335	1.8161
40	2.3041	2.3347	2.4060	2.2336	2.2937	2.2618
50	2.9204	2.9791	3.0748	2.8067	2.9053	2.8496
60	3.7341	3.8480	3.9626	3.5578	3.7267	3.6318
70	4.8288	5.0146	5.1275	4.5313	4.8336	4.6744
80	6.1229	6.5540	6.6152	5.7674	6.3108	6.0499
90	7.7946	8.5266	8.4369	7.2994	8.2282	7.8175
100	9.7080	10.966	10.555	9.1591	10.610	9.9997
110	11.742	13.880	12.891	11.378	13.427	12.575
120	13.897	17.264	15.356	13.971	16.612	15.493
130	16.103	21.106	17.879	16.895	20.099	18.700
140	18.469	25.399	20.413	20.033	23.830	22.150
150	20.846	30.139	22.932	23.246	27.765	25.806
160	23.525	35.326	25.419	26.438	31.874	29.645
170	26.022	40.961	27.866	29.569	36.135	33.648
180	28.640	47.051	30.270	32.628	40.534	37.800
190	31.188	53.602	32.627	35.618	45.056	42.092
200	33.837	60.622	34.934	38.545	49.695	46.516

TABLE XVI (CONT'D)

E. Enhancement Factor at 180°K ($P_1^0 = 0.274$ atm)

P, atm	EXP	SBB	SBWR	SMH	SLJ	SK
5	1.0858	1.0924	1.0794	1.0915	1.0915	1.0901
10	1.1861	1.1875	1.1767	1.1836	1.1859	1.1821
20	1.3869	1.4101	1.4039	1.3980	1.4057	1.3952
30	1.6533	1.6853	1.6839	1.6614	1.6754	1.6549
40	2.0000	2.0270	2.0300	1.9862	2.0078	1.9726
50	2.4270	2.4516	2.4575	2.3866	2.4187	2.3619
60	2.9562	2.9776	2.9830	2.8775	2.9260	2.8380
70	3.6022	3.6232	3.6216	3.4731	3.5493	3.4167
80	4.3796	4.4042	4.3843	4.1850	4.3067	4.1122
90	5.2833	5.3305	5.2739	5.0210	5.2114	4.9342
100	6.3869	6.4050	6.2835	5.9865	6.2679	5.8848
110	7.6277	7.6243	7.3967	7.0857	7.4713	6.9584
120	8.9781	8.9805	8.5913	8.3229	8.8080	8.1431
130	10.485	10.464	9.8421	9.7012	10.260	9.4231
140	12.058	12.064	11.126	11.219	11.810	10.783
150	13.741	13.772	12.425	12.866	13.439	12.207
160	15.591	15.581	13.726	14.619	15.133	13.683
170	17.496	17.482	15.017	16.447	16.881	15.201
180	19.380	19.470	16.293	18.320	18.673	16.753
190	21.219	21.542	17.550	20.210	20.501	18.333
200	22.993	23.691	18.785	22.096	22.360	19.934

P, atm	EXP	GBB	GBWR	GMH	GLJ	GK
5	1.0858	1.0838	1.0884	1.0766	1.0828	1.0818
10	1.1861	1.1792	1.1879	1.1679	1.1774	1.1741
20	1.3869	1.4033	1.4212	1.3811	1.3980	1.3887
30	1.6533	1.6822	1.7111	1.6444	1.6703	1.6518
40	2.0000	2.0318	2.0731	1.9709	2.0085	1.9764
50	2.4270	2.4720	2.5265	2.3767	2.4310	2.3786
60	2.9562	3.0266	3.0937	2.8797	2.9601	2.8778
70	3.6022	3.7226	3.7981	3.4982	3.6224	3.4962
80	4.3796	4.5875	4.6605	4.2488	4.4457	4.2567
90	5.2833	5.6453	5.6933	5.1459	5.4552	5.1792
100	6.3869	6.9142	6.8952	6.2024	6.6671	6.2760
110	7.6277	8.4043	8.2491	7.4324	8.0840	7.5491
120	8.9781	10.120	9.7244	8.8525	9.6946	8.9899
130	10.485	12.060	11.284	10.479	11.478	10.582
140	12.058	14.222	12.892	12.317	13.409	12.308
150	13.741	16.605	14.520	14.345	15.464	14.147
160	15.591	19.206	16.149	16.508	17.622	16.084
170	17.496	22.024	17.763	18.734	19.867	18.105
180	19.380	25.058	19.355	20.961	22.185	20.200
190	21.219	28.309	20.919	23.154	24.566	22.359
200	22.993	31.776	22.452	25.298	27.001	24.575

TABLE XVI (CONT'D)

F. Enhancement Factors at 190°K ($P_1^0 = 0.679$ atm)

P_2 , atm	EXP	SBB	SBWR	SMH	SLJ	SK
10	1.1458	1.1688	1.1364	1.1697	1.1680	1.1640
20	1.3042	1.3529	1.3189	1.3510	1.3514	1.3402
30	1.5641	1.5719	1.5353	1.5665	1.5684	1.5472
40	1.8321	1.8325	1.7921	1.8223	1.8254	1.7903
50	2.1576	2.1421	2.0959	2.1254	2.1292	2.0753
60	2.5184	1.5080	2.4536	2.4826	2.4875	2.4081
70	2.9249	2.9371	2.8709	2.8997	2.9075	2.7943
80	3.4521	3.4350	3.3521	3.3815	3.3954	3.2385
90	4.0427	4.0049	3.8987	3.9309	3.9557	3.7433
100	4.7717	4.6477	4.5086	4.5495	4.5900	4.3092
110	5.6215	5.3619	5.1768	5.2384	5.2970	4.9341
120	6.6274	6.1443	5.8952	5.9983	6.0725	5.6140
130	7.6583	6.9906	6.6540	6.8306	6.9100	6.3429
140	8.7629	7.8960	7.4428	7.7362	7.8021	7.1145
150	9.8969	8.8559	8.2513	8.7157	8.7412	7.9218
160	11.075	9.8660	9.0708	9.7674	9.7199	8.7591
170	12.268	10.922	9.8937	10.887	10.732	9.6205
180	13.520	12.022	10.714	12.065	11.771	10.502
190	14.831	13.161	11.528	13.290	12.832	11.398
200	16.141	14.337	12.331	14.550	13.912	12.307

P_2 , atm	EXP	GBB	GBWR	GMH	GLJ	GK
10	1.1458	1.1504	1.1572	1.1375	1.1491	1.1460
20	1.3402	1.3367	1.3494	1.3164	1.3337	1.3243
30	1.5641	1.5605	1.5800	1.5306	1.5540	1.5356
40	1.8321	1.8307	1.8577	1.7873	1.8178	1.7867
50	2.1576	2.1575	2.1928	2.0952	2.1346	2.0857
60	2.5184	2.5527	2.5967	2.4632	2.5150	2.4415
70	2.9249	3.0293	3.0813	2.9006	2.9712	2.8638
80	3.4521	3.6004	3.6581	3.4159	3.5156	3.3623
90	4.0427	4.2738	4.3358	4.0164	4.1598	3.9455
100	4.7717	5.0751	5.1185	4.7090	4.9126	4.6195
110	5.6215	5.9979	6.0040	5.5010	5.7785	5.3870
120	6.6274	7.0530	6.9830	6.4016	6.7564	6.2468
130	7.6583	8.2442	8.0397	7.4226	7.8402	7.1937
140	8.7629	9.5737	9.1542	8.5778	9.0196	8.2203
150	9.8969	11.043	10.306	9.8794	10.823	9.3177
160	11.075	12.653	11.476	11.330	11.617	10.477
170	12.268	14.404	12.650	12.911	13.011	11.689
180	13.520	16.296	13.817	14.578	14.455	12.947
190	14.831	18.330	14.970	16.276	15.940	14.244
200	16.141	20.506	16.101	17.960	17.459	15.574

used in the analytical solutions are those obtained using the c_j of Kihara⁽⁶⁷⁾.

The equations of state employed here are intended for application below various reduced densities, where the term reduced density has its usual definition of the ratio of actual density to critical density. For the virial equation of state through the third virial and the Beattie-Bridgeman equation, the upper limit of applicability is generally considered to be a reduced density of approximately 0.8. On the other hand, the Benedict-Webb-Rubin and Martin-Hou equations are applicable to at least a reduced density of 1.5 and in some instances have been extended to 2.0 times the critical density with good results. Due to the wide range of applicability of these equations, it will be convenient to briefly discuss the comparison of the analytical and experimental results at each temperature while noting the pressures at which certain values of reduced densities occur.

At 140°K a reduced density of 0.8 occurs at approximately 50 atmospheres while a value of 1.0 is reached at approximately 55 atmospheres, 1.5 at 80 and a maximum of just over 2 is attained at 200 atmospheres. From Figure 27 it will be noted that the solutions based on the virial and Beattie-Bridgeman equations begin to deviate sharply from the experimental results at approximately 50 atmospheres. This is as expected since this pressure corresponds to the upper limit of applicability of these equations. Figure 27 also shows that the solutions based on the Benedict-Webb-Rubin

TABLE XVII

EFFECT OF QUADRUPOLE CORRECTION ON ENHANCEMENT FACTORS
CALCULATED FROM VIRIAL EQUATION OF STATE
USING THE LENNARD-JONES (6-12) POTENTIAL

T °K	P atm	Enhancement Factor				
		Exp.	General Solution		Simplified Solution	
			Without Quadrupole Correction	With Quadrupole Correction	Without Quadrupole Correction	With Quadrupole Correction
190	50	2.1576	2.1346	2.1706	2.1292	2.1628
	100	4.7717	4.9126	5.1763	4.5900	4.7822
	150	9.8969	10.283	11.342	8.741	9.3241
	200	16.141	17.459	19.871	13.912	15.042
180	50	2.4270	2.4310	2.4831	2.4187	2.4682
	100	6.3869	6.6671	7.1572	6.2679	6.6483
	150	13.741	15.464	17.463	13.439	14.692
	200	22.993	27.001	31.320	22.360	24.768
170	50	2.9204	2.9053	2.9793	2.8899	2.9613
	100	9.7080	10.610	11.730	9.9657	10.852
	150	20.846	27.765	32.123	24.352	27.279
	200	33.837	49.695	58.526	41.884	47.232
160	50	3.8694	3.6950	3.8772	3.6780	3.8560
	70	7.4013	7.4587	8.2993	7.3107	8.0829
	100	17.006	21.333	26.205	19.870	23.725
	150	37.452	58.939	75.267	52.056	63.423
150	50	5.5886	5.5640	6.0216	5.5379	5.9853
	70	14.316	19.211	25.885	18.379	23.968
	90	27.717	51.120	69.832	46.808	61.294
	110	42.509	90.213	122.28	80.588	103.84
140	20	1.9565	2.0008	2.0381	2.0007	2.0380
	40	5.4348	5.5665	6.1089	5.5577	6.0954
	60	25.435	92.737	167.68	82.602	138.11
	80	57.391	157.15	- *	137.92	- *

* Solution failed to converge

TABLE XVIII

EFFECT OF QUADRUPOLE CORRECTION ON ENHANCEMENT FACTORS
CALCULATED FROM VIRIAL EQUATION OF STATE
USING THE KIHARA POTENTIAL

T °K	P atm	Enhancement Factor				
		Exp.	General Solution		Simplified Solution	
			Without Quadrupole Correction	With Quadrupole Correction	Without Quadrupole Correction	With Quadrupole Correction
190	50	2.1576	2.0857	2.1803	2.0753	2.1592
	100	4.7717	4.6195	5.2586	4.3092	4.7378
	150	9.8969	9.3177	12.089	7.9218	9.1737
	200	16.141	15.574	23.458	12.307	14.755
180	50	2.4270	2.3786	2.5125	2.3619	2.4855
	100	6.3869	6.2760	7.4298	5.8848	6.7237
	150	13.741	14.147	19.430	12.207	14.944
	200	22.993	24.575	38.806	19.934	25.359
170	50	2.9204	2.8496	3.0639	2.8303	3.0388
	100	9.7080	9.9997	12.880	9.3600	11.497
	150	20.846	25.806	39.809	22.295	29.763
	200	33.837	46.516	84.705	37.921	52.553
160	50	3.8694	3.7084	4.1218	3.6864	4.0857
	70	7.4013	7.4680	9.3151	7.2955	8.9423
	100	17.006	21.458	33.597	19.728	28.117
	150	37.452	61.656	121.05	52.194	79.412
150	50	5.5886	5.7397	6.9458	5.7050	6.8729
	70	14.316	20.461	40.547	19.370	34.009
	90	27.717	57.011	132.57	50.535	92.232
	110	42.509	104.40	301.26	88.321	160.93
140	20	1.9565	2.0548	2.1723	2.0546	2.1720
	40	5.4348	6.0408	7.6572	6.0268	7.6210
	60	25.435	140.60	- *	113.99	332.50
	80	57.391	377.03	- *	273.39	- *

* Solution failed to converge

and Martin-Hou equations compare quite well with the experimental results to 200 atmospheres, the maximum pressure investigated, with the latter giving extremely good results. This behavior is consistent with the expected range of applicability of these equations. It will also be noted that the Benedict-Webb-Rubin solution runs consistently higher but parallel to the experimental results. This would tend to indicate that a slight change in the constants for this equation might improve the comparison between the experimental and analytical results.

A reduced density of 0.8 occurs at approximately 65 atmospheres at 150°K with a value of 1.0 being reached near 75 atmospheres, 1.5 at 120 and a maximum of approximately 1.8 occurring at 200 atmospheres. At this temperature, the point at which the virial and Beattie-Bridgeman solutions begin to deviate noticeably from the experimental results is not as pronounced as at 140°K . Nevertheless, it would appear from Figure 28 that the results of these solutions begin to differ appreciably from those determined experimentally at approximately 0.8 reduced density (65 atmospheres) which is consistent with the range of applicability of the two equations. Once again the results based on the Benedict-Webb-Rubin and Martin-Hou equations compare reasonably well with the experimental values over the entire pressure range covered with the latter still being the better. As at 140°K , the Benedict-Webb-Rubin solution runs higher but parallel to the experimental results.

A maximum reduced density of approximately 1.7 occurs at 200 atmospheres for the 160°K isotherm with a value of 0.8 being reached at approximately 75 atmospheres, 1.0 at 100 and 1.5 at 150. At this temperature it becomes increasingly more difficult to detect the point at which the results based on the virial and Beattie-Bridgeman equations begin to deviate appreciably from the experimental. However, from Figure 29 it would seem valid to state that these solutions compare favorably with the experimental results to a reduced density of 0.8 (75 atmospheres). As before, the solutions based on the other two equations of state give results comparable to those obtained experimentally over the entire pressure range. However, as seen in Figure 29, the enhancement factor based on the Martin-Hou equation is slightly larger than that for the Benedict-Webb-Rubin from 150 atmospheres and above, this being the first point at which these two solutions cross. The Benedict-Webb-Rubin still lies above the experimental results but the two tend to approach each other at the higher pressures.

A reduced density of 0.8 occurs at approximately 85, 100, and 110 atmospheres, a value of 1.0 at 105, 125 and 140 and a maximum value of approximately 1.5, 1.4 and 1.3 is reached at 200 atmospheres for the three remaining isotherms, 170° , 180° and 190°K , respectively. The analytical solutions are closer to one another at these three temperatures than at the lower temperatures and in some cases the results cross one another as evidenced in Figures 30-32. These figures also show that the solutions

based on the virial and Beattie-Bridgeman equations lie consistently above the experimental results at the higher pressures as would be expected since this region is beyond the range of applicability of the equations. There is one exception to the foregoing statement in that the General Kihara solution at 190°K lies below the experimental results. The Benedict-Webb-Rubin and Martin-Hou solutions still give results comparable to those obtained experimentally over the entire pressure range covered at the three temperatures under consideration. It might also be noted from Figures 30-32 that these latter solutions cross one another at the higher pressures.

Examination of Table XVI reveals that at higher pressures the simplified solutions produce consistently lower values than do the general solutions. This may be attributed to the fact that at high pressures the equations of state begin to diverge in such a direction that the calculated volumes are too small at a given temperature and pressure, resulting in enhancement factors too large. The simplification to the solution introduces an error in the opposite direction such that the simplified solutions lie below the general. This behavior causes the simplified solutions to be closer to the experimental results in some instances than are the general solutions, especially at high densities.

Based upon the results presented in Figures 27-32 and Table XVI, it appears that the solutions based on the Benedict-Webb-Rubin and Martin-Hou equations are superior to all others. This is reasonable since both equations

are intended for higher density application than the other equations of state used here. Regarding the relative merit of the two equations, the Martin-Hou equation predicts values closer to the experimental than does the Benedict-Webb-Rubin equation for the entire pressure and temperature range, with the exception of the higher pressures at 170⁰-190⁰K. It may be noted that the Martin-Hou equation predicts values less than the experimental data over a wide temperature and pressure range. However, the Benedict-Webb-Rubin equation gives values which are so much larger than the experimental data over this same pressure and temperature range that the results based on the Martin-Hou equation are better. Thus, after considering the entire pressure and temperature range covered in this investigation, it is concluded that the Martin-Hou equation of state is superior to that of Benedict-Webb-Rubin in ability to analytically predict the experimental data of the system investigated here.

It should be recalled that the analytical solutions were made under the assumption that the solid phase was pure carbon dioxide. If the nitrogen dissolved in the solid is taken into account, the analytical solutions would result in smaller enhancement factors. Unfortunately, there is presently no solubility data available to estimate the magnitude of this correction. If and when such data become available, it would be of interest to make the necessary corrections in the analytical solutions in order to determine the effect on the comparison between the analytical and experimental results.

APPENDIX A

EXPERIMENTAL DATA

The data presented here is given with the following notation and units:

TIME:	Minutes
TEMP:	Nominal Equilibrium Temperature, °K
PRESS:	Nominal Equilibrium Pressure, ATM
ANAL:	Gas Analyzer Reading, Percent CO ₂
ERR:	Analyzer Error Limits, ± Percent CO ₂
TCL:	Temperature Control Limits, ± °K
SPR:	System Flow Rate, FT/SEC through equilibrium coil
AFR:	Analyzer Flow Rate, SCFH
P1:	Equilibrium Pressure, PSIA
P2:	Main Stream Exhaust Pressure, PSIG
P3:	Analyzer Exit Pressure, PSIG
T ₀ :	Resistance Thermometer Reading, °K
T ₁ :	Equilibrium Cell Temperature, °K
T ₂ :	Temperature 2 inches Below Top Plate, °K
T ₃ :	Temperature ½ inch Below Top Plate, °K
T ₄ :	Temperature 4 inches Below Top Plate, °K
T ₅ :	Temperature in Center of Cryostat, °K
T ₆ :	Temperature at Bottom of Equilibrium Coil, °K
T ₇ :	Temperature Adjacent to Resistance Thermometer, °K
T ₈ :	Inlet Temperature of Heat Exchanger Coolant, °K
T ₉ :	Test Gas Exit Temperature From Heat Exchanger, °K
T ₁₀ :	Equilibrium Pressure Gauge Temperature, °F
T ₁₁ :	Analyzer Exit Temperature, °F
T ₁₂ :	Main Stream Exhaust Temperature, °F

Note: Runs marked (*) were performed by picking up solid carbon dioxide in equilibrium cell with nitrogen gas.

RUN	A1	A2*	A3	A4	A5*	A6	A7	A8	A9*	A10
TIME	45	30	30	30	30	30	30	30	30	30
TEMP	190	190	190	190	190	190	190	190	190	190
PRESS	100	80	100	200	200	180	180	200	180	180
ANAL	3.24	2.92	3.22	5.48	5.48	5.10	5.10	5.50	5.10	5.08
ERR	0.02	0.02	0.02	0.03	0.03	0.03	0.03	0.03	0.03	0.03
TCL	0.05	0.05	0.05	0.05	0.05	0.04	0.05	0.05	0.04	0.05
SFR	0.0500	0.0355	0.0222	0.0470	0.0201	0.0269	0.0269	0.0201	0.0236	0.0303
AFR	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5
P1	1470	1176	1470	2940	2940	2646	2646	2940	2646	2646
P2	3.0	3.0	3.0	3.0	3.0	3.0	3.0	3.0	3.0	3.0
P3	3.0	3.0	3.0	3.0	3.0	3.0	3.0	3.0	3.0	3.0
T ₀	190.00	190.00	190.00	190.00	190.00	190.00	190.00	190.00	190.00	190.00
T ₁	Thermocouple not functioning. See section VII-C.									
T ₂	190.30	190.03	190.20	190.40	190.13	190.17	190.27	190.27	190.20	190.23
T ₃	199.20	200.30	198.50	197.90	198.90	198.00	198.70	202.40	202.40	201.60
T ₄	190.03	189.93	190.03	190.00	190.00	190.00	190.03	190.03	190.03	190.07
T ₅	190.00	189.93	190.00	190.07	190.03	190.00	190.00	190.03	190.03	190.03
T ₆	190.03	189.93	190.00	190.00	190.03	190.03	189.93	190.03	190.00	190.03
T ₇	190.00	190.00	190.00	190.00	190.00	190.00	190.00	190.00	190.00	190.00
T ₈	113.5	120.5	191.5	213.5	220.5	224.5	225.5	82.0	82.0	146.5
T ₉	191.5	192.0	192.0	192.5	193.0	193.5	195.0	200.0	199.5	201.0
T ₁₀	74.5	76.5	78.0	80.0	80.0	82.5	81.0	86.0	86.0	86.0
T ₁₁	77.0	79.0	81.0	83.0	81.0	84.0	84.0	87.5	88.0	89.0
T ₁₂	55.0	63.0	62.0	62.0	61.5	61.5	62.5	65.0	66.5	66.5

RUN	A11	A12	A13	A14	A15	A16	A17*	A18*	A19*	A20
TIME	25	35	40	30	45	30	35	30	25	45
TEMP	190	190	190	190	190	190	190	190	190	190
PRESS	190	190	110	120	170	160	110	60	80	150
ANAL	5.30	5.30	3.47	3.75	4.90	4.70	3.47	2.85	2.94	4.48
ERR	0.03	0.03	0.03	0.03	0.03	0.03	0.02	0.02	0.02	0.03
TCL	0.04	0.04	0.04	0.05	0.05	0.05	0.05	0.05	0.05	0.04
SFR	0.0290	0.0225	0.0504	0.0508	0.0486	0.0288	0.0252	0.0301	0.0496	0.0492
AFR	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5
P1	2793	2793	1617	1764	2499	2352	1617	822	1176	2205
P2	3.0	3.0	3.0	3.0	3.0	3.0	3.0	3.0	3.0	3.0
P3	3.0	3.0	3.0	3.0	3.0	3.0	3.0	3.0	3.0	3.0
T ₀	190.00	190.00	190.00	190.00	190.00	190.00	190.00	190.00	190.00	190.00
T ₁	Thermocouple not functioning. See section VII-C.									
T ₂	190.07	190.13	190.27	190.33	190.57	190.23	190.30	190.07	190.13	190.47
T ₃	202.40	206.40	204.90	204.60	205.40	207.60	206.90	211.2	209.10	206.20
T ₄	190.03	190.03	190.00	190.00	190.03	190.03	189.97	190.00	189.97	190.03
T ₅	190.03	190.00	190.00	190.00	190.00	190.03	190.00	190.00	189.97	190.00
T ₆	190.00	190.00	189.97	189.97	190.00	190.00	189.97	189.93	189.97	190.00
T ₇	190.00	190.00	190.00	190.00	190.00	190.00	190.00	190.00	190.00	190.00
T ₈	151.0	83.0	129.5	111.0	116.5	126.5	117.0	110.0	112.0	175.5
T ₉	199.5	199.0	199.0	198.5	198.0	200.5	198.0	199.0	200.0	199.0
T ₁₀	86.0	84.5	78.0	76.0	84.0	86.0	88.0	87.5	88.0	87.5
T ₁₁	89.0	86.0	81.0	78.5	86.0	88.5	90.0	90.0	91.0	90.5
T ₁₂	66.5	64.5	60.5	58.0	59.0	66.0	66.5	66.5	66.5	66.5

RUN	A21	A22	A23	A24	A25	A26	A27	A28	A29*	A30*
TIME	35	40	35	30	30	25	40	35	30	35
TEMP	190	190	190	190	190	180	180	180	180	180
PRESS	140	130	160	140	50	100	190	190	200	180
ANAL	4.25	4.00	4.70	4.25	2.93	1.75	3.07	3.05	3.15	2.95
ERR	0.03	0.03	0.03	0.03	0.02	0.02	0.02	0.02	0.02	0.02
TCL	0.04	0.04	0.05	0.04	0.04	0.03	0.04	0.04	0.05	0.05
SFR	0.0481	0.0476	0.0504	0.0281	0.0495	0.0498	0.0447	0.0298	0.0290	0.0481
AFR	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5
P1	2058	1911	2532	2058	735	1470	2793	2793	2940	2646
P2	3.0	3.0	3.0	3.0	3.0	3.0	3.0	3.0	3.0	3.0
P3	3.0	3.0	3.0	3.0	3.0	3.0	3.0	3.0	3.0	3.0
T ₀	190.00	190.00	190.00	190.00	190.00	180.00	180.00	180.00	180.00	180.00
T ₁			Thermocouple not functioning. See section VII-C.							
T ₂	190.27	190.40	190.30	190.23	190.07	180.37	180.63	180.27	180.21	180.47
T ₃	207.00	207.20	206.60	207.90	212.20	199.80	197.80	197.40	196.90	196.40
T ₄	189.97	190.00	189.97	189.97	190.00	179.93	180.07	180.10	180.00	180.07
T ₅	190.00	189.97	189.97	189.97	190.00	179.97	180.03	180.00	180.00	180.00
T ₆	190.00	190.00	189.97	189.97	190.00	179.90	179.97	179.93	179.97	180.00
T ₇	190.00	190.00	190.00	190.00	190.00	180.00	180.00	180.00	180.00	180.00
T ₈	100.5	96.0	124.0	120.00	123.0	81.5	81.0	115.5	81.0	82.0
T ₉	197.5	192.5	189.5	189.5	186.0	187.0	187.5	188.0	189.5	190.0
T ₁₀	83.5	83.0	83.0	82.5	82.5	83.0	78.5	78.0	77.0	76.0
T ₁₁	86.0	85.5	85.5	85.5	85.5	85.5	81.5	81.0	80.5	80.5
T ₁₂	64.0	64.0	64.5	64.0	64.5	58.0	62.0	59.0	61.5	60.5

RUN	A31	A32	A33	A34*	A35*	A36	A37	A38	A39*	A40*
TIME	40	35	35	30	35	30	35	30	25	25
TEMP	180	180	180	180	180	180	180	180	180	180
PRESS	200	170	160	150	160	140	130	120	140	120
ANAL	3.15	2.82	2.67	2.51	2.67	2.36	2.21	2.05	2.35	2.04
ERR	0.02	0.02	0.02	0.02	0.02	0.02	0.02	0.02	0.02	0.02
TCL	0.04	0.04	0.05	0.04	0.05	0.05	0.04	0.04	0.04	0.05
SFR	0.0493	0.0449	0.0493	0.0484	0.0266	0.0475	0.0466	0.0464	0.0366	0.0337
AFR	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5
P1	2940	2499	2352	2205	2352	2058	1911	1764	2058	1764
P2	3.0	3.0	3.0	3.0	3.0	3.0	3.0	3.0	3.0	3.0
P3	3.0	3.0	3.0	3.0	3.0	3.0	3.0	3.0	3.0	3.0
T ₀	180.00	180.00	180.00	180.00	180.00	180.00	180.00	180.00	180.00	180.00
T ₁	Thermocouple not functioning. See section VII-C.									
T ₂	180.73	180.50	180.33	180.50	180.20	180.40	180.40	180.27	180.30	180.27
T ₃	196.80	197.00	202.60	199.10	199.50	198.80	198.80	198.80	199.00	198.90
T ₄	180.07	180.03	180.03	180.03	180.03	180.10	180.07	180.10	180.07	180.00
T ₅	180.03	180.07	180.00	179.97	180.00	180.03	180.07	180.00	180.00	180.03
T ₆	179.90	179.90	179.93	179.93	179.93	179.97	179.87	179.93	179.93	179.93
T ₇	180.00	180.00	180.00	180.00	180.00	180.00	180.00	180.00	180.00	180.00
T ₈	124.5	119.5	115.5	119.5	91.0	81.5	82.0	82.0	82.0	81.5
T ₉	190.0	189.0	182.0	189.0	189.0	188.0	187.5	187.0	185.5	185.0
T ₁₀	77.0	76.5	76.5	74.0	76.0	76.0	75.5	75.5	75.5	74.5
T ₁₁	80.5	80.0	80.0	79.5	79.5	79.0	79.0	78.5	78.0	78.0
T ₁₂	61.0	61.0	58.5	58.5	55.5	56.5	58.5	58.0	56.0	56.5

RUN	A41*	B42	B43	B44	B45	B46	B47	B48	B49	B50*
TIME	30	25	30	30	35	35	30	25	30	35
TEMP	180	180	180	180	170	170	170	170	170	170
PRESS	110	120	110	100	100	200	190	180	170	200
ANAL	1.90	2.05	1.90	1.75	0.964	1.68	1.63	1.58	1.52	1.68
ERR	0.02	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01
TCL	0.04	0.05	0.04	0.05	0.03	0.02	0.02	0.03	0.02	0.03
SFR	0.0451	0.0464	0.0451	0.0398	0.0497	0.0480	0.0493	0.0475	0.0483	0.0266
AFR	0.5	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0
P1	1617	1764	1617	1470	1470	2940	2793	2646	2499	2940
P2	3.0	3.0	3.0	3.0	3.0	3.0	3.0	3.0	3.0	3.0
P3	3.0	3.0	3.0	3.0	3.0	3.0	3.0	3.0	3.0	3.0
T ₀	180.00	180.00	180.00	180.00	170.00	170.00	170.00	170.00	170.00	170.00
T ₁	Thermocouple not functioning. See section VII-C.									
T ₂	180.30	180.57	180.00	180.33	170.50	170.70	170.77	170.73	170.73	170.37
T ₃	198.70	202.90	207.10	203.20	196.60	195.40	195.10	194.80	194.40	195.80
T ₄	180.07	179.97	179.97	179.97	169.97	170.03	170.03	170.00	170.03	169.97
T ₅	180.03	180.00	180.03	180.00	169.97	170.00	170.00	170.00	170.00	169.97
T ₆	179.93	179.90	179.90	179.97	169.93	169.97	169.97	169.97	170.00	169.93
T ₇	180.00	180.00	180.00	180.00	170.00	170.00	170.00	170.00	170.00	170.00
T ₈	82.0	84.0	83.0	83.0	138.0	90.0	83.0	83.0	83.0	83.0
T ₉	184.0	183.0	185.0	182.0	174.0	177.5	177.5	172.0	172.0	172.5
T ₁₀	74.5	85.5	85.0	86.5	88.5	88.5	88.0	87.5	87.5	89.5
T ₁₁	78.0	87.0	87.5	88.5	90.0	90.0	90.0	90.0	89.5	91.0
T ₁₂	57.5	65.0	67.5	65.0	65.0	65.0	65.0	65.0	65.0	65.0

RUN	B51*	B52*	B53	B54	B55*	B56*	B57*	B58	B59	B60
TIME	30	25	30	35	25	30	25	30	30	35
TEMP	170	170	170	170	170	170	170	170	170	170
PRESS	180	160	160	150	140	120	100	140	120	130
ANAL	1.58	1.46	1.45	1.38	1.31	1.16	0.965	1.31	1.15	1.23
ERR	0.01	0.01	0.01	0.01	0.01	0.01	0.05	0.01	0.01	0.01
TCL	0.03	0.03	0.03	0.03	0.02	0.02	0.02	0.02	0.02	0.02
SFR	0.0278	0.0235	0.0499	0.0494	0.0498	0.0494	0.0346	0.0321	0.0366	0.0428
AFR	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0
P1	2646	2352	2352	2205	2058	1764	1470	2058	1764	1911
P2	3.0	3.0	3.0	3.0	3.0	3.0	3.0	3.0	3.0	3.0
P3	3.0	3.0	3.0	3.0	3.0	3.0	3.0	3.0	3.0	3.0
T ₀	170.00	170.00	170.00	170.00	170.00	170.00	170.00	170.00	170.00	170.00
T ₁	Thermocouple not functioning. See section VII-C.									
T ₂	170.33	170.40	170.70	170.63	170.63	170.60	170.50	170.50	170.60	170.70
T ₃	195.60	196.00	193.10	193.10	194.10	194.10	194.60	194.60	194.00	195.10
T ₄	169.93	169.97	170.03	170.03	170.03	170.00	170.03	170.00	170.00	170.00
T ₅	169.97	170.00	170.00	170.00	170.03	170.03	170.03	170.00	170.00	170.03
T ₆	169.90	169.93	169.93	169.90	169.93	169.90	169.90	169.93	169.93	169.93
T ₇	170.00	170.00	170.00	170.00	170.00	170.00	170.00	170.00	170.00	170.00
T ₈	135.0	133.5	102.0	101.5	82.0	82.0	82.0	82.0	82.0	84.0
T ₉	172.0	172.5	180.5	181.0	172.5	172.0	172.0	172.0	172.0	172.0
T ₁₀	89.5	89.5	89.5	89.5	89.0	88.5	88.5	88.0	88.0	86.0
T ₁₁	91.0	91.5	91.5	91.5	91.0	91.0	91.0	90.5	90.5	88.0
T ₁₂	66.0	66.5	63.0	64.0	66.5	64.0	61.5	64.0	65.0	65.0

RUN	B61	B62	B63*	B64	B65	B66	B67	B68*	B69*	B70*
TIME	30	30	35	30	30	35	30	35	30	35
TEMP	170	160	160	160	160	160	160	160	160	160
PRESS	110	100	200	200	190	180	170	180	160	140
ANAL	1.06	0.534	0.937	0.937	0.912	0.883	0.856	0.885	0.821	0.740
ERR	0.01	0.005	0.005	0.005	0.005	0.005	0.005	0.005	0.005	0.005
TCL	0.02	0.04	0.04	0.04	0.04	0.04	0.04	0.04	0.03	0.03
SFR	0.0492	0.0490	0.0411	0.0291	0.0322	0.0355	0.0365	0.0456	0.0479	0.0401
AFR	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0
P1	1617	1470	2940	2940	2793	2646	2499	2646	2352	2058
P2	3.0	3.0	3.0	3.0	3.0	3.0	3.0	3.0	3.0	3.0
P3	3.0	3.0	3.0	3.0	3.0	3.0	3.0	3.0	3.0	3.0
T ₀	170.00	160.00	160.00	160.00	160.00	160.00	160.00	160.00	160.00	160.00
T ₁	Thermocouple not functioning. See section VII-C.									
T ₂	170.60	161.50	161.46	161.85	161.92	160.77	161.19	161.62	161.50	161.04
T ₃	195.00	184.40	182.80	180.50	180.20	180.20	180.10	179.80	179.50	179.50
T ₄	170.00	160.08	160.08	160.04	160.08	160.04	159.96	160.08	160.11	160.00
T ₅	170.03	160.00	160.00	160.00	159.96	160.00	160.00	160.00	160.00	160.00
T ₆	169.93	160.04	160.00	160.04	160.04	160.00	160.00	160.00	160.00	160.04
T ₇	170.00	160.00	160.00	160.00	160.00	160.00	160.00	160.00	160.00	160.00
T ₈	84.0	97.5	97.5	85.0	84.5	97.0	89.0	84.0	94.0	84.0
T ₉	172.0	162.0	162.0	165.5	165.0	164.5	165.0	158.0	155.0	153.0
T ₁₀	85.0	75.5	78.0	73.0	73.0	69.0	68.5	68.5	68.5	68.0
T ₁₁	88.0	78.0	80.0	75.0	75.0	72.0	72.0	71.5	71.0	71.0
T ₁₂	65.0	59.0	61.0	58.0	58.0	57.5	57.5	57.0	56.5	56.5

RUN	B71*	B72	B73*	B74*	B75*	B76*	B77	B78	B79	B80*
TIME	35	30	35	35	30	25	30	30	35	30
TEMP	160	160	160	160	160	160	160	160	150	150
PRESS	170	160	150	130	120	110	120	140	100	100
ANAL	0.856	0.823	0.784	0.698	0.650	0.595	0.650	0.741	0.295	0.295
ERR	0.005	0.005	0.005	0.005	0.005	0.005	0.005	0.005	0.003	0.003
TCL	0.04	0.04	0.05	0.04	0.04	0.03	0.04	0.04	0.03	0.03
SFR	0.0469	0.0320	0.0336	0.0425	0.0413	0.0445	0.0510	0.0459	0.0364	0.0455
AFR	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0
P1	2499	2352	2205	1911	1764	1617	1764	2058	1470	1470
P2	3.0	3.0	3.0	3.0	3.0	3.0	3.0	3.0	3.0	3.0
P3	3.0	3.0	3.0	3.0	3.0	3.0	3.0	3.0	3.0	3.0
T ₀	160.00	160.00	160.00	160.00	160.00	160.00	160.00	160.00	150.00	150.00
T ₁	Thermocouple not functioning. See section VII-C.									
T ₂	161.61	160.92	160.96	161.03	160.96	160.92	161.35	161.42	150.92	150.96
T ₃	179.80	179.60	179.80	179.30	179.30	179.40	179.00	178.90	171.70	171.90
T ₄	160.08	160.04	160.08	160.08	160.00	159.96	160.04	160.04	150.08	150.04
T ₅	159.96	160.00	159.96	159.96	159.96	159.92	159.96	159.96	149.96	150.00
T ₆	159.96	159.96	160.00	159.96	160.00	160.00	160.00	160.00	150.00	150.00
T ₇	160.00	160.00	160.00	160.00	160.00	160.00	160.00	160.00	150.00	150.00
T ₈	97.5	98.0	206.0	122.0	116.0	105.0	113.5	113.0	121.0	125.5
T ₉	151.5	162.0	166.5	167.0	167.5	167.0	167.0	167.0	158.5	160.0
T ₁₀	67.5	67.5	67.5	67.5	67.5	67.0	67.0	67.0	71.5	72.0
T ₁₁	70.5	70.5	71.0	71.5	72.0	72.0	72.0	72.0	74.5	76.0
T ₁₂	55.5	56.0	56.0	56.5	57.0	57.0	57.0	57.5	57.0	56.0

RUN	B81*	B82*	B83*	B84*	B85*	B86*	B87*	B88*	B89*	E90*
TIME	25	30	35	30	25	30	35	25	25	30
TEMP	150	150	150	150	150	150	150	150	150	150
PRESS	80	200	190	180	170	160	150	140	130	120
ANAL	0.219	0.469	0.458	0.446	0.434	0.419	0.405	0.388	0.370	0.348
ERR	0.003	0.003	0.003	0.003	0.003	0.003	0.003	0.003	0.003	0.003
TCL	0.04	0.04	0.04	0.04	0.03	0.04	0.04	0.04	0.04	0.04
SFR	0.0446	0.0455	0.0461	0.0468	0.0355	0.0362	0.0420	0.0436	0.0448	0.0467
AFR	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0
P1	1176	2940	2793	2646	2499	2352	2205	2058	1911	1764
P2	3.0	3.0	3.0	3.0	3.0	3.0	3.0	3.0	3.0	3.0
P3	3.0	3.0	3.0	3.0	3.0	3.0	3.0	3.0	3.0	3.0
T ₀	150.00	150.00	150.00	150.00	150.00	150.00	150.00	150.00	150.00	150.00
T ₁	Thermocouple not functioning. See section VII-C.									
T ₂	151.20	152.04	151.96	151.68	151.28	151.40	152.08	151.64	152.08	151.44
T ₃	170.80	170.60	170.50	170.60	170.70	170.70	171.40	171.20	171.00	170.90
T ₄	150.08	150.12	150.08	150.12	150.12	150.08	150.12	150.08	150.12	150.04
T ₅	150.00	150.00	150.00	150.00	150.00	150.00	149.96	150.00	150.00	150.04
T ₆	150.04	150.00	150.04	150.00	150.04	150.00	150.04	150.04	150.04	150.04
T ₇	150.00	150.00	150.00	150.00	150.00	150.00	150.00	150.00	150.00	150.00
T ₈	115.0	110.0	84.0	84.0	83.5	83.5	83.5	83.5	90.0	90.0
T ₉	157.0	156.5	149.0	144.0	143.0	141.0	139.5	138.5	137.5	136.0
T ₁₀	73.5	74.0	74.5	74.5	75.0	75.5	76.5	76.5	77.0	77.0
T ₁₁	77.0	78.0	78.0	78.5	79.0	79.0	80.0	80.0	80.5	80.5
T ₁₂	58.5	56.0	56.0	58.0	60.0	59.5	61.0	61.0	61.0	61.0

RUN	B91*	B92	B93	B94	B95	B96	B97*	C98	C99*	C100*
TIME	25	25	25	30	25	30	30	30	25	25
TEMP	150	150	150	150	150	150	150	150	150	150
PRESS	110	120	140	160	180	200	90	110	100	70
ANAL	0.324	0.349	0.388	0.420	0.447	0.468	0.259	0.325	0.295	0.172
ERR	0.003	0.003	0.003	0.003	0.003	0.003	0.003	0.002	0.002	0.002
TCL	0.04	0.04	0.04	0.03	0.04	0.04	0.04	0.04	0.04	0.04
SFR	0.0430	0.0385	0.0359	0.0435	0.0351	0.0340	0.0396	0.0515	0.0455	0.0675
AFR	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0
P1	1617	1764	2058	2352	2646	2940	1323	1617	1470	1029
P2	3.0	3.0	3.0	3.0	3.0	3.0	3.0	3.0	3.0	3.0
P3	3.0	3.0	3.0	3.0	3.0	3.0	3.0	3.0	3.0	3.0
T ₀	150.00	150.00	150.00	150.00	150.00	150.00	150.00	150.00	150.00	150.00
T ₁	Thermocouple not functioning. See section VII-C.									
T ₂	151.68	151.32	151.68	152.24	151.52	151.72	151.64	151.48	150.84	151.08
T ₃	170.90	170.90	171.10	170.90	171.20	171.10	171.00	174.00	172.40	171.50
T ₄	150.08	150.08	150.04	150.04	150.04	150.08	150.04	150.08	150.04	150.04
T ₅	150.00	150.00	149.96	149.96	150.00	149.96	150.00	150.00	150.04	150.00
T ₆	150.04	150.04	150.00	150.00	150.04	150.00	150.00	149.96	150.00	150.00
T ₇	150.00	150.00	150.00	150.00	150.00	150.00	150.00	150.00	150.00	150.00
T ₈	110.0	146.0	146.0	106.0	106.0	107.0	107.0	126.5	83.0	85.0
T ₉	137.5	153.0	157.0	157.5	157.0	156.4	156.5	157.0	156.0	155.0
T ₁₀	77.0	76.5	76.5	76.5	76.5	76.5	76.5	69.0	67.5	67.5
T ₁₁	81.0	80.0	80.0	80.0	80.0	80.0	80.0	71.0	71.0	71.0
T ₁₂	60.0	61.0	61.0	61.0	61.0	61.0	61.0	56.0	56.5	56.5

RUN	C101*	C102*	C103*	C104*	C105*	C106*	C107*	C108*	C109*	C110*
TIME	30	25	35	25	25	30	25	30	25	30
TEMP	140	140	140	140	140	140	140	140	140	140
PRESS	90	80	200	190	180	170	160	150	140	130
ANAL	0.148	0.132	0.237	0.232	0.227	0.221	0.215	0.208	0.200	0.192
ERR	0.002	0.002	0.002	0.002	0.002	0.002	0.002	0.002	0.002	0.002
TCL	0.04	0.04	0.03	0.04	0.03	0.03	0.04	0.04	0.04	0.04
SFR	0.0392	0.0424	0.0394	0.0377	0.0381	0.0367	0.0372	0.0374	0.0359	0.0367
AFR	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0
P1	1323	1176	2940	2793	2646	2499	2352	2205	2058	1911
P2	3.0	3.0	3.0	3.0	3.0	3.0	3.0	3.0	3.0	3.0
P3	3.0	3.0	3.0	3.0	3.0	3.0	3.0	3.0	3.0	3.0
T ₀	140.00	140.00	140.00	140.00	140.00	140.00	140.00	140.00	140.00	140.00
T ₁	Thermocouple not functioning. See section VII-C.									
T ₂	141.40	141.76	141.64	141.44	141.48	141.52	141.52	141.56	141.28	141.48
T ₃	164.70	163.80	164.70	164.30	164.10	163.70	163.50	163.30	163.30	163.20
T ₄	140.08	140.04	140.04	140.08	140.04	140.04	140.00	140.04	140.04	140.00
T ₅	140.00	140.04	140.00	140.04	139.96	140.00	139.96	140.00	139.96	140.00
T ₆	140.00	140.00	140.04	140.00	140.00	140.00	140.00	140.00	139.96	139.96
T ₇	140.00	140.00	140.00	140.00	140.00	140.00	140.00	140.00	140.00	140.00
T ₈	83.0	82.5	82.0	82.5	82.0	82.0	82.0	82.0	82.0	82.0
T ₉	141.0	140.0	138.5	137.5	136.0	134.0	132.5	131.0	129.5	128.0
T ₁₀	67.5	67.5	67.5	67.0	67.0	66.5	66.5	66.5	66.0	66.0
T ₁₁	71.5	71.5	71.5	71.5	71.5	71.0	71.0	71.0	71.0	71.0
T ₁₂	56.0	57.0	55.5	57.5	57.5	57.0	57.0	57.0	57.0	57.0

RUN	C111*	C112*	C113	C114	C115	C116	C117	C118	C119	C120*
TIME	25	30	25	25	25	25	30	25	25	25
TEMP	140	140	140	140	140	140	140	140	140	140
PRESS	120	110	160	180	200	70	100	120	140	100
ANAL	0.133	0.173	0.215	0.227	0.236	0.112	0.162	0.184	0.200	0.163
ERR	0.002	0.002	0.002	0.002	0.002	0.001	0.001	0.001	0.001	0.001
TCL	0.04	0.04	0.03	0.04	0.04	0.04	0.03	0.04	0.03	0.03
SFR	0.0353	0.0340	0.0284	0.0275	0.0269	0.0440	0.0377	0.0283	0.0269	0.0302
AFR	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0
P1	1764	1617	2352	2646	2940	1029	1470	1764	2058	1470
P2	3.0	3.0	3.0	3.0	3.0	3.0	3.0	3.0	3.0	3.0
P3	3.0	3.0	3.0	3.0	3.0	3.0	3.0	3.0	3.0	3.0
T ₀	140.00	140.00	140.00	140.00	140.00	140.00	140.00	140.00	140.00	140.00
T ₁	Thermocouple not functioning. See section VII-C.									
T ₂	141.44	141.36	140.88	141.00	140.96	141.28	141.32	141.20	141.28	140.92
T ₃	162.80	162.80	163.40	163.20	163.40	161.50	161.90	162.20	162.70	162.20
T ₄	140.04	140.04	140.04	140.00	140.04	140.08	140.04	140.04	140.04	140.00
T ₅	140.00	139.96	139.96	140.00	139.96	140.04	140.04	140.00	139.96	139.96
T ₆	140.00	140.00	139.96	140.00	140.00	140.00	140.00	140.00	140.00	140.00
T ₇	140.00	140.00	140.00	140.00	140.00	140.00	140.00	140.00	140.00	140.00
T ₈	132.5	132.0	132.5	115.0	121.0	116.0	116.0	116.0	116.0	116.0
T ₉	134.5	139.0	145.0	146.0	147.5	147.0	147.0	146.5	146.5	146.5
T ₁₀	66.0	66.0	65.5	65.5	65.0	64.5	64.5	64.5	64.5	64.0
T ₁₁	70.5	70.5	70.0	70.0	70.0	70.0	70.0	70.0	70.0	70.0
T ₁₂	57.0	56.5	56.0	56.5	57.0	56.5	56.5	56.0	55.5	56.0

APPENDIX B
ITERATIVE METHOD

1. Newton Method for One Equation in One Unknown

Consider a function of one variable,

$$F(v) = 0 \quad (B-1)$$

In order to find the value of v satisfying (B-1), assume some initial value, say v_1 . In general, $F_1(v)$ will be unequal to zero, since v_1 is not the true solution. Then it will be necessary to correct $F_1(v)$ by some amount $\Delta F_1(v)$, in order to equate it to zero. That is,

$$F_1(v) + \Delta F_1(v) = 0 \quad (B-2)$$

The problem now reduces to that of specifying $\Delta F_1(v)$ in terms of a correction Δv_1 in v which will give a closer approximation to the solution of (B-1) than did the first trial v_1 . This relation can be given by

$$\Delta F_1(v) \approx \left[\frac{dF(v)}{dv} \right]_1 \Delta v_1 = F'_1(v) \cdot \Delta v_1 \quad (B-3)$$

Now, equating (B-2) and (B-3),

$$\Delta v_1 = \frac{-F_1(v)}{F'_1(v)} \quad (B-4)$$

or, the value v_2 for the next trial is given by

$$v_2 = v_1 - \frac{F_1(v)}{F'_1(v)} \quad (B-5)$$

In general, for the i -th trial,

$$v_{i+1} = v_i - \frac{F_i(v)}{F'_i(v)} \quad (B-6)$$

The iterative process is repeated until the error Δv_i is less than the desired amount.

2. The Newton-Raphson Method for Two Equations in Two Unknowns

Consider two non-linear functions in two variables,

$$F_A(y, v) = 0 \quad (B-7)$$

$$F_B(y, v) = 0 \quad (B-8)$$

In order to simultaneously solve the above set for y, v , assume some initial values y_1, v_1 . Since these values will, in general not satisfy (B-7), (B-8), the functions $F_{A_1}(y, v)$ and $F_{B_1}(y, v)$ must be corrected by some amounts to equal zero, or

$$F_{A_1}(y, v) + \Delta F_{A_1}(y, v) = 0 \quad (B-9)$$

$$F_{B_1}(y, v) + \Delta F_{B_1}(y, v) = 0 \quad (B-10)$$

Corrections to y_1 and v_1 to give improved values can be related to the function corrections in a manner analogous to that for one equation in one unknown, since the corrections can be expressed as:

$$\Delta F_{A_1}(y, v) \approx \left[\frac{\partial F_A(y, v)}{\partial y} \right]_1 \Delta y_1 + \left[\frac{\partial F_A(y, v)}{\partial v} \right]_1 \Delta v_1 \quad (B-11)$$

$$\Delta F_{B_1}(y, v) \approx \left[\frac{\partial F_B(y, v)}{\partial y} \right]_1 \Delta y_1 + \left[\frac{\partial F_B(y, v)}{\partial v} \right]_1 \Delta v_1 \quad (B-12)$$

Substituting (B-9) into (B-11) and (B-10) into (B-12) results in

$$\left[\frac{\partial F_A(y,v)}{\partial y} \right]_1 \Delta y_1 + \left[\frac{\partial F_A(y,v)}{\partial v} \right]_1 \Delta v_1 = -F_{A_1}(y,v) \quad (B-13)$$

$$\left[\frac{\partial F_B(y,v)}{\partial y} \right]_1 \Delta y_1 + \left[\frac{\partial F_B(y,v)}{\partial v} \right]_1 \Delta v_1 = -F_{B_1}(y,v) \quad (B-14)$$

The resulting set of linear correction Equations (B-13), (B-14), can be solved simultaneously for the corrections Δy_1 , Δv_1 , by matrix inversion. Therefore,

$$\Delta y_1 = \frac{D_{y_1}}{D_1}, \quad \Delta v_1 = \frac{D_{v_1}}{D_1} \quad (B-15)$$

where

$$D_1 = \begin{vmatrix} \left[\frac{\partial F_A(y,v)}{\partial y} \right]_1 & \left[\frac{\partial F_A(y,v)}{\partial v} \right]_1 \\ \left[\frac{\partial F_B(y,v)}{\partial y} \right]_1 & \left[\frac{\partial F_B(y,v)}{\partial v} \right]_1 \end{vmatrix}$$

$$D_{y_1} = \begin{vmatrix} -F_{A_1}(y,v) & \left[\frac{\partial F_A(y,v)}{\partial v} \right]_1 \\ -F_{B_1}(y,v) & \left[\frac{\partial F_B(y,v)}{\partial v} \right]_1 \end{vmatrix} \quad (B-16)$$

$$D_{v_1} = \begin{vmatrix} \left[\frac{\partial F_A(y,v)}{\partial y} \right]_1 & -F_{A_1}(y,v) \\ \left[\frac{\partial F_B(y,v)}{\partial y} \right]_1 & -F_{B_1}(y,v) \end{vmatrix}$$

the values y_2, v_2 for the second trial are thus given by

$$\begin{aligned}y_2 &= y_1 + D_{y_1}/D_1 \\v_2 &= v_1 + D_{v_1}/D_1\end{aligned}\tag{B-17}$$

In general, for the i -th trial

$$\begin{aligned}y_{i+1} &= y_i + D_{y_i}/D_i \\v_{i+1} &= v_i + D_{v_i}/D_i\end{aligned}\tag{B-18}$$

in which the D 's represent the determinants of the i -th trial, as in (B-16). The iterative procedure is repeated until both of the errors $\Delta y_i, \Delta v_i$, are as small as desired.

This method can be converted quite easily to a logarithmic form if it is found necessary to improve the convergence or preclude the possibility of y or v becoming negative during the course of the solution.

NOTE: The material presented in this appendix has been taken from reference 25 with permission of the author.

APPENDIX C

FORM OF ANALYTICAL EQUATIONS USED IN NEWTON AND NEWTON-RAPHSON ITERATIVE METHODS OF SOLUTION

In order to solve a single equation in one unknown by the Newton method, the equation must be of the form

$$F(v) = 0 \quad (C-1)$$

as indicated in Appendix B. Similarly, in order to solve two equations in two unknowns by the Newton-Raphson method, the equations must be of the form

$$F_A(y,v) = 0 \quad (C-2)$$

$$F_B(y,v) = 0 \quad (C-3)$$

this also being indicated in Appendix B. However, the sets of equations presented in Chapter V for the simplified and general solutions using the Beattie-Bridgeman, Benedict-Webb-Rubin, Martin-Hou and virial equations of state are not in this form. Therefore, they must be transformed into the form indicated in equations (C-1) thru (C-3) since these equations are to be solved by the Newton and Newton-Raphson method. This may be accomplished by transposing all terms in a given equation to one side of the equal sign and equating the resulting expression to zero. The equations resulting from this transposition on the expressions presented in Chapter V are included in the following sections. These equations shall be referred to in Appendix D where the

computer flow diagrams for the solutions of the equations are presented.

1. Beattie-Bridgeman Equation of State

For the simplified case, equation (5-48) is expressed as

$$F(v) = \frac{RT}{v} + \frac{\beta_2}{v^2} + \frac{\gamma_2}{v^3} + \frac{\delta_2}{v^4} - P = 0 \quad (C-4)$$

and for the general case, equation (5-13) is rewritten as

$$F_A(y_1, v) = \frac{RT}{v} + \frac{\beta_m}{v^2} + \frac{\gamma_m}{v^3} + \frac{\delta_m}{v^4} - P = 0 \quad (C-5)$$

and (5-42) as

$$\begin{aligned} F_B(y_1, v) = & RT \ln \frac{y_1 v_1^0}{v} - v_1^s (P - P_1^0) - \frac{2 \beta_{11}}{v_1^0} \\ & - \frac{3 \gamma_{11}}{2(v_1^0)^2} - \frac{4 \delta_{11}}{3(v_1^0)^3} + 2 y_1 [\beta_{11} - \beta_{12}] / v \\ & + [2y_1(\overline{\gamma}_{m,11} - \overline{\gamma}_{m,12}) + 2\overline{\gamma}_{m,12} - \gamma_m] / 2v^2 \\ & + 2 [y_1(\overline{\delta}_{m,m,11} - \overline{\delta}_{m,m,12}) + \overline{\delta}_{m,m,12} - \delta_m] \\ & / 3v^3 = 0 \end{aligned} \quad (C-6)$$

2. Benedict-Webb-Rubin Equation of State

For the simplified case, equation (5-29) is rewritten as

$$\begin{aligned} F(v) = & \frac{RT}{v} + \frac{B_0 - A_0 - C_0/T^2}{v^2} + \frac{b_2 RT - a_2}{v^3} + \frac{a_2 \alpha_2}{v^6} \\ & + \frac{C_2(1 + \gamma_2/v^2) \exp(-\gamma_2/v^2)}{T^2 v^3} - P = 0 \end{aligned} \quad (C-7)$$

and for the general case, equation (5-56) is expressed as

$$F_A(y_1, v) = \frac{RT}{v} + \frac{B_{0m} - A_{0m} - C_{0m}/T^2}{v^2} + \frac{b_m RT - a_m}{v^3} + \frac{a_m \alpha_m}{v^6} \\ + \frac{c_m(1 + \gamma_m/v^2) \exp(-\gamma_m/v^2)}{T^2 v^3} - P = 0 \quad (C-8)$$

and (5-91) as

$$F_B(y_1, v) = RT \ln \frac{y_1 v_1^0}{v} - v_1^s (P - P_1^0) - \frac{2A_{11}}{v_1^0} - \frac{3B_{111}}{2(v_1^0)^2} \\ - \frac{6a_{111}\alpha_{111}}{5(v_1^0)^5} - \frac{c_{111}}{T^2} - \frac{c_{111}}{T^2} \exp[-\gamma_{11}/(v_1^0)^2] \\ \left[\frac{\gamma_{11}}{(v_1^0)^4} + \frac{1}{2(v_1^0)^2} - \frac{1}{\gamma_{11}} \right] + \frac{2[y_1(A_{11} - A_{12}) + A_{12}]}{v} \\ + \frac{3[y_1^2(B_{111} - 2B_{112} + B_{122}) + 2y_1(B_{112} - B_{122}) + B_{122}]}{2v^2} \\ + \frac{3(a_m \alpha_{1mm} + \alpha_m a_{1mm})}{5v^5} + \frac{3c_{1mm}}{T^2 \gamma_m} + \frac{2c_m \gamma_{1m}}{T^2 \gamma_m^2} \\ - \frac{\exp(-\gamma_m/v^2)}{T^2} \left[\frac{3c_{1mm}}{\gamma_m} - \frac{2c_m \gamma_{1m}}{\gamma_m^2} + \frac{3c_{1mm}}{2v^2} \right. \\ \left. - \frac{2c_m \gamma_{1m}}{\gamma_m v^2} - \frac{c_m \gamma_{1m}}{v^4} \right] = 0 \quad (C-9)$$

3. Martin-Hou Equation of State

For the simplified case, equation (5-120) is rewritten as

$$F(v) = \frac{RT}{v-b_{22}} + \frac{F_{222}}{(v-b_{22})^2} + \frac{F_{322}}{(v-b_{22})^3} + \frac{A_{422}}{(v-b_{22})^4} + \frac{F_{522}}{(v-b_{22})^5} - P = 0 \quad (C-10)$$

and for the general case, equation (5-102) is expressed as

$$F_A(y_1, v) = \frac{RT}{v-b_m} + \frac{F_{2m}}{(v-b_m)^2} + \frac{F_{3m}}{(v-b_m)^3} + \frac{A_{4m}}{(v-b_m)^4} + \frac{F_{5m}}{(v-b_m)^5} - P = 0 \quad (C-11)$$

and (5-125) as

$$\begin{aligned} F_B(y_1, v) = & RT \ln \frac{y_1 v_1^0}{v} - v_1^s (P - P_1^0) - RT \ln \frac{v_1^0}{v_1^0 - b_{11}} \\ & + RT \ln \frac{v}{v-b_m} - P_1^0 v_1^0 - \frac{F_{211}}{(v_1^0 - b_{11})} - \frac{F_{311}}{2(v_1^0 - b_{11})^2} \\ & - \frac{A_{411}}{3(v_1^0 - b_{11})^3} - \frac{F_{511}}{4(v_1^0 - b_{11})^4} + P [v + (1-y_1)(b_{11} - b_{22})] \\ & + \frac{\bar{F}_{2m}}{(v-b_m)} + \frac{\bar{F}_{3m}}{2(v-b_m)^2} + \frac{y_1(2-y_1)(A_{411} - 2A_{412} + A_{422})}{3(v-b_m)^3} \\ & + \frac{2A_{412} - A_{422}}{3(v-b_m)^3} + \frac{\bar{F}_{5m}}{4(v-b_m)^4} = 0 \quad (C-12) \end{aligned}$$

4. Virial Equation of State

For the simplified case, equation (5-146) is expressed as

$$F(v) = \frac{RT}{v} + \frac{RTB_{22}}{v^2} + \frac{RTC_{222}}{v^3} - P = 0 \quad (C-13)$$

and for the general case, equation (5-135) is rewritten as

$$F_A(y_1, v) = \frac{RT}{v} + \frac{RTB_m}{v^2} + \frac{RTC_m}{v^3} - P = 0 \quad (C-14)$$

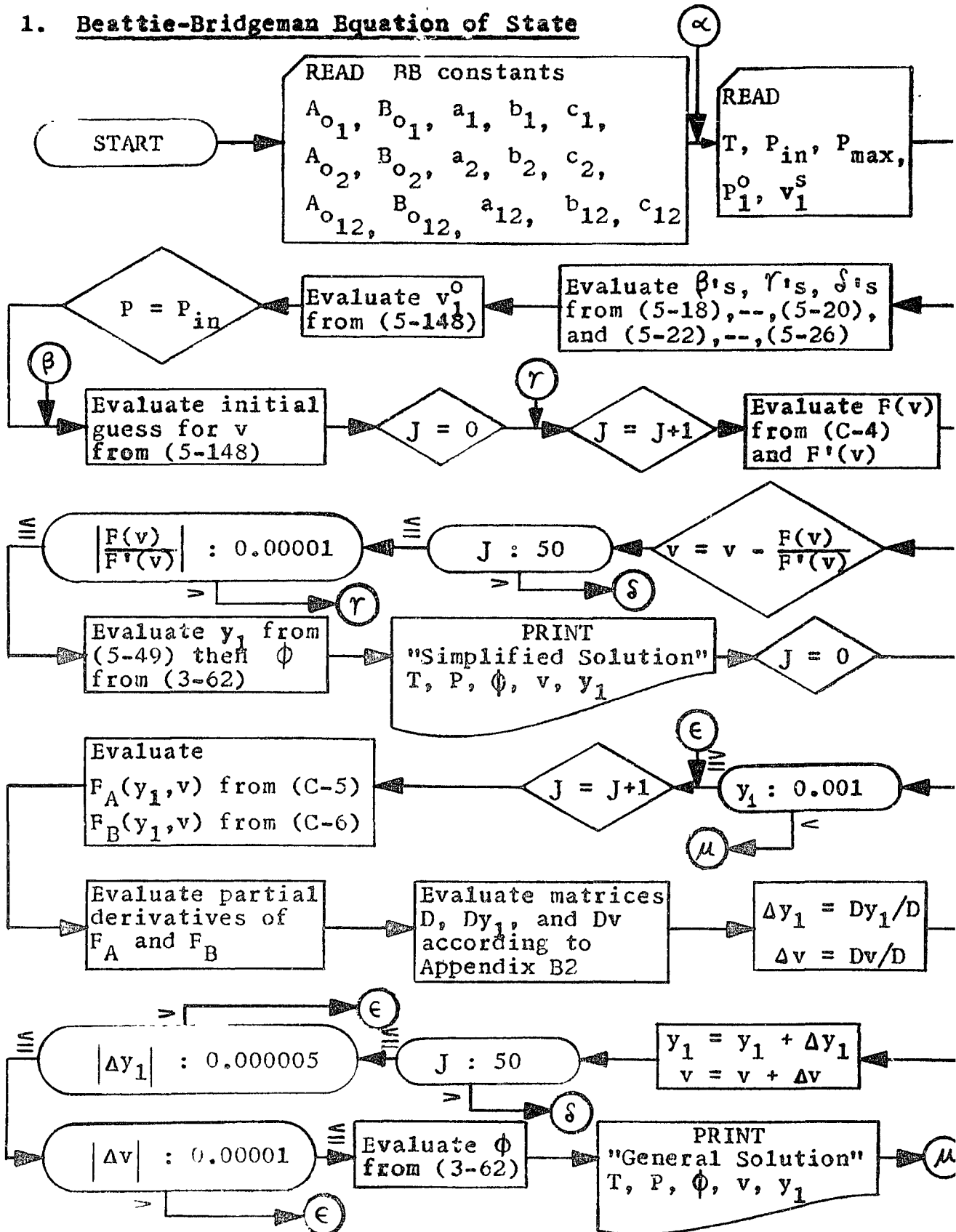
and (5-145) as

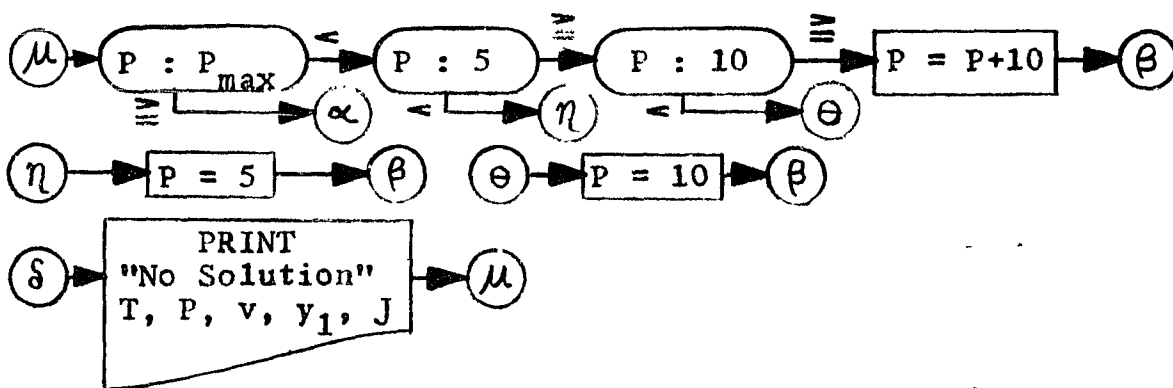
$$\begin{aligned} F_B(y_1, v) = & RT \ln \frac{y_1 v_1^0}{v} - v_1^s (P - P_1^0) - \frac{2RTB_{11}}{v_1^0} \\ & - \frac{3RTC_{111}}{2(v_1^0)^2} + \frac{2RT}{v} \left[y_1 (B_{11} - B_{12}) + B_{12} \right] - \frac{3RT}{2v^2} \\ & \left[y_1^2 (C_{111} - 2C_{112} + C_{122}) + 2y_1 (C_{112} - C_{122}) + C_{122} \right] = 0 \end{aligned} \quad (C-15)$$

APPENDIX D

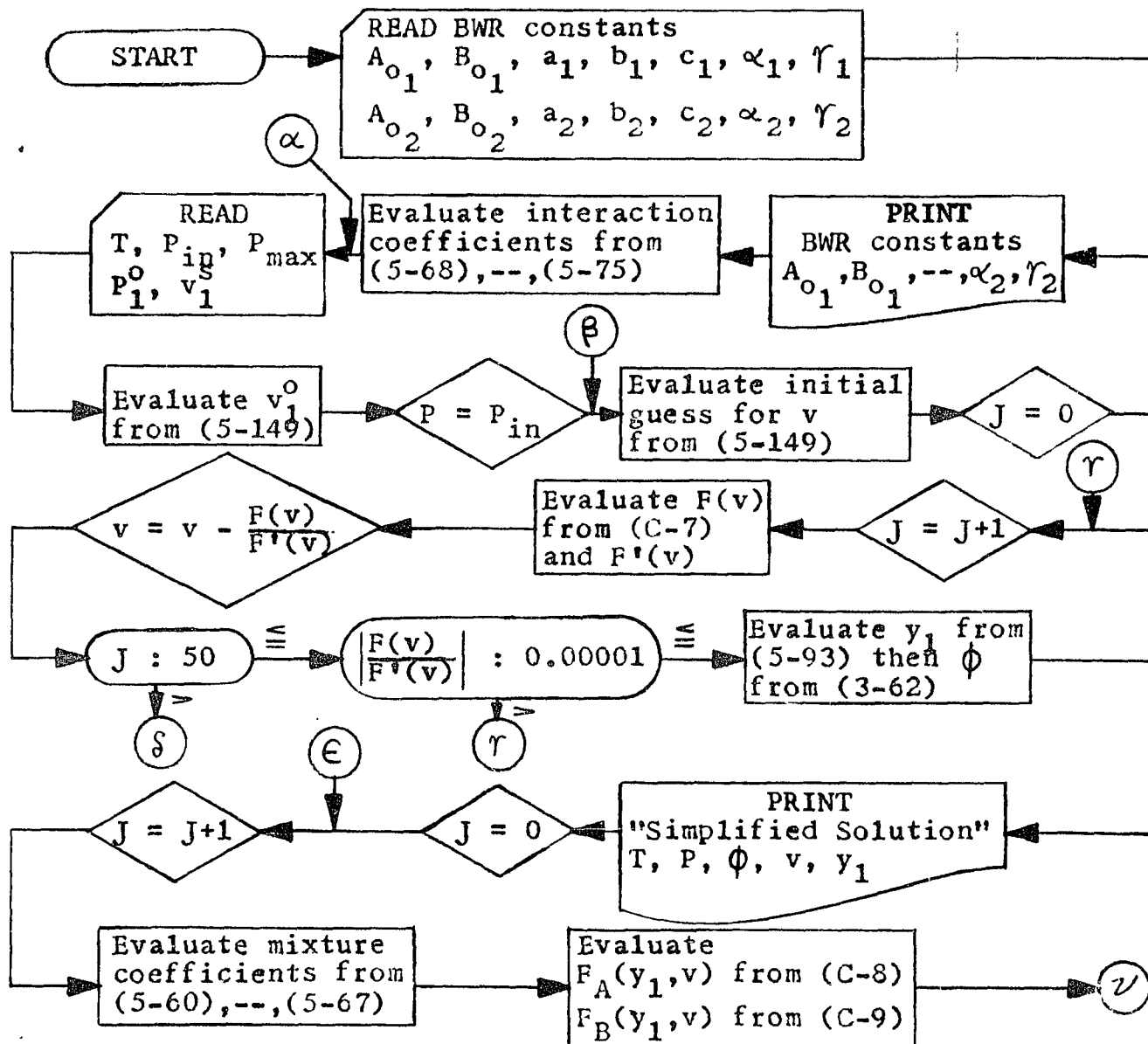
COMPUTER FLOW DIAGRAMS FOR ANALYTICAL SOLUTIONS

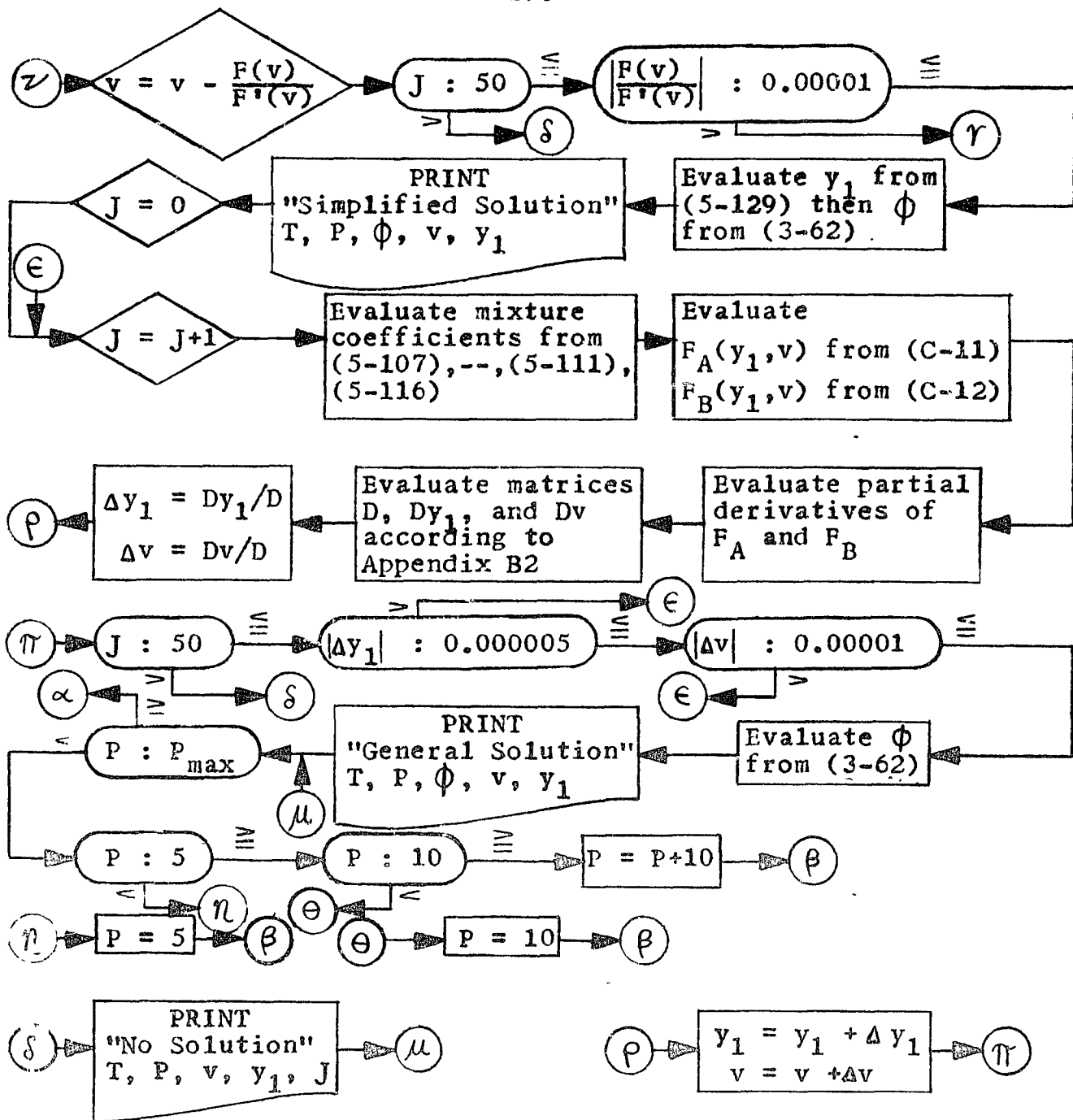
1. Beattie-Bridgeman Equation of State



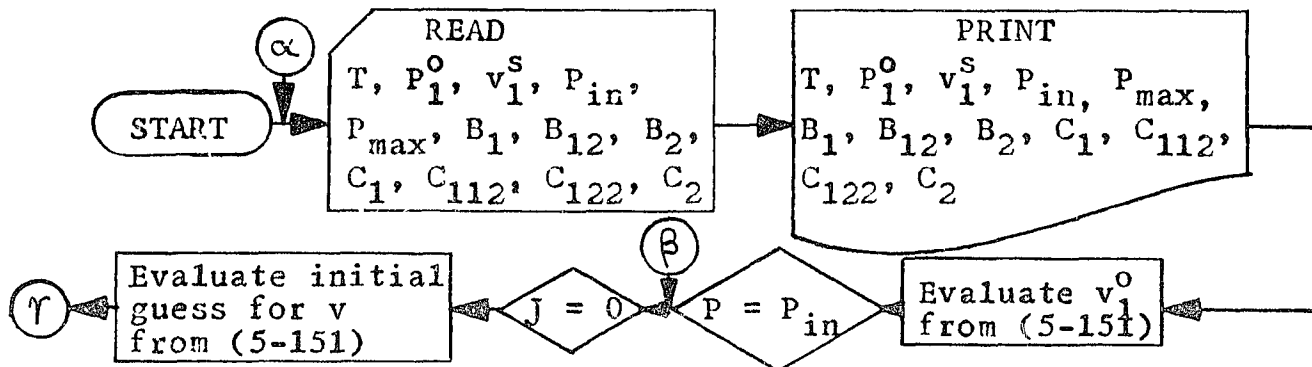


2. Benedict-Webb-Rubin Equation of State





4. Virial Equation of State



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